The working (6,7): a verified graphical identity

A forward derivation with connected braid strips, every sphere move identified, and an independent boundary-action ending.

BRIEF SUMMARY: (6,7) -> fifty -> an actual standard twenty plus thirty -> two verified sphere identities.

The new 154-letter representative is our working (6,7). Join it to the saved calculation chart, follow the checked named substitutions to fifty positive units, and isolate a standard twenty in seven Hurwitz moves and one sphere rechart. Rewrite the remaining thirty on a five-point disk; its exact based action is the double outer-boundary twist. Cap that double boundary and close the twenty independently.

Proof scope. The ending is an exact factor-level identity certificate using a faithful disk-action calculation. It is not a purely elementary Hurwitz normal form W0 W1. The isolated thirty has permutation orbit sizes 3,2,1, so it cannot alone be Hurwitz-equivalent to W1. The classification theorem is not used as an identity proof.

How to read the figures. Each support is drawn with the shared Arc/Loop library and paired with the braid that represents that pictured support. Blue dots separate factors, and braid strand colors continue downward. A yellow SPHERE SUPPORT CHART label means the display is an explicitly certified sphere representative. Actual sphere derivation moves are labeled SPHERE MOVE. Raw factors, disk discrepancies, global frames and lift checks are retained in the online certificates.

Navigation. This detailed book retains every old atomic substitution instead of hiding it in a group. Its front section is a brief explanation; the subsequent pages give the complete diagrams. The online view offers chapter/group navigation and side-scrolling columns. The final disk cut-system comparison is jointly certified; the other drawings are individual factor supports.

Contents

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All atomic moves retained. Sphere moves labeled explicitly; certificates in Section VI.

What the steps establish

1. Start from the working (6,7). Three inverse adjacent Hurwitz moves, an explicit global change of chart, and exact disk support presentations join the working seed to the saved source. This does not use a sphere identity.

2. Reach the actual positive fifty. The preserved 145 atomic moves use transported PC lantern/half-lantern substitutions, disk Hurwitz moves and explicitly recorded sphere changes. The previous inverse pair is replaced using a squared sphere relation, preserving the genus-two lift.

3. Extract the standard twenty. Move F3 upward past the transported {5,6} pair. Rechart that pair as round T56 on the sphere, then move it unchanged past six neighbors. The resulting Q A Q A block is a disk conjugate of W0, where A=43211234 and Q=55. Its closure is checked independently.

4. Close the thirty in a five-point chart. Put point 6 at infinity. One pair loop becomes its complementary four-point loop; its expanded disk-core length increases by ten. The exact ordered five-point model has forty positive core letters and the same action on all five based meridians as the double outer-boundary twist. That is a faithful disk braid equality.

5. Cap the double boundary. The double cap is a separate sphere move with trivial genus-two lift discrepancy. A single five-point boundary twist has lift -I and would be wrong. No closure hypothesis for the starting word is used in the final disk-action calculation.

I. The working (6,7) and the calculation chart

The working 154-letter factorization joins the literal saved source by checked disk operations and one explicit global conjugation. No sphere substitution occurs in this joining phase.

The working (6,7): complete factorization

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. F1
T3^2



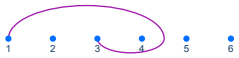
2. F2
H



3. F4
T3^2



4. F3
H



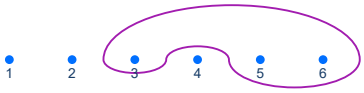
5. F5
T3^2



6. F6
H



7. F8
T3^2



8. F9
H



9. F7
H



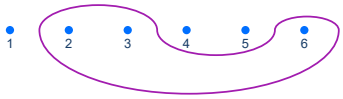
10. F10
T3^2
SPHERE CHART



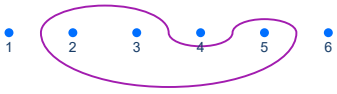
11. F11
H

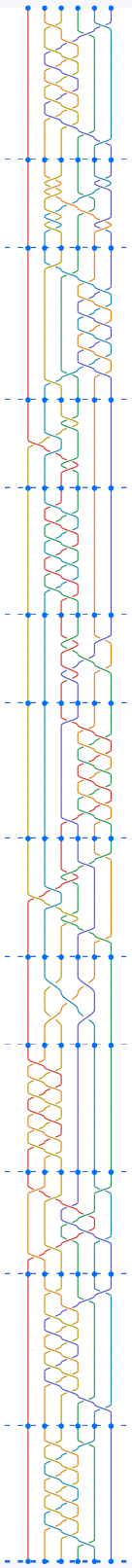


12. F12
T3^2
SPHERE CHART



13. F13
T3^2





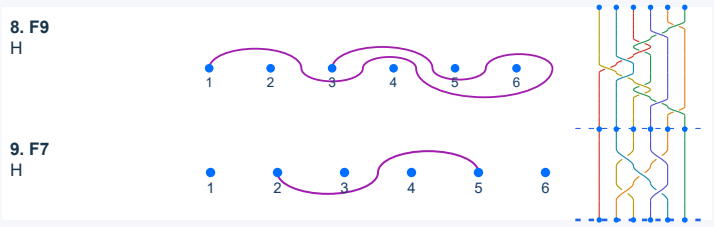
I - step 001

Change the working ordering by an inverse Hurwitz move

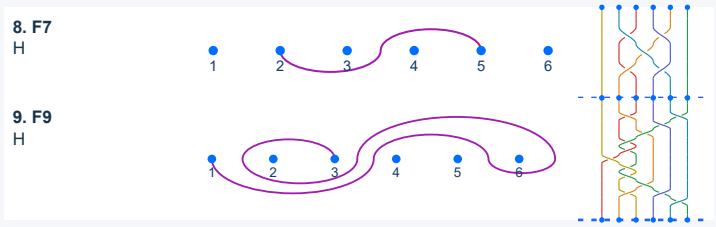
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



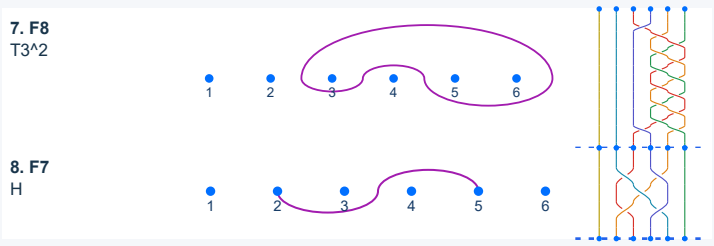
I - step 002

Change the working ordering by an inverse Hurwitz move

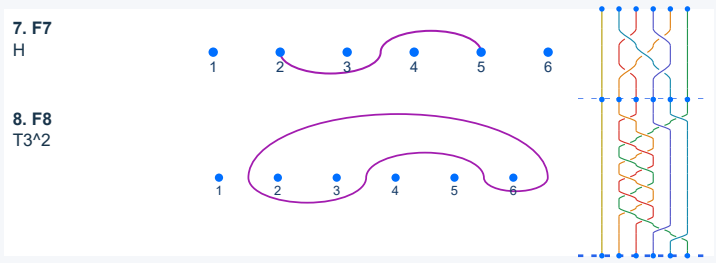
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



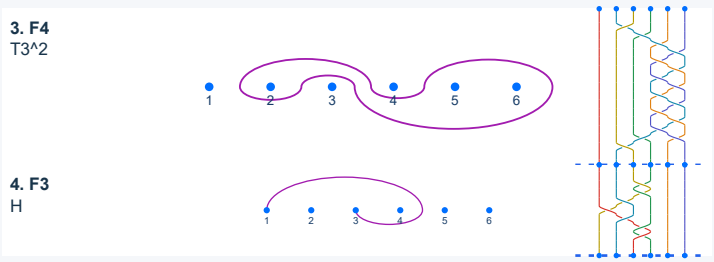
I - step 003

Change the working ordering by an inverse Hurwitz move

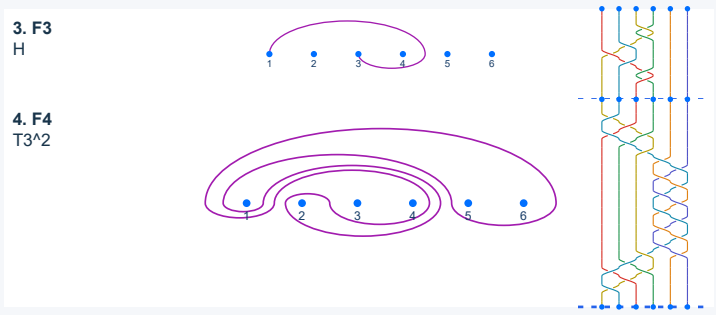
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before



After



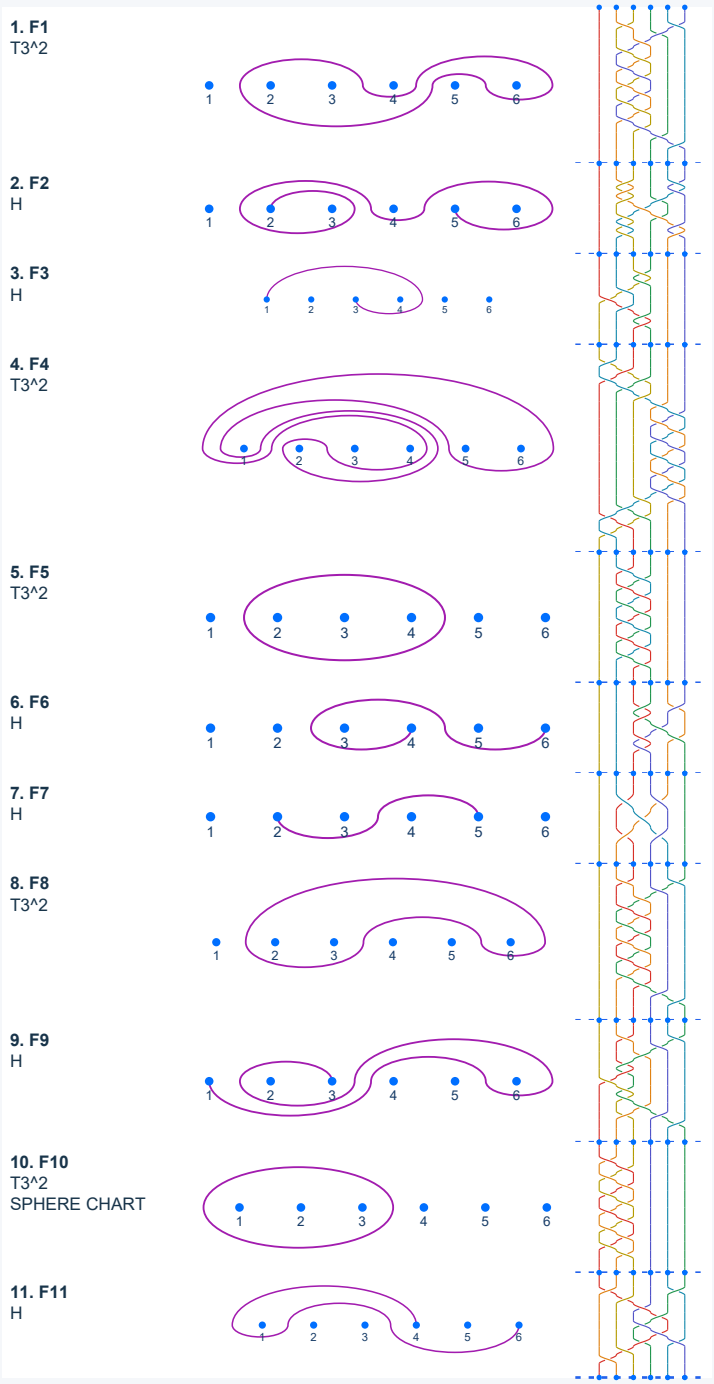
I - step 004

Global conjugation by 3 -5 -2 -3 -2 into the saved calculation chart

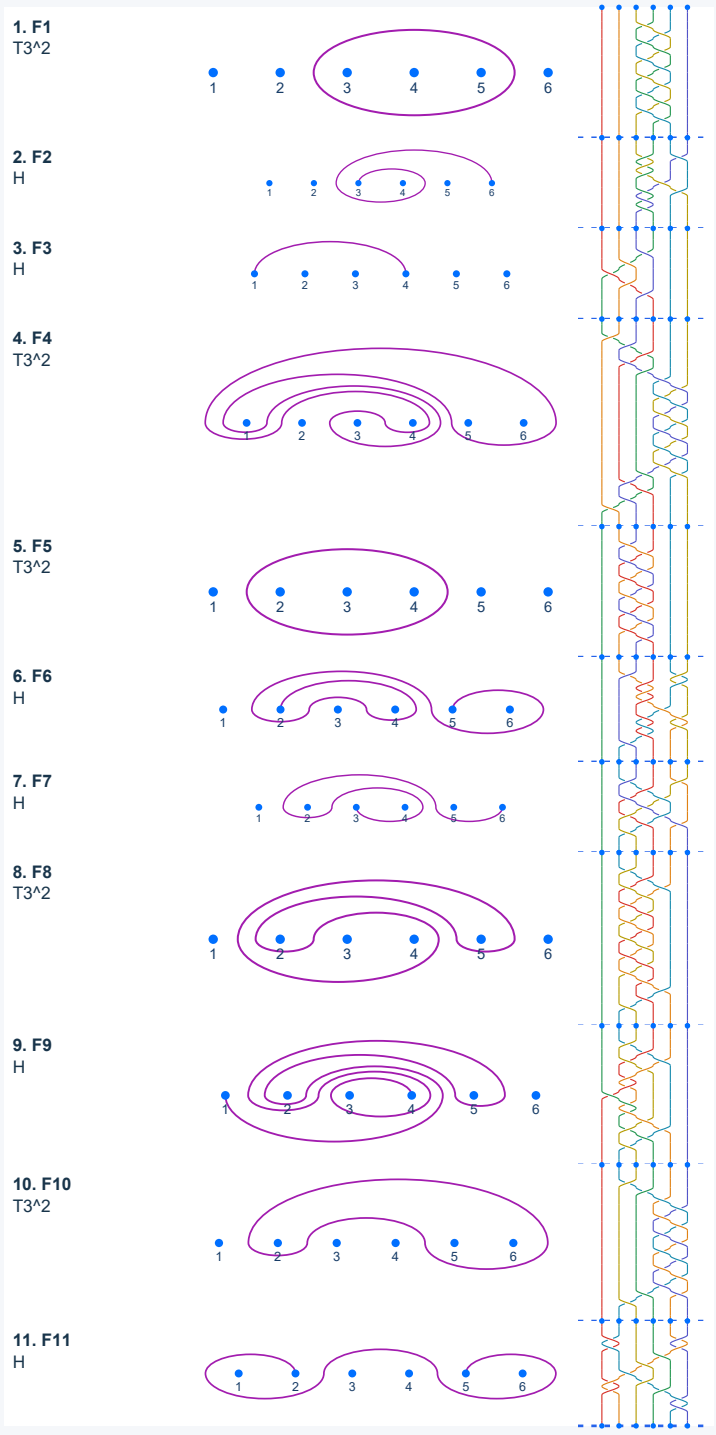
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



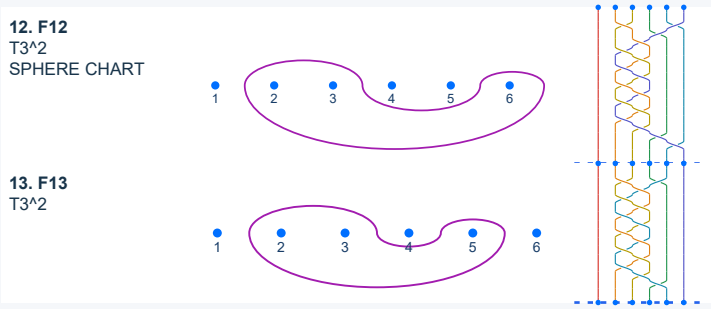
I - step 004 - continued

Global conjugation by 3 -5 -2 -3 -2 into the saved calculation chart

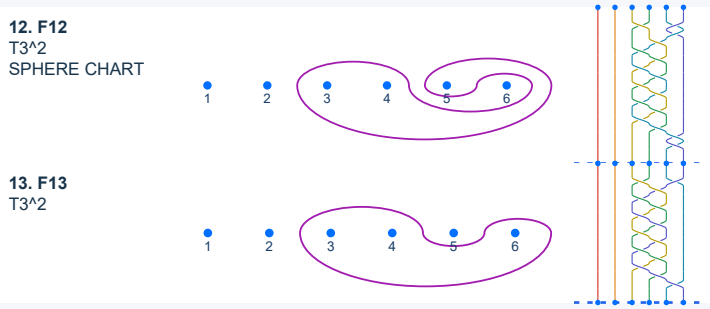
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



I - step 005

Use the literal source presentations of the saved route

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

II. The named-substitution route to fifty

All 145 saved atomic moves are retained in their forward order. Column chart frames and display-only sphere replacements are retained separately from the actual derivation moves.

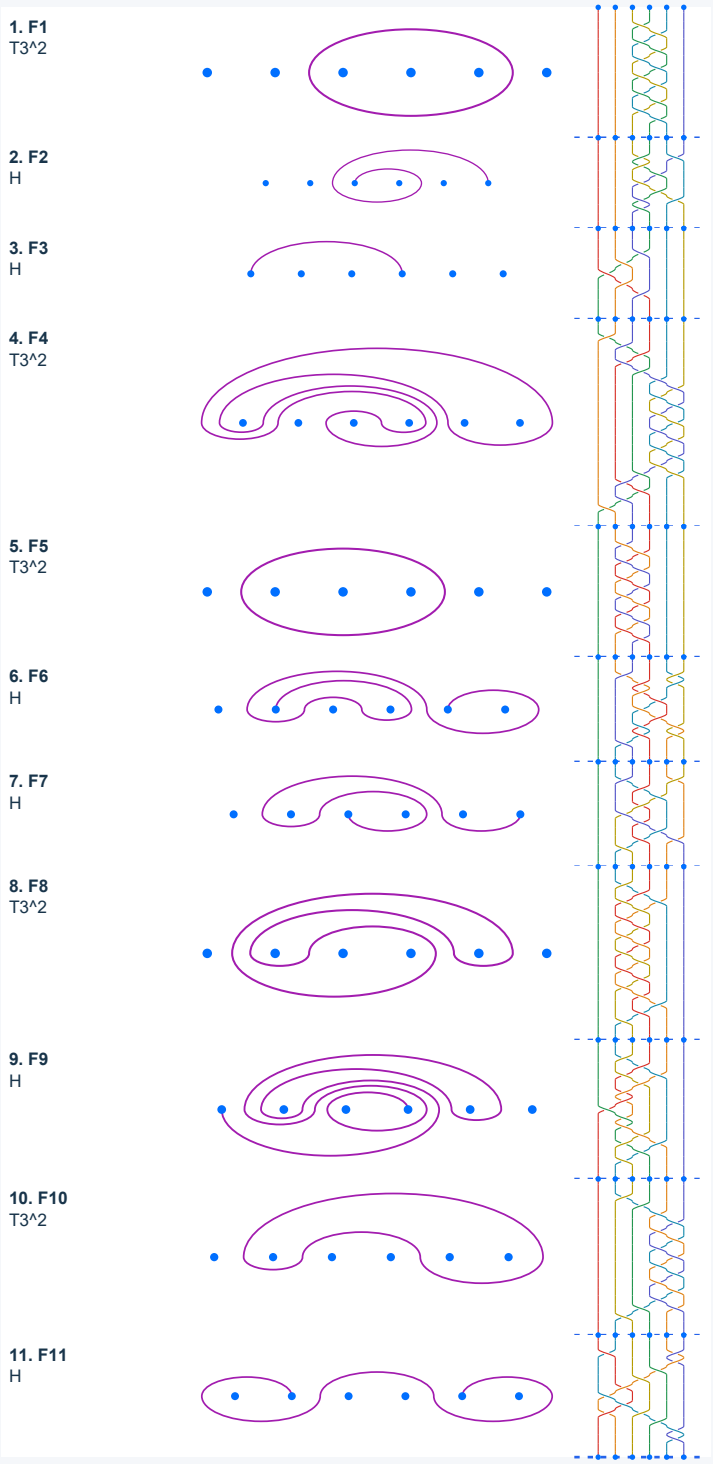
II - step 001

Split F1 into ordinary roots

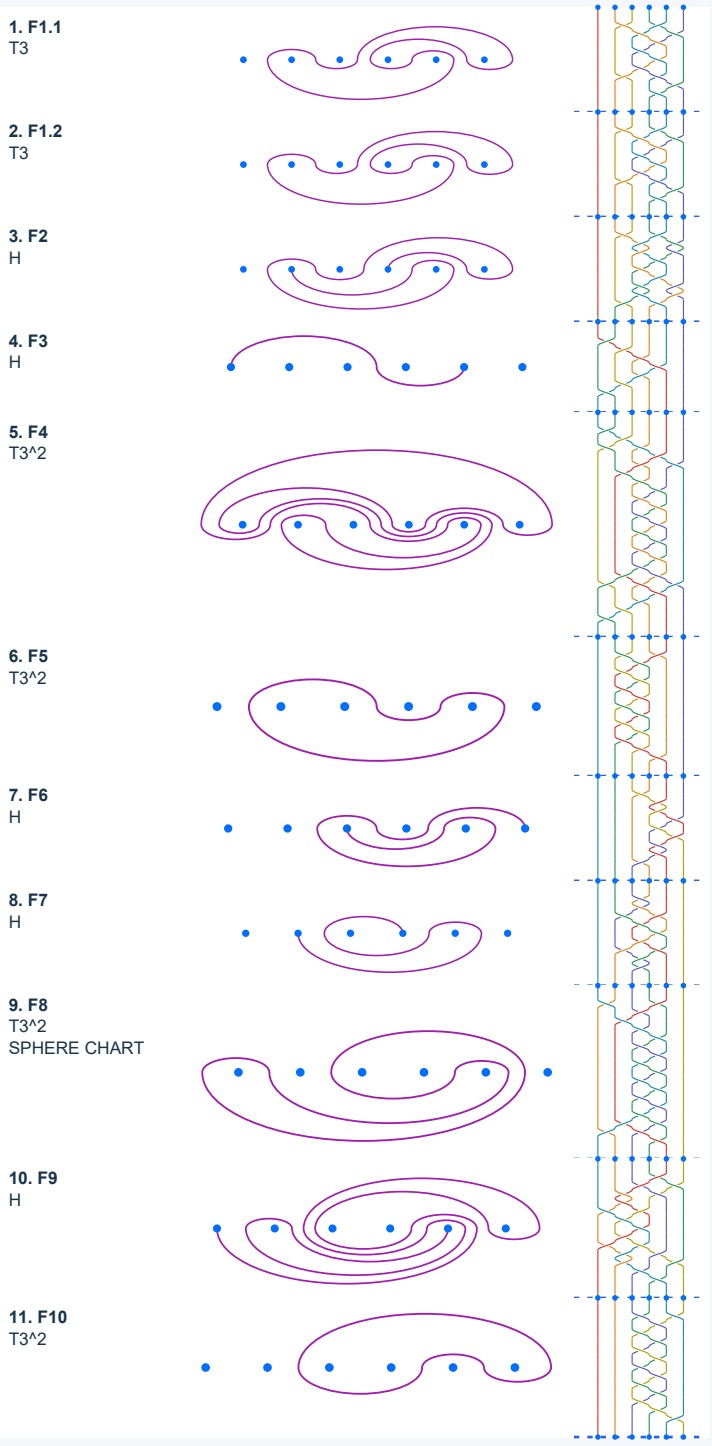
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=identity; after g=2 -4 5.

Before



After



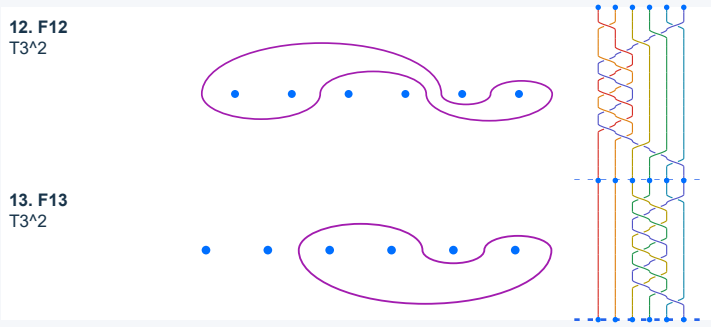
II - step 001 - continued

Split F1 into ordinary roots

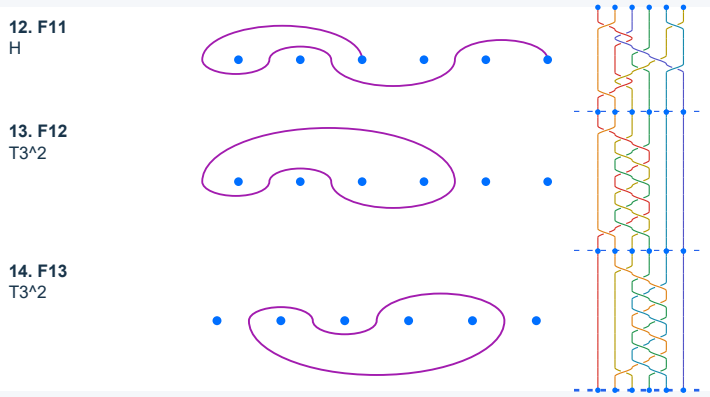
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=identity; after g=2 -4 5.

Before



After



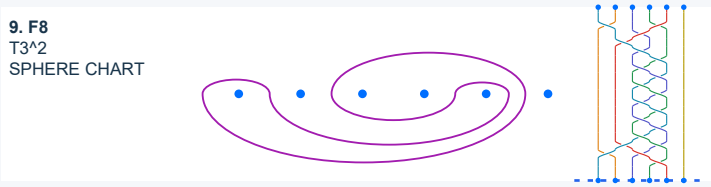
II - step 002

Split F8 into ordinary roots

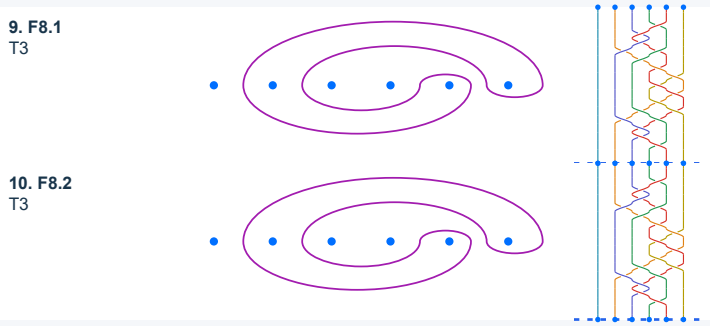
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=2 -4 5.

Before



After



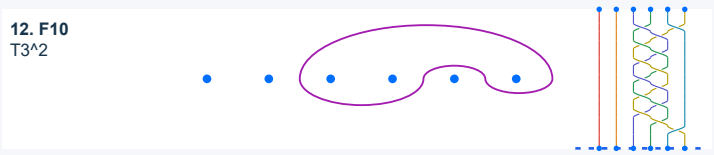
II - step 003

Split F10 into ordinary roots

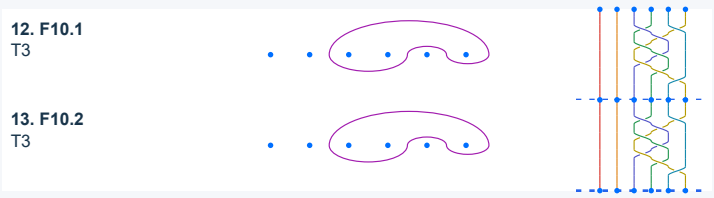
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=2 -4 5.

Before



After



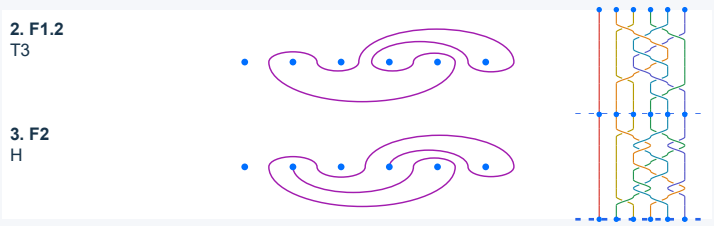
II - step 004

F1.2 down past F2

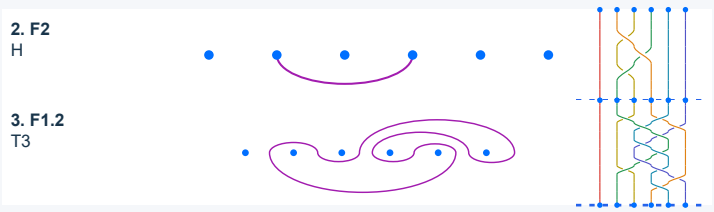
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=2 -4 5.

Before



After



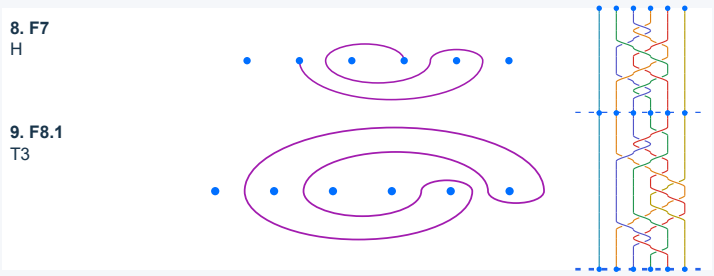
II - step 005

F8.1 up past F7

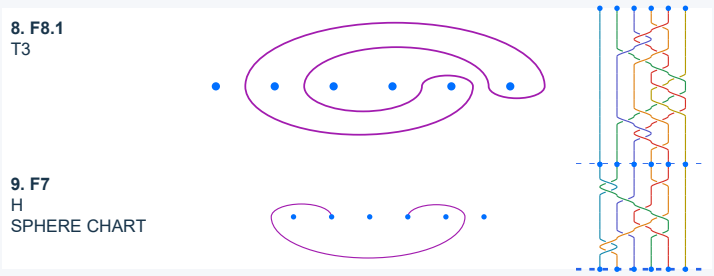
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=2 -4 5.

Before



After



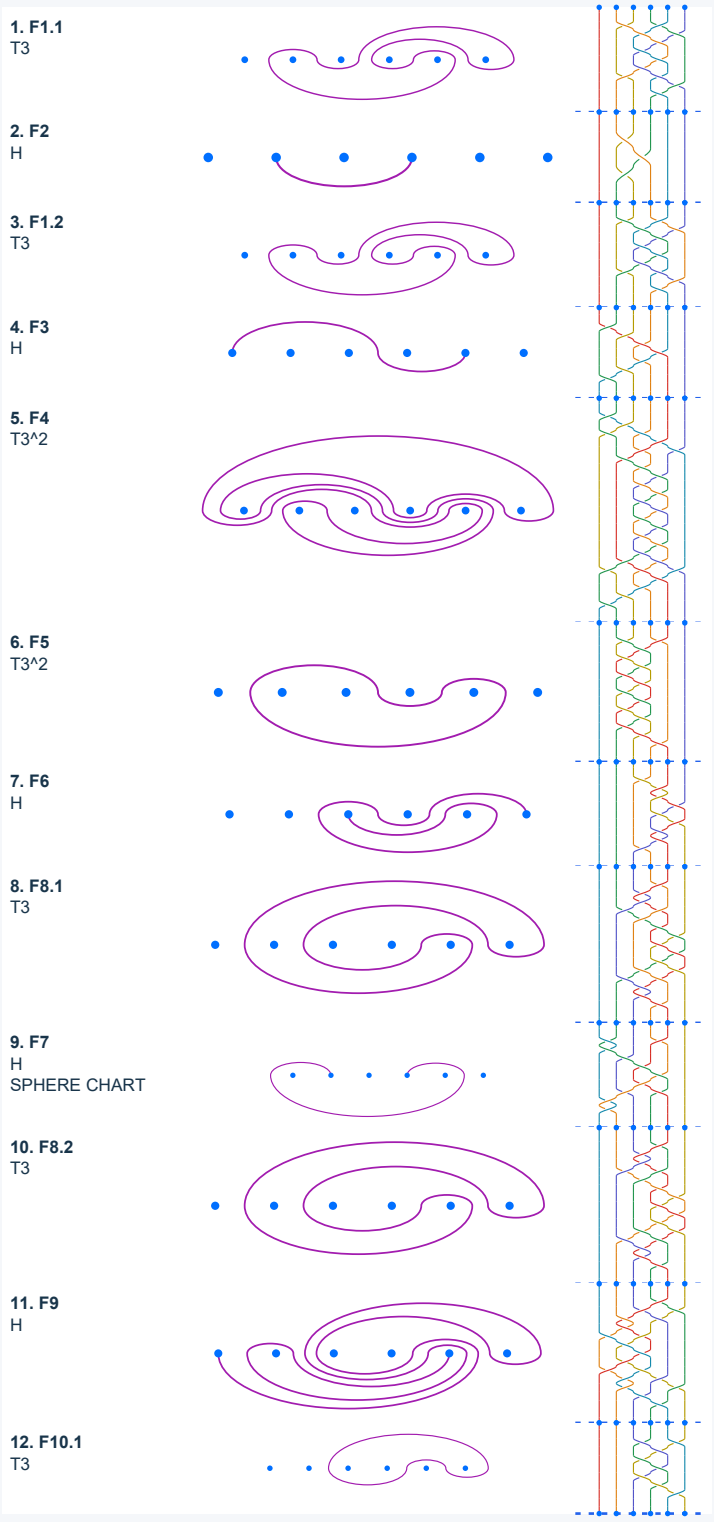
II - step 006

F10.1 up past F9

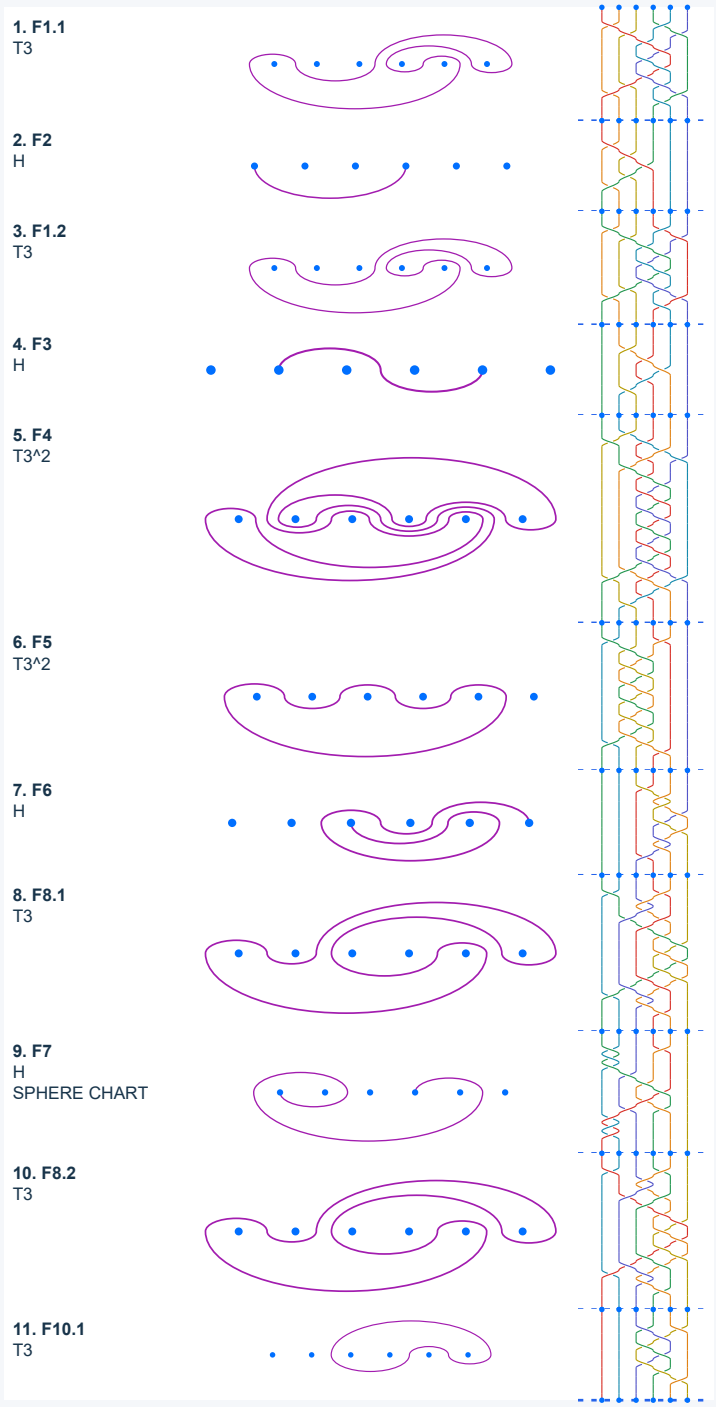
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=1 2 -4 5.

Before



After



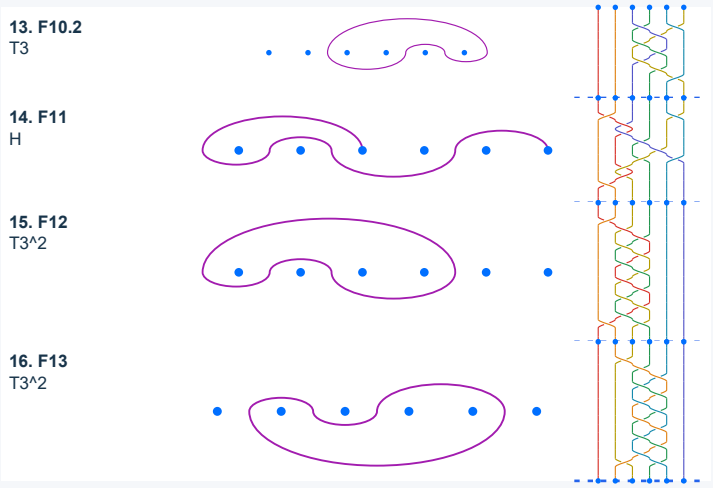
II - step 006 - continued

F10.1 up past F9

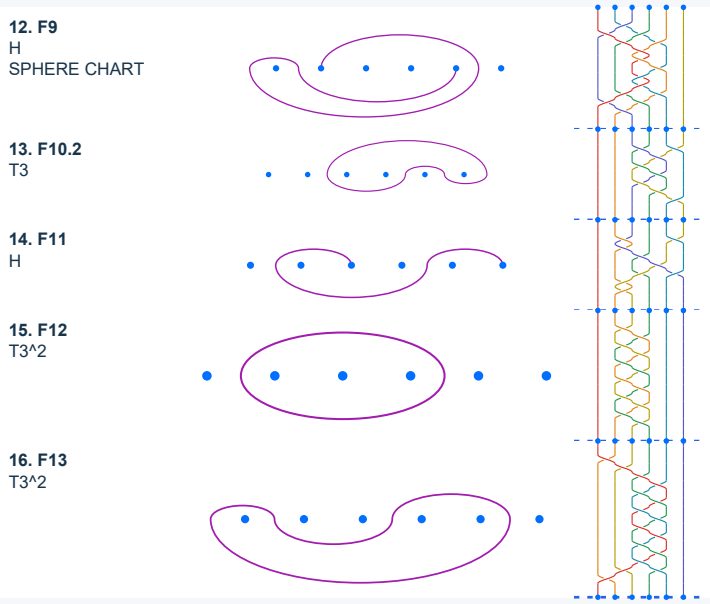
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 -4 5; after g=1 2 -4 5.

Before



After



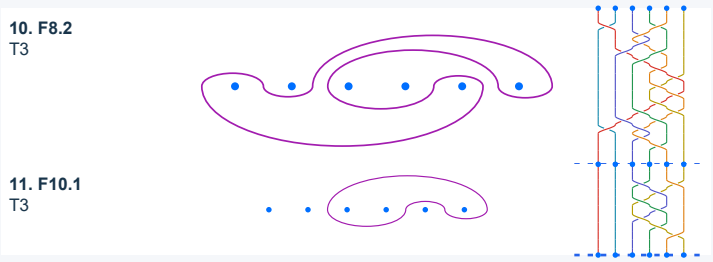
II - step 007

F10.1 up past F8.2

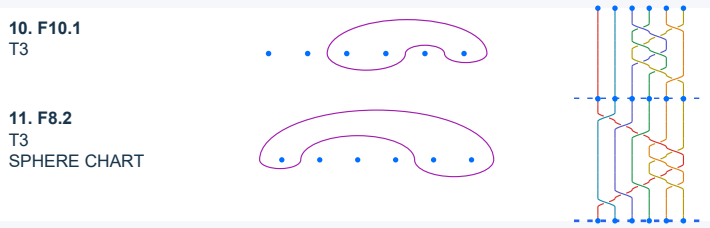
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



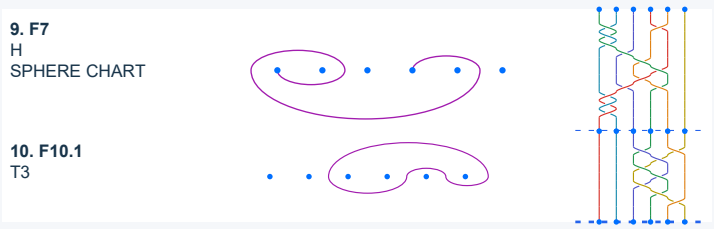
II - step 008

F10.1 up past F7

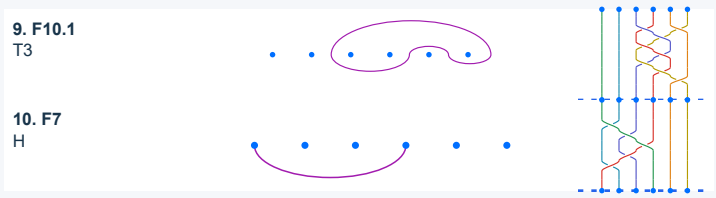
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



II - step 009

Use the exact disk equality of the two simple arcs

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



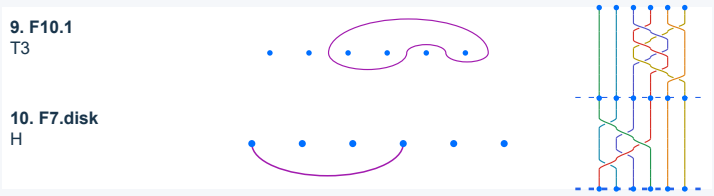
II - step 010

F7.disk up past F10.1

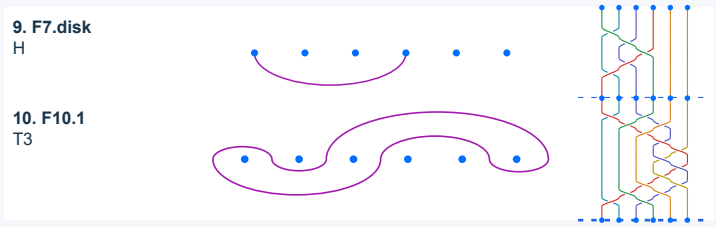
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



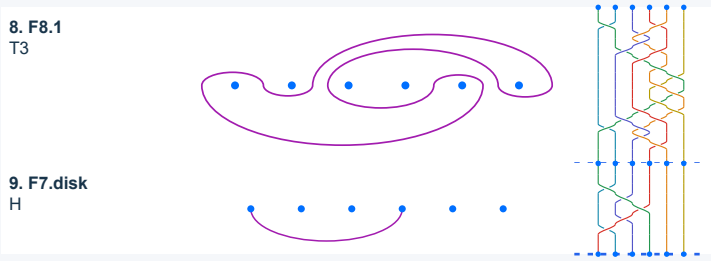
II - step 011

F7.disk up past F8.1

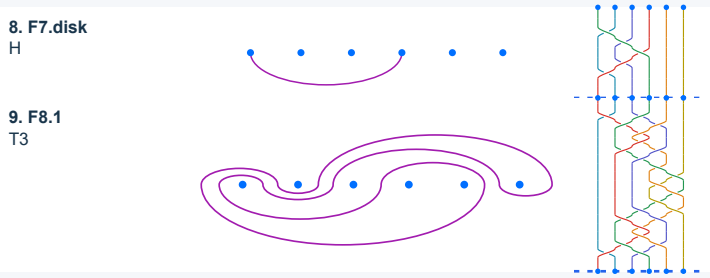
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=1 2 -4 5.

Before



After



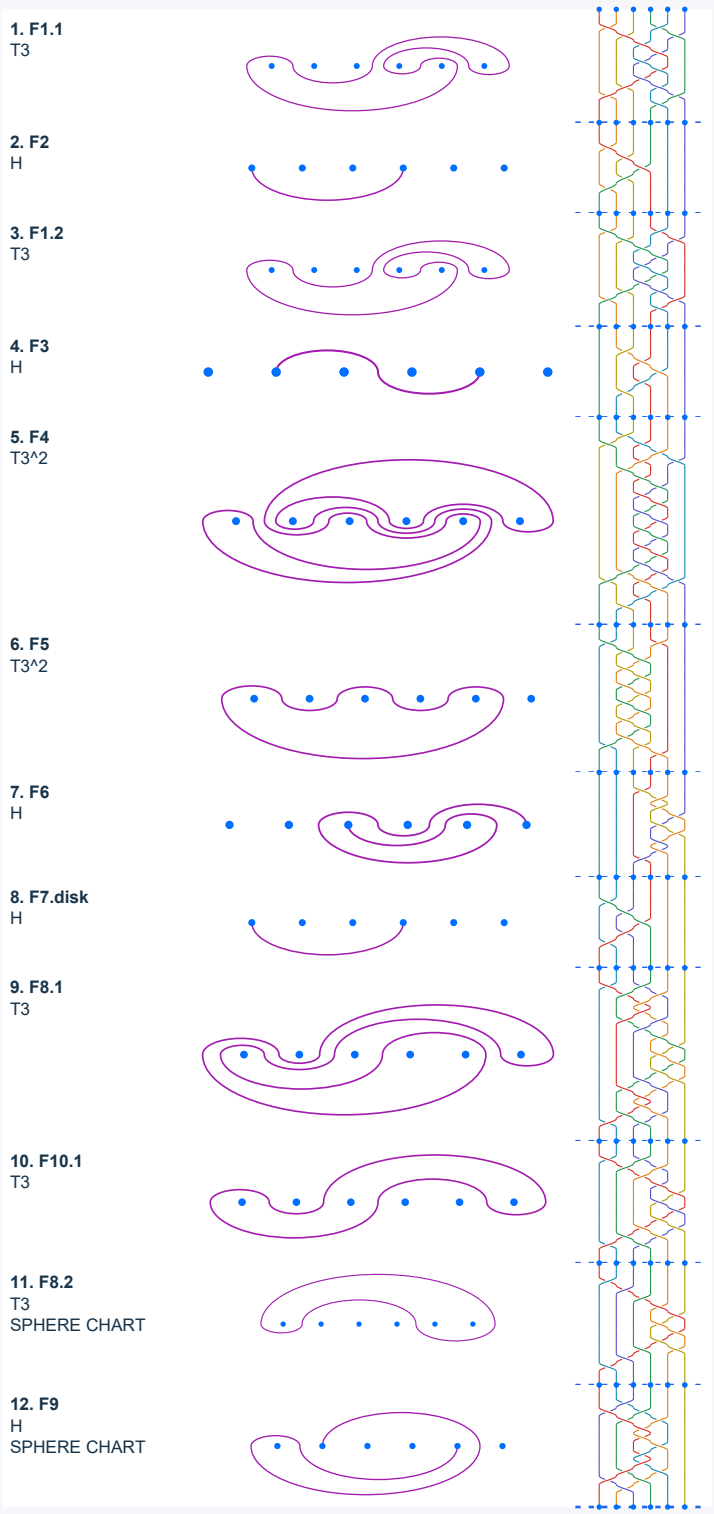
II - step 012

F7.disk up past F6

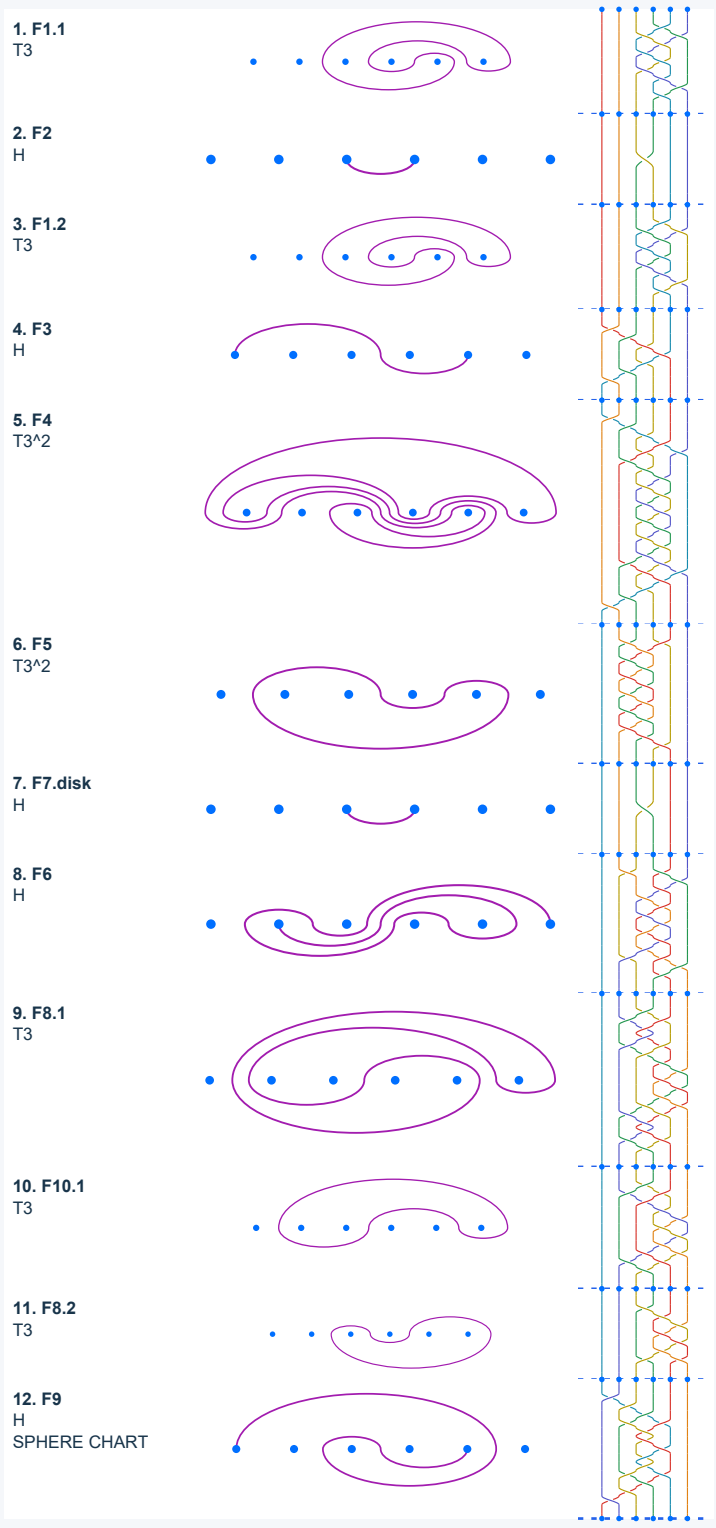
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=-4 5.

Before



After



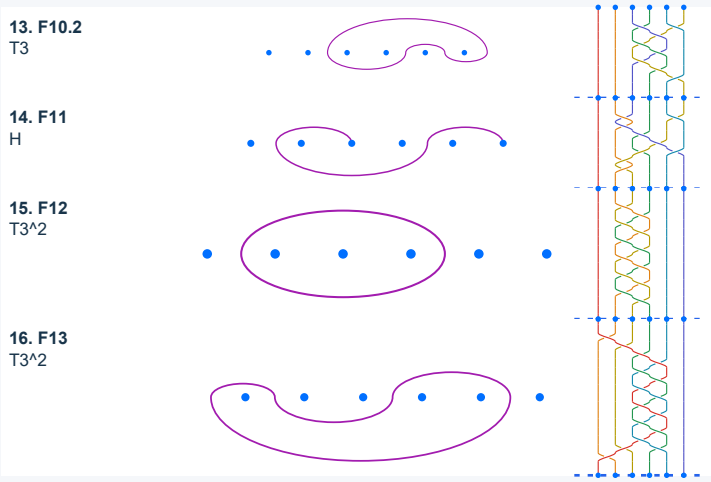
II - step 012 - continued

F7.disk up past F6

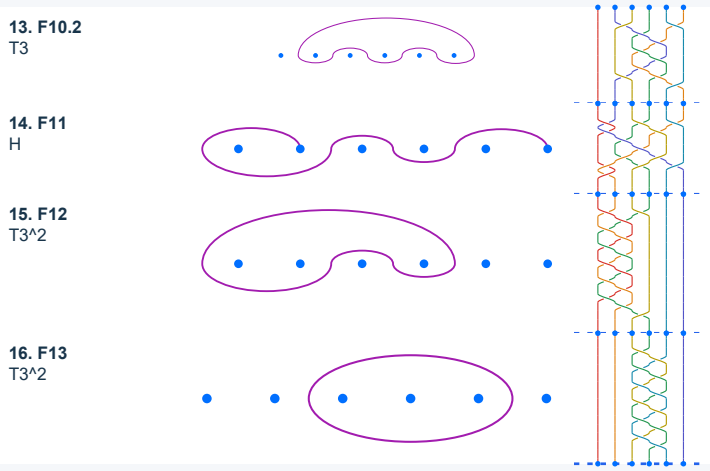
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 5; after g=-4 5.

Before



After



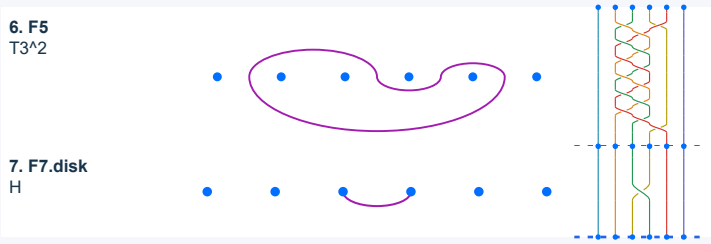
II - step 013

F7.disk up past F5

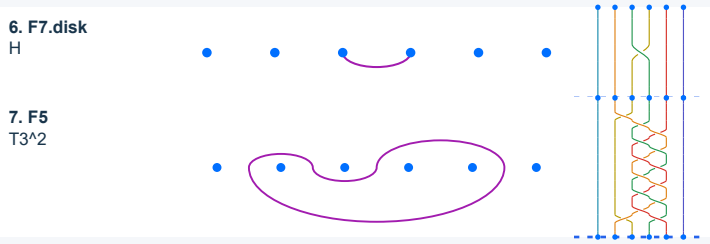
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



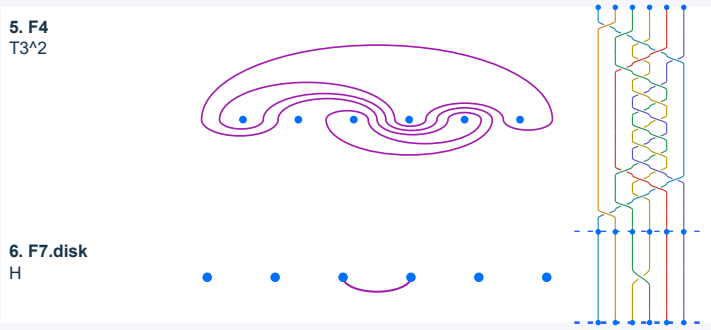
II - step 014

F7.disk up past F4

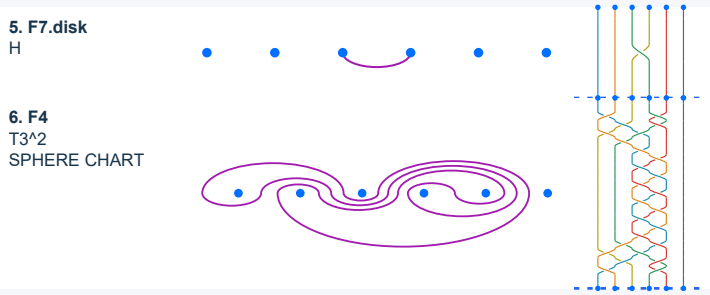
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



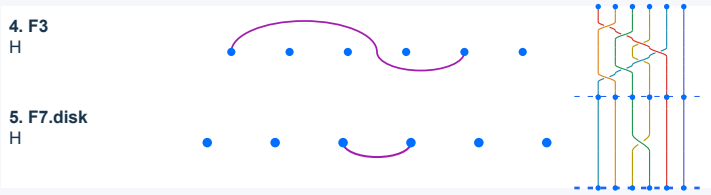
II - step 015

F7.disk up past F3

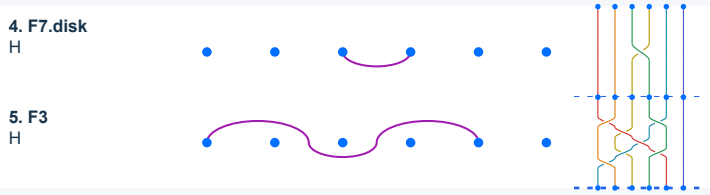
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



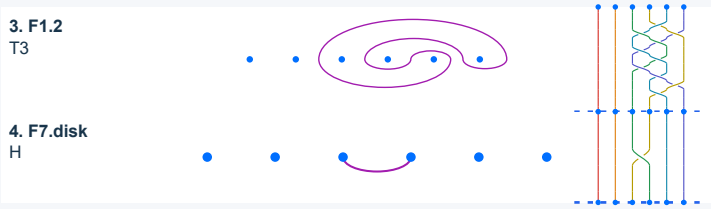
II - step 016

F7.disk up past F1.2

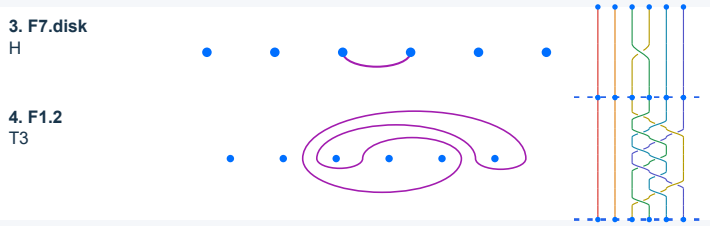
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



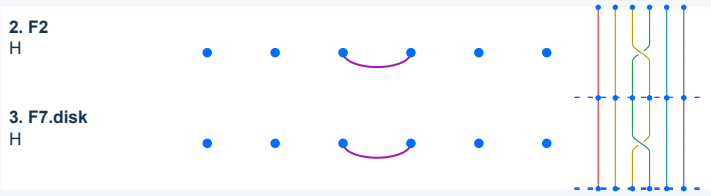
II - step 017

Combine transported F2 and F7 into a pair twist

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



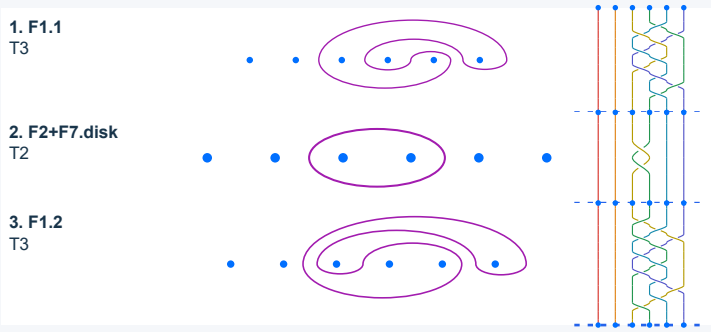
II - step 018

Replace that prefix by its two lantern boundary twists

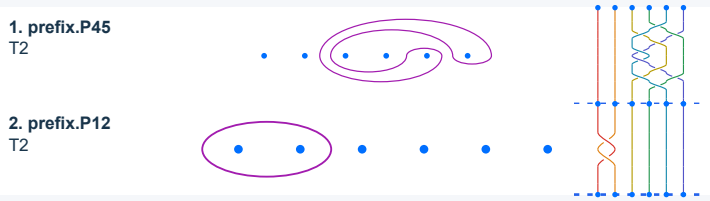
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



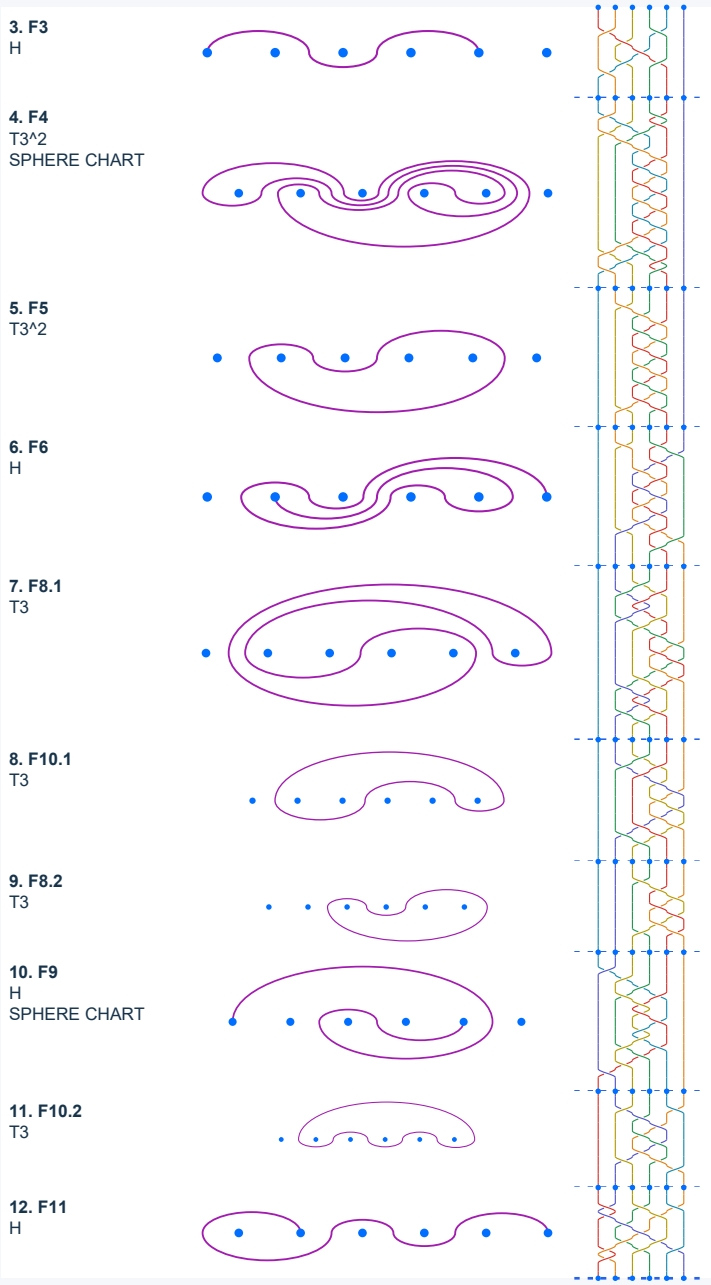
II - step 019

Apply the {34}, {26}, {5} cluster lantern

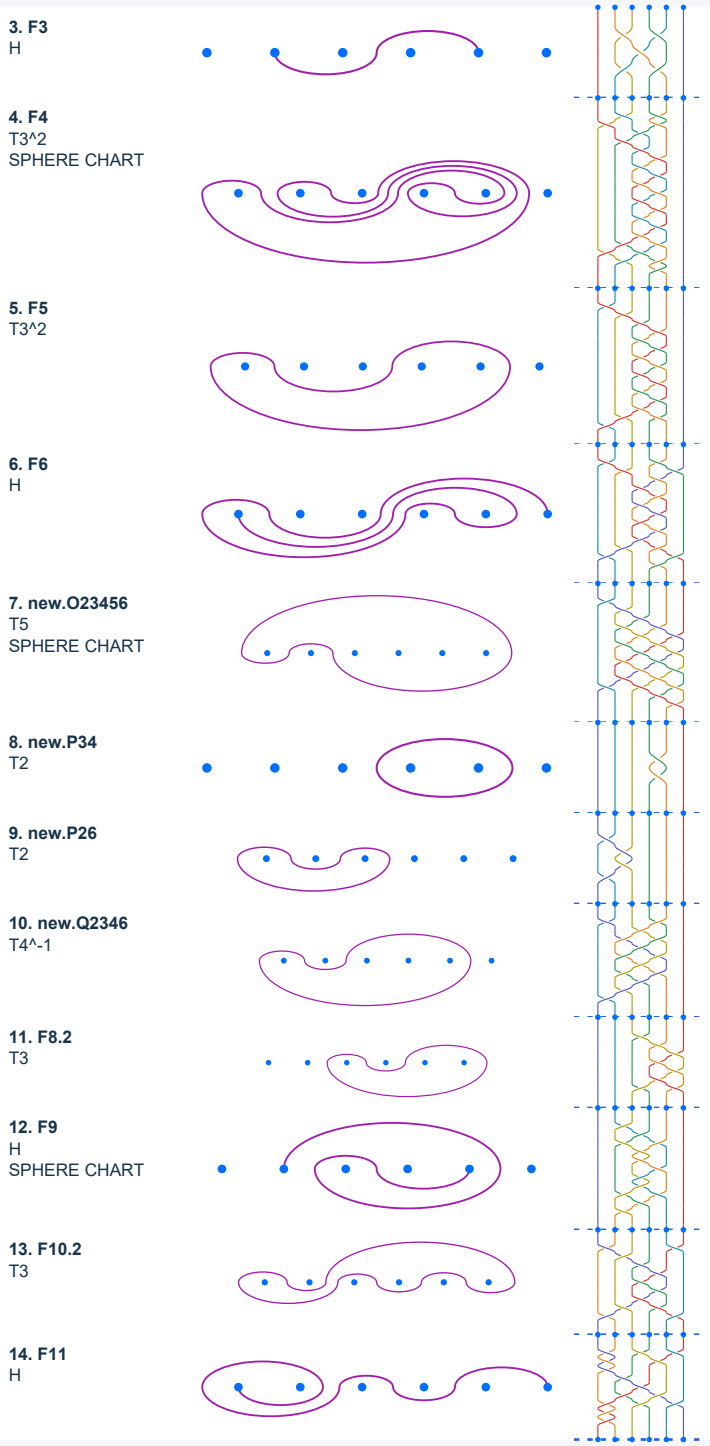
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=1 -4 5.

Before



After



II - step 020

Pair the peripheral omission with the complementary-root rechart

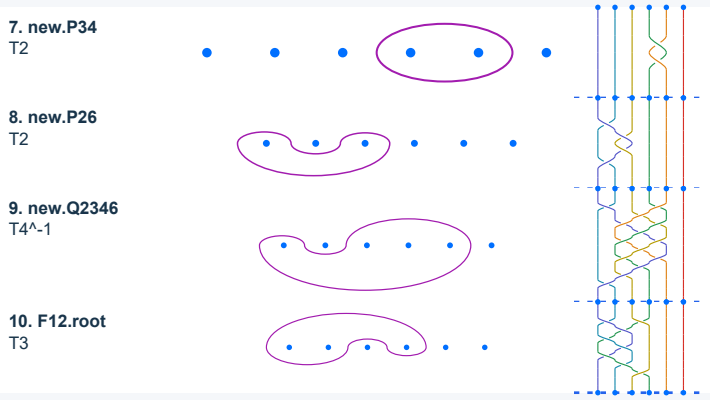
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

6 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=1 -4 5; after g=1 -4 5.

Before



After



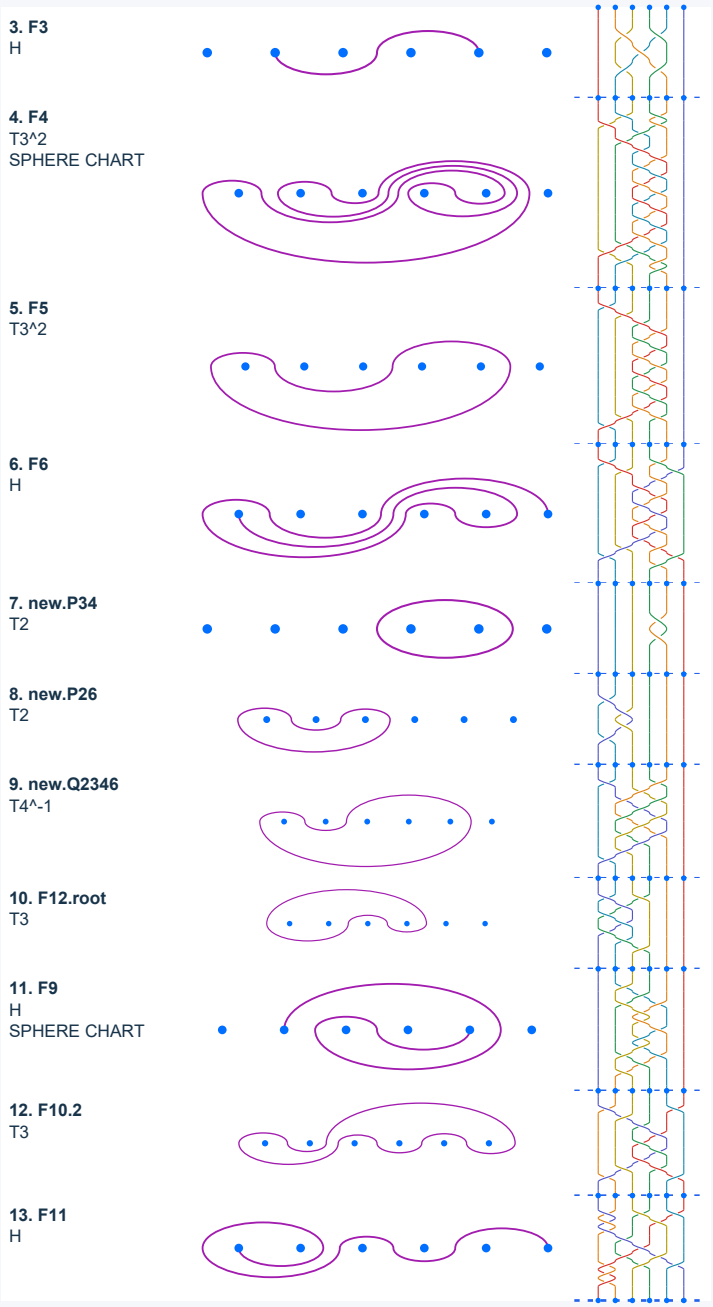
II - step 021

F12.root down past F9

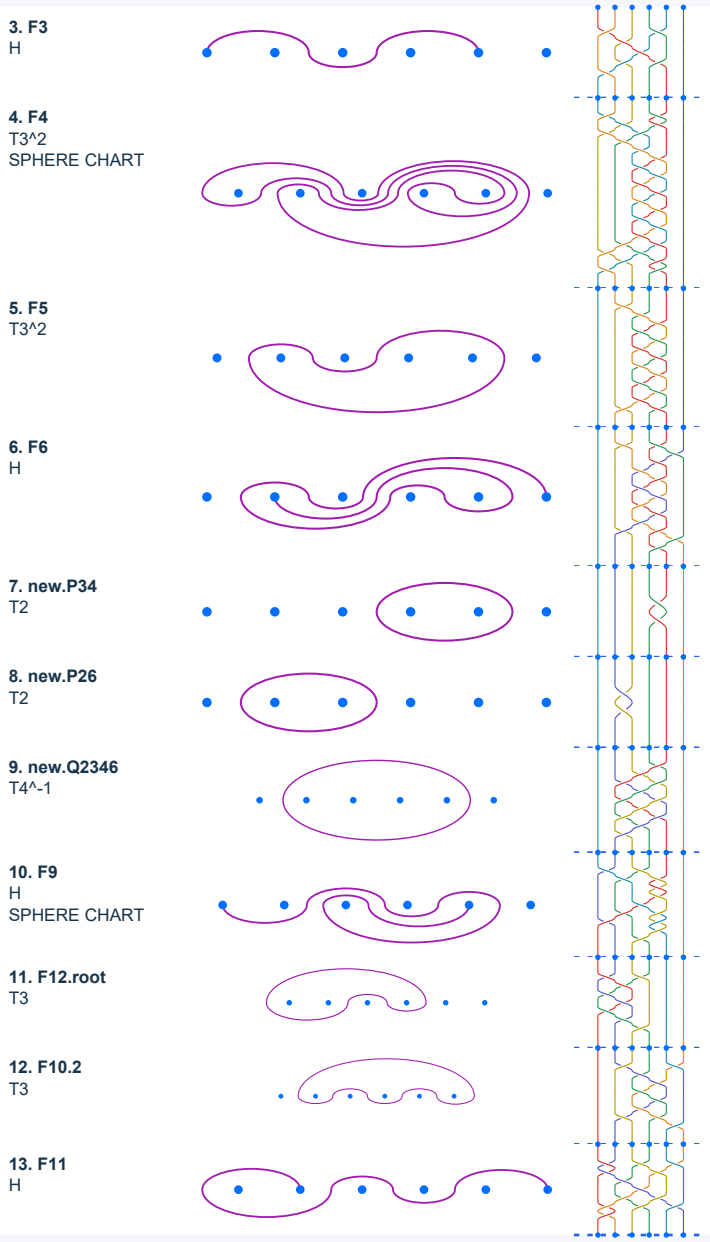
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=1 -4 5; after g=-4 5.

Before



After



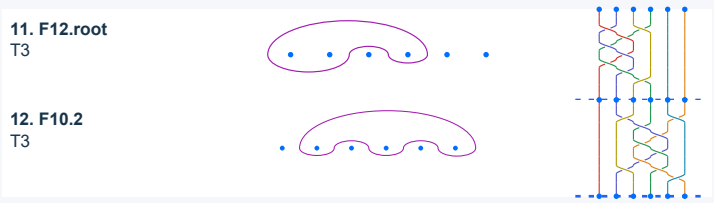
II - step 022

F12.root down past F10.2

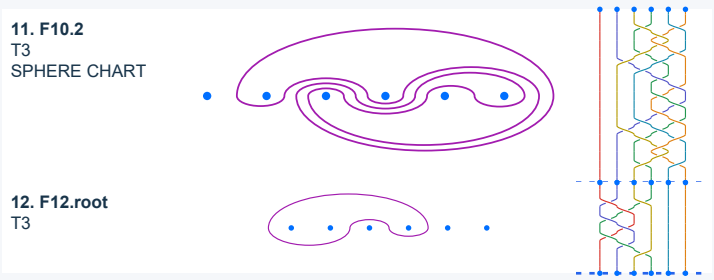
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



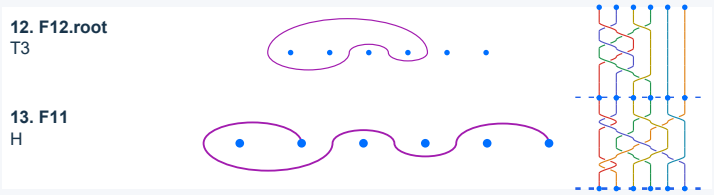
II - step 023

F12.root down past F11

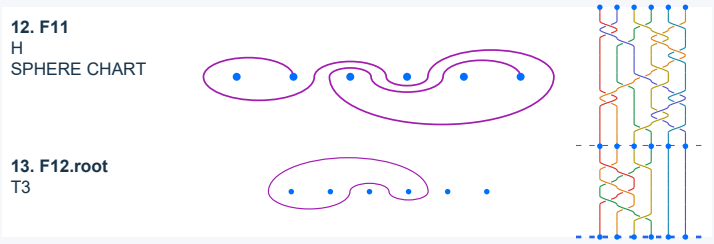
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



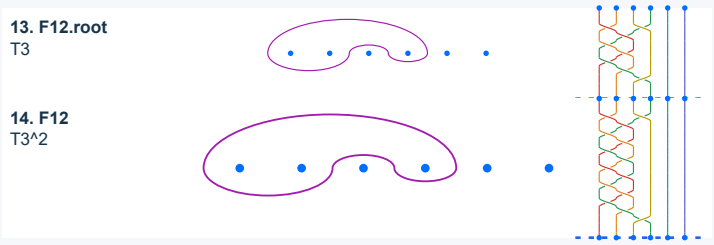
II - step 024

Combine the complementary root with F12 into a cube

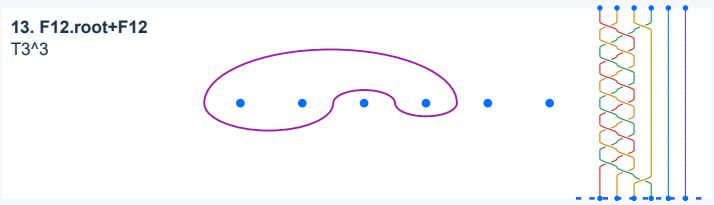
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



II - step 025

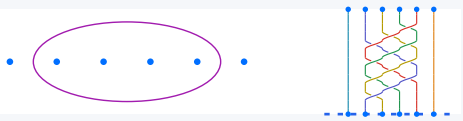
Draw the inverse Q along its complementary {1,5} pair

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

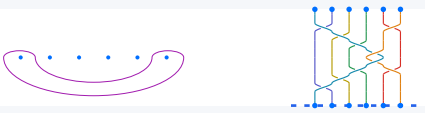
Before

9. new.Q2346
T4^-1



After

9. new.Q.complement
T2^-1



II - step 026

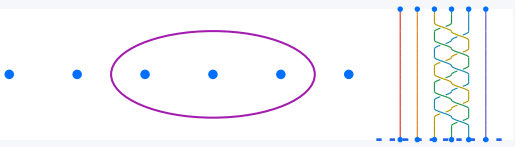
Rechart the whole C13 square along its complementary triple

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

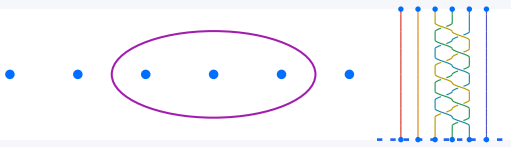
Before

14. F13
T3^2



After

14. C13.complement. square
T3^2
SPHERE CHART



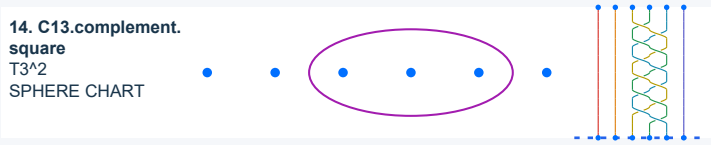
II - step 027

Apply the chosen PC half lantern to each C13 unit

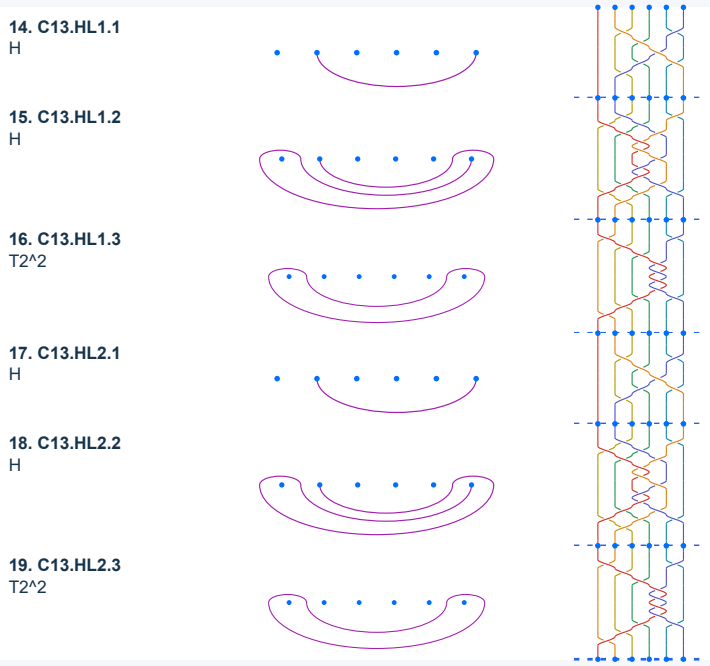
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



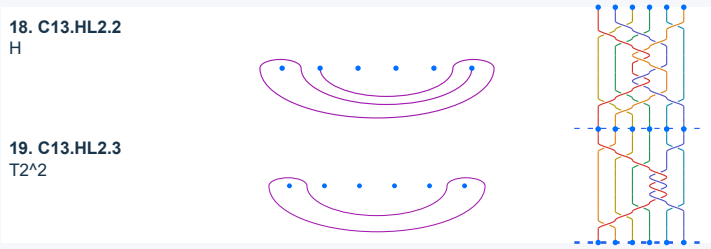
II - step 028

C13.HL2.3 up past C13.HL2.2

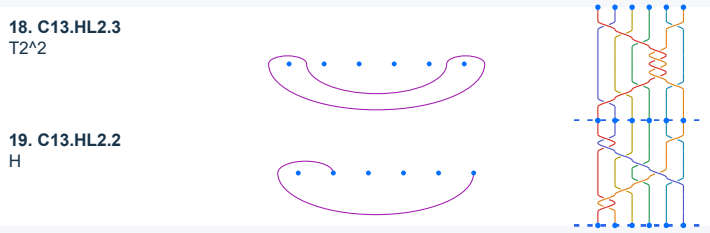
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

17 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



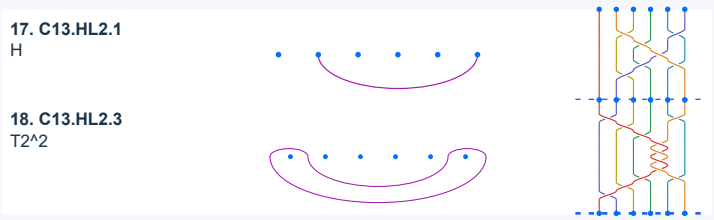
II - step 029

C13.HL2.3 up past C13.HL2.1

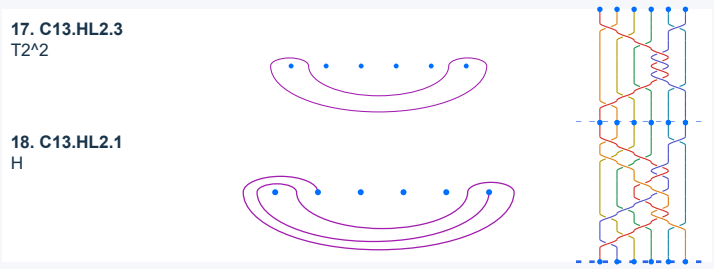
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



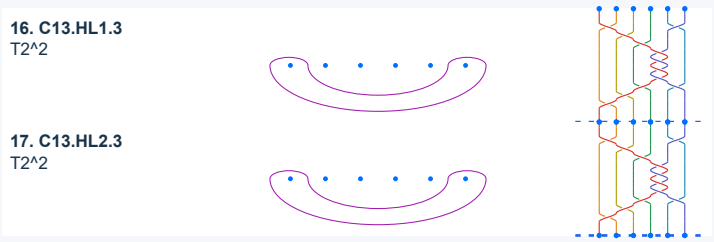
II - step 030

C13.HL2.3 up past C13.HL1.3

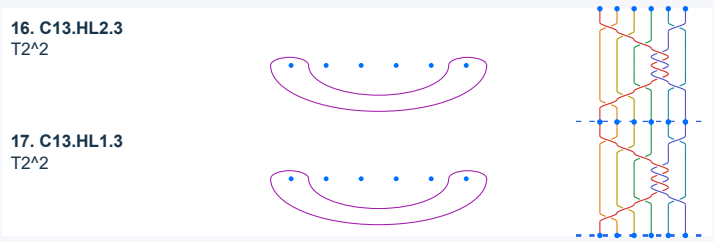
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



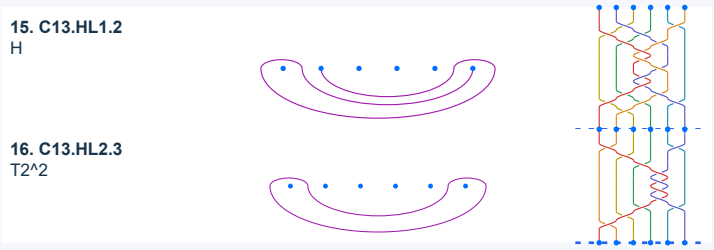
II - step 031

C13.HL2.3 up past C13.HL1.2

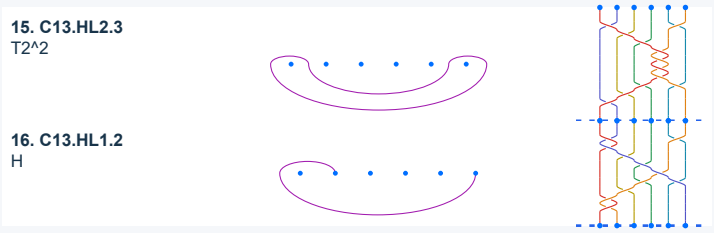
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



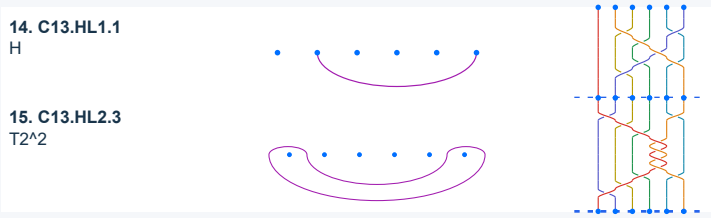
II - step 032

C13.HL2.3 up past C13.HL1.1

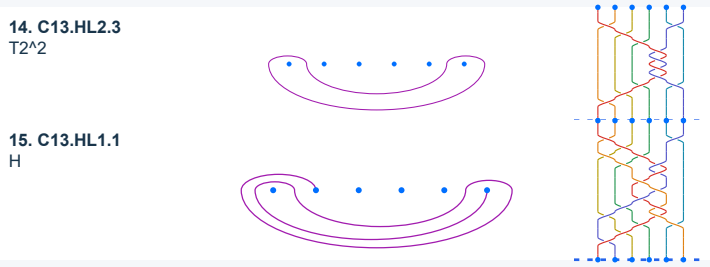
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



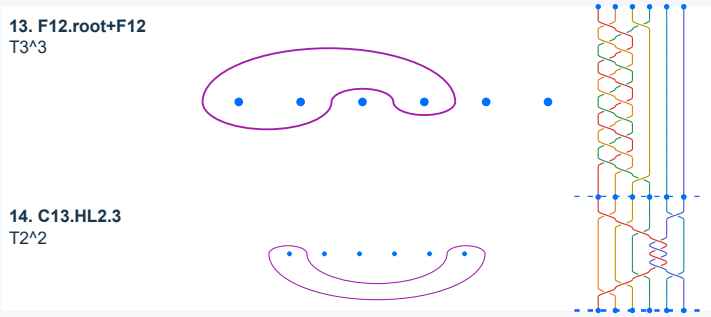
II - step 033

C13.HL2.3 up past F12.root+F12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=-4 5.

Before



After



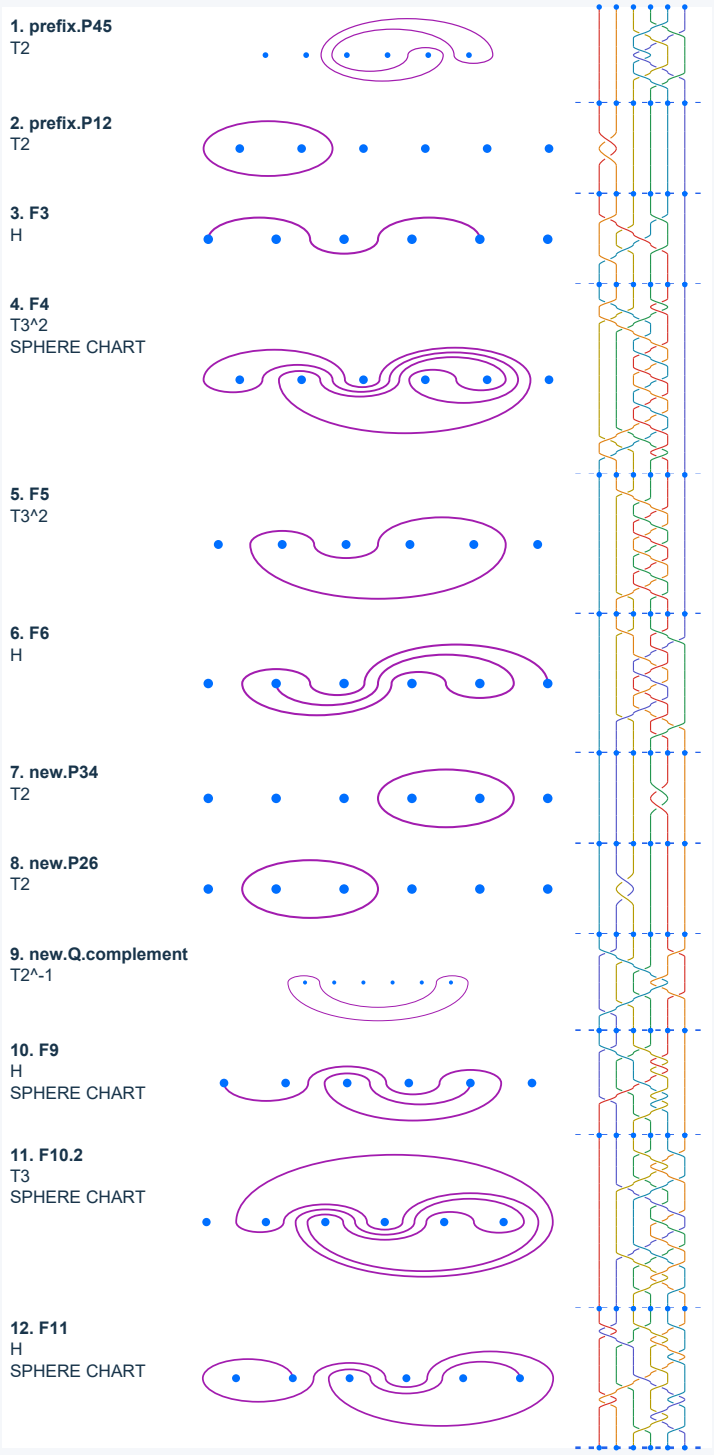
II - step 034

C13.HL2.3 up past F11

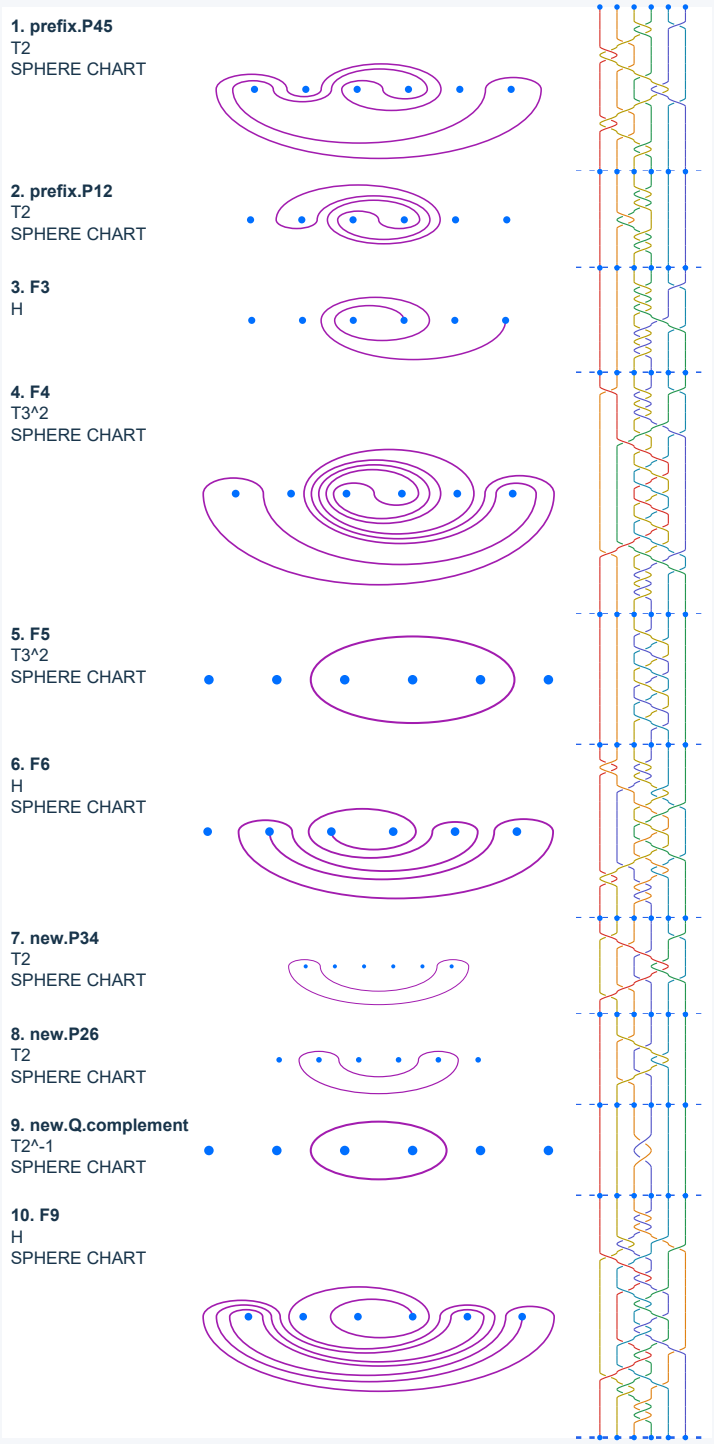
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



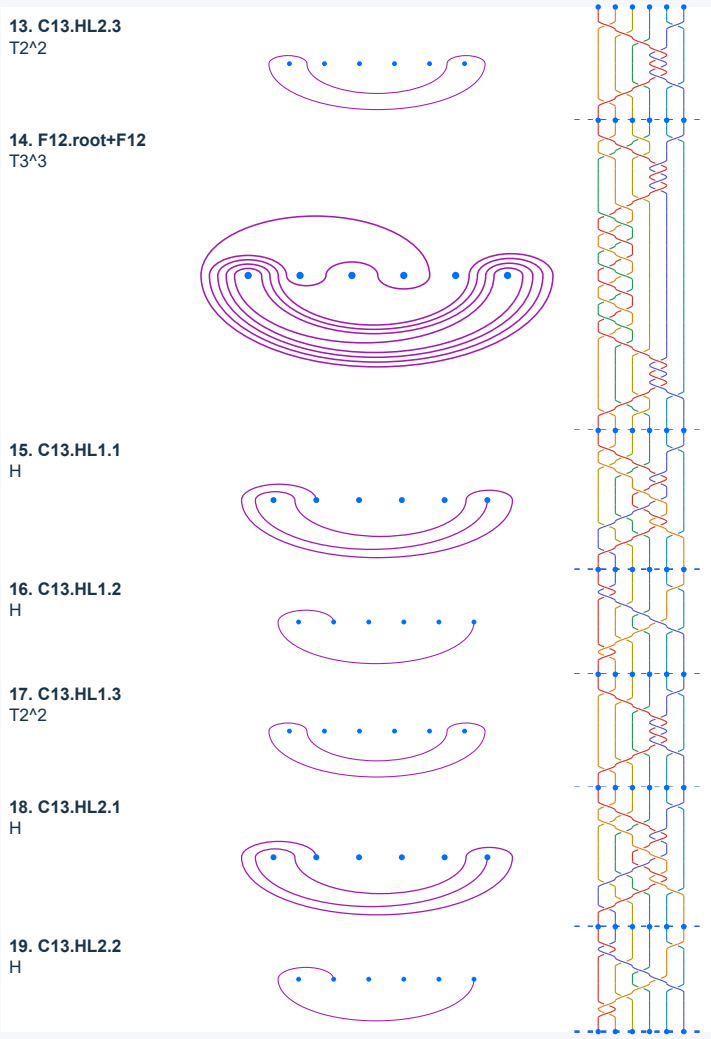
II - step 034 - continued

C13.HL2.3 up past F11

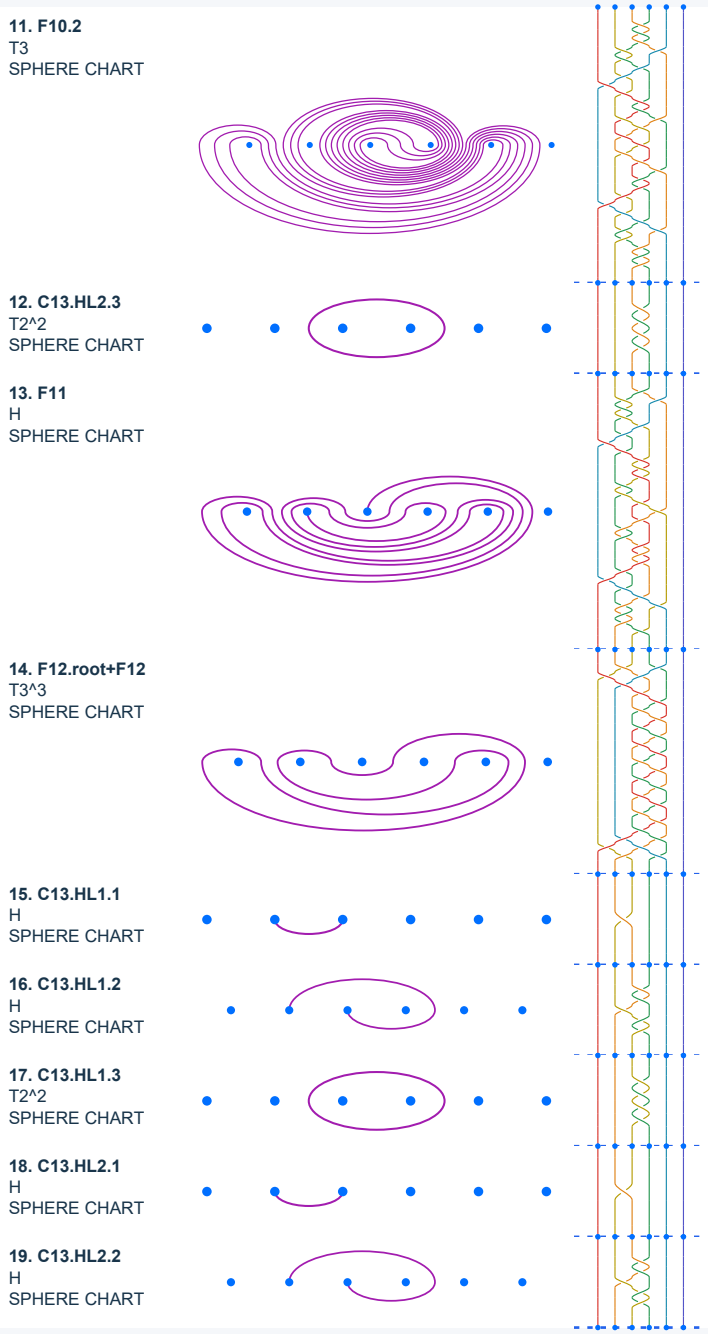
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



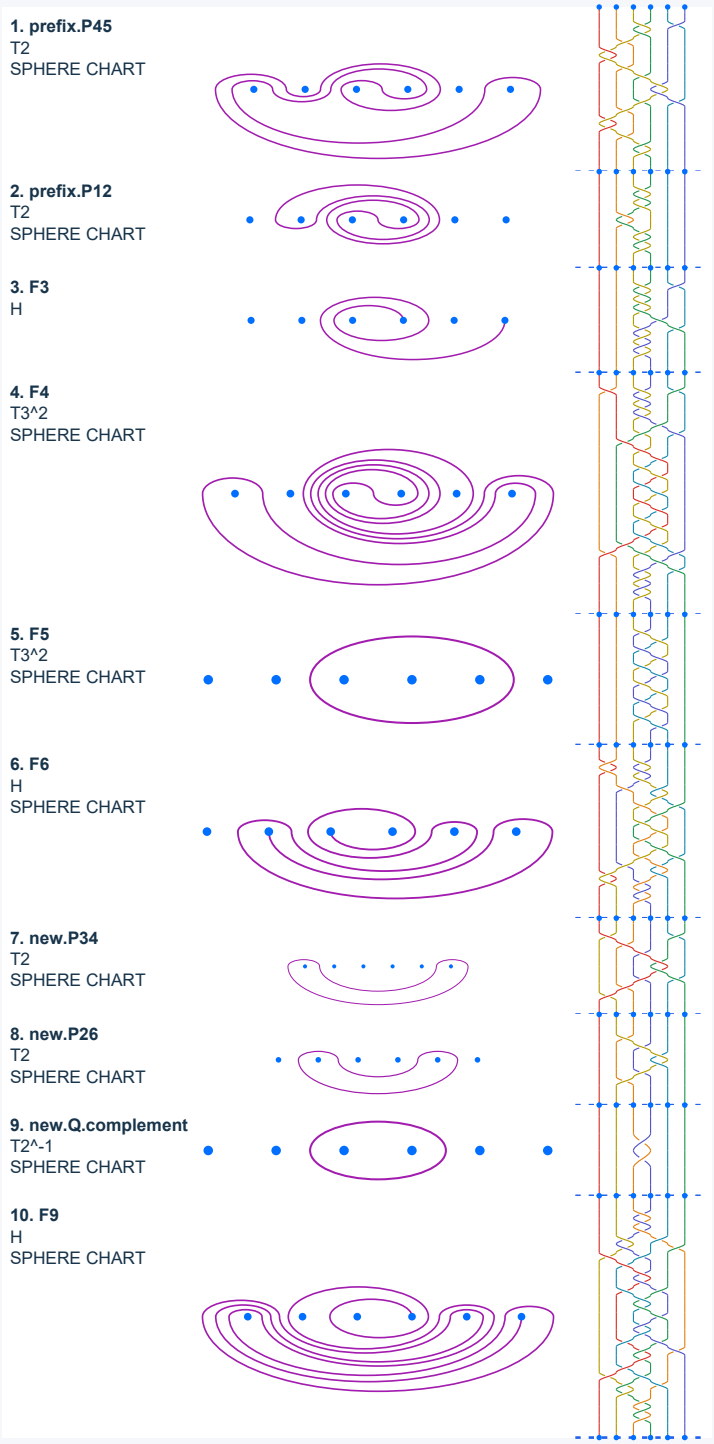
II - step 035

C13.HL2.3 up past F10.2

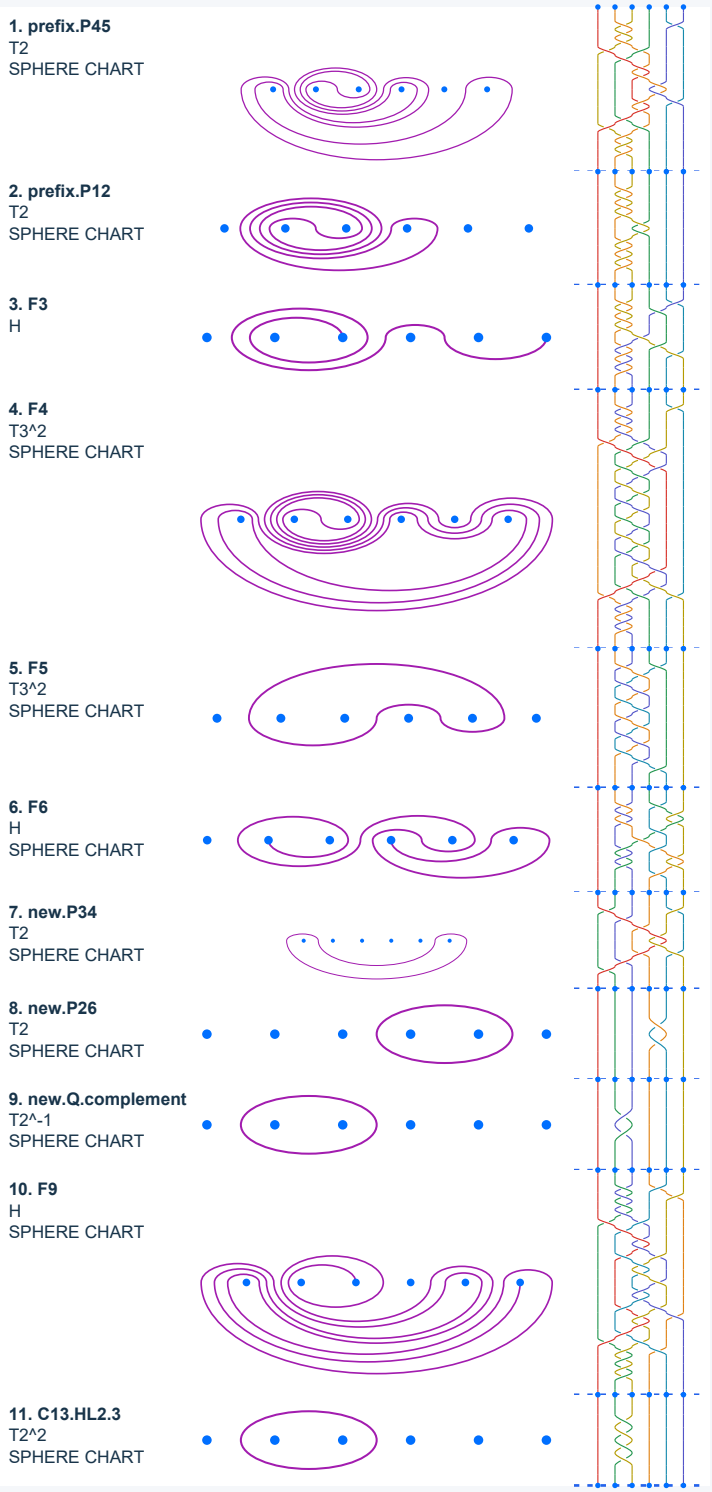
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



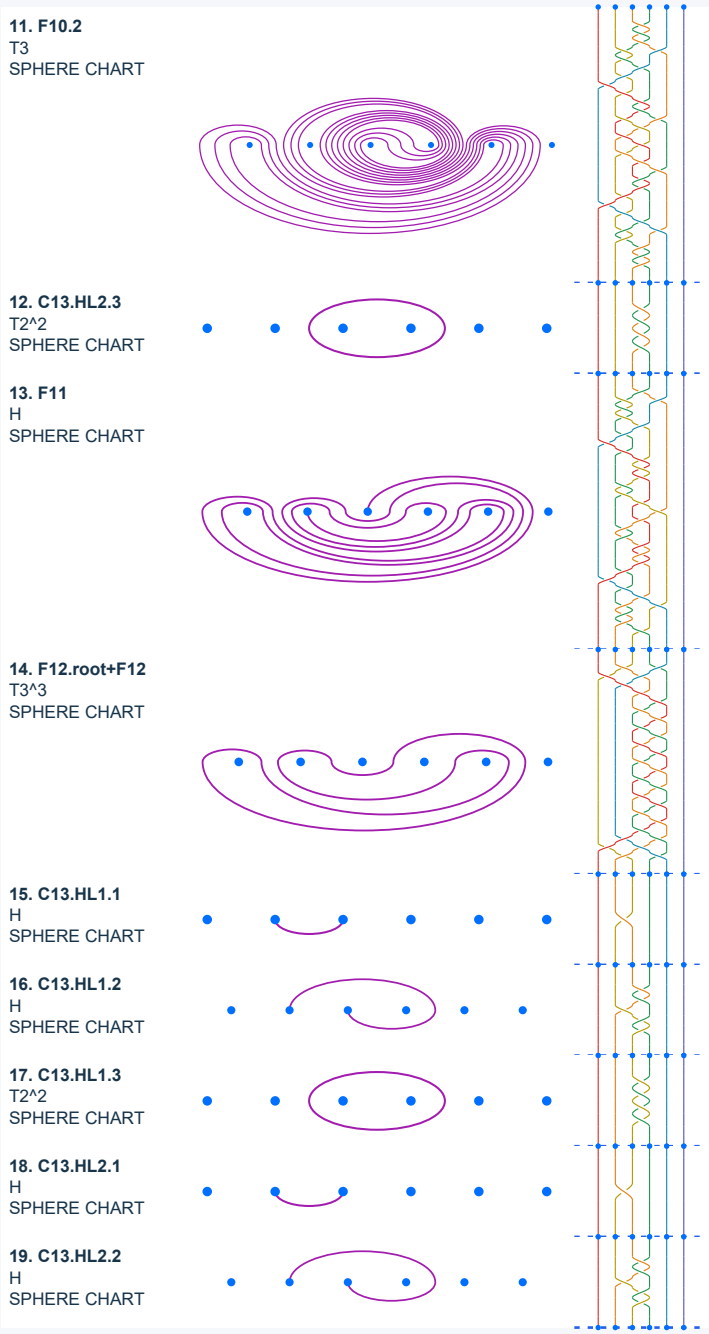
II - step 035 - continued

C13.HL2.3 up past F10.2

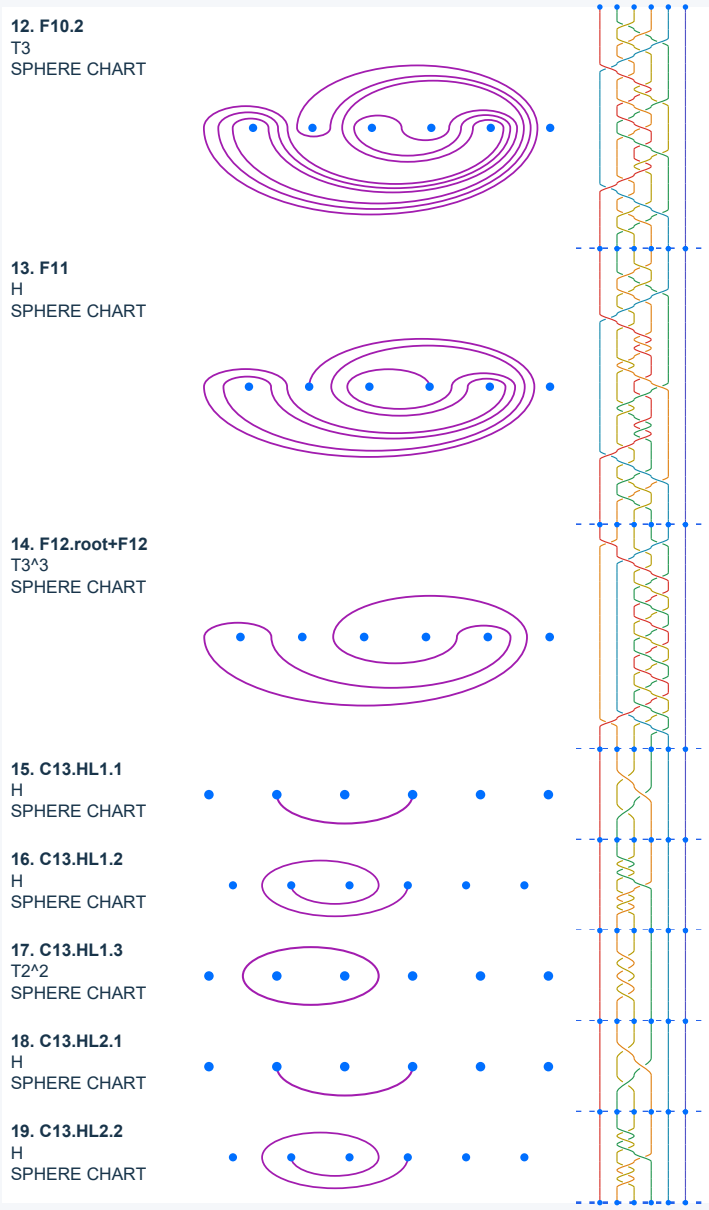
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



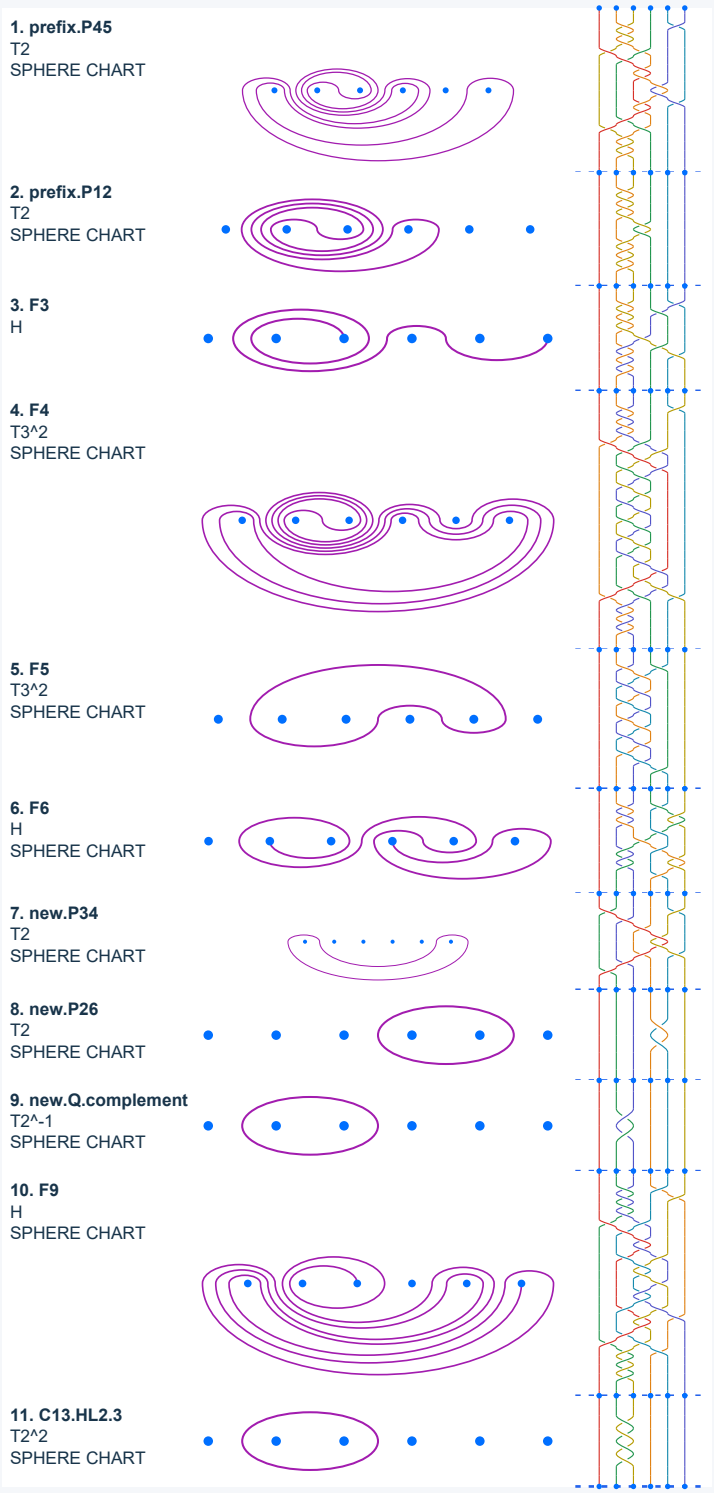
II - step 036

C13.HL2.3 up past F9

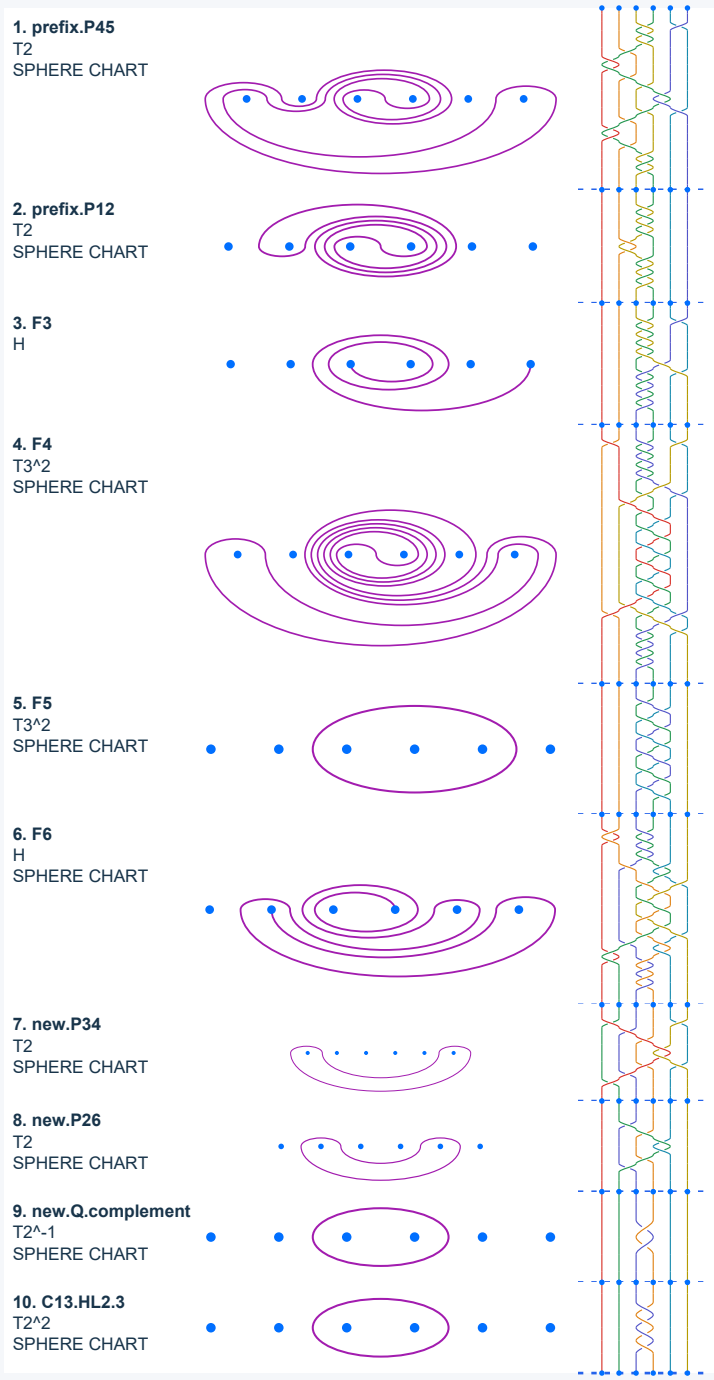
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



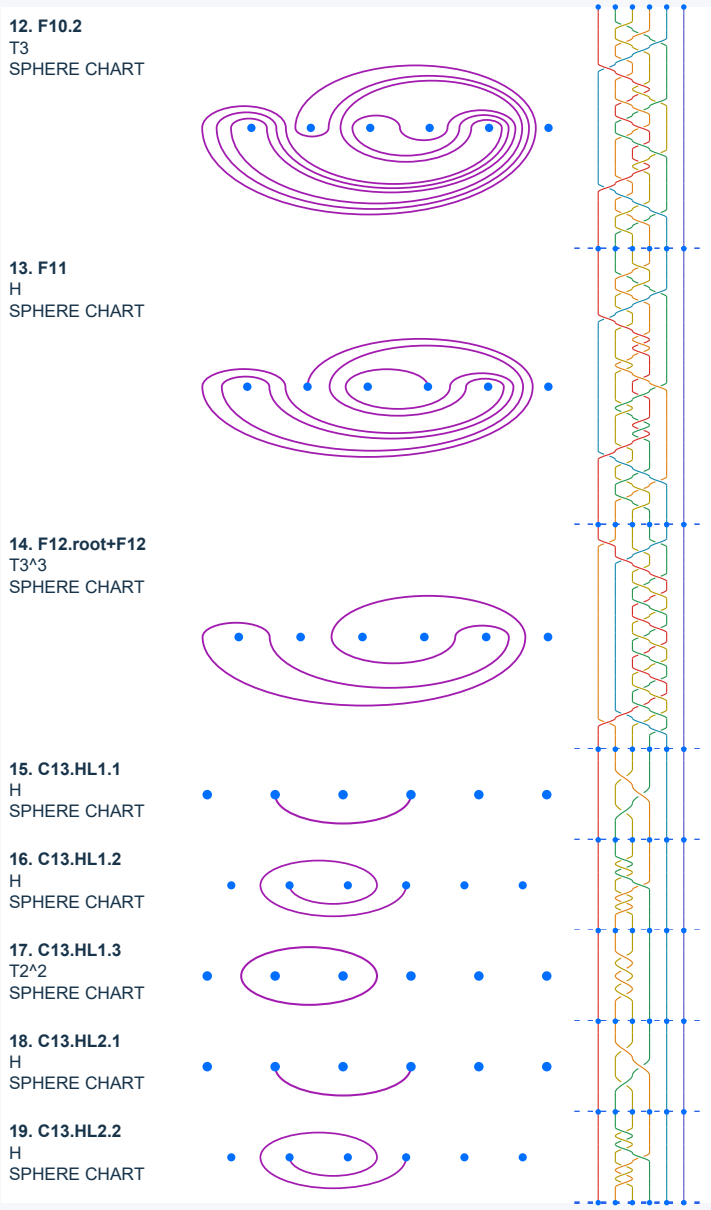
II - step 036 - continued

C13.HL2.3 up past F9

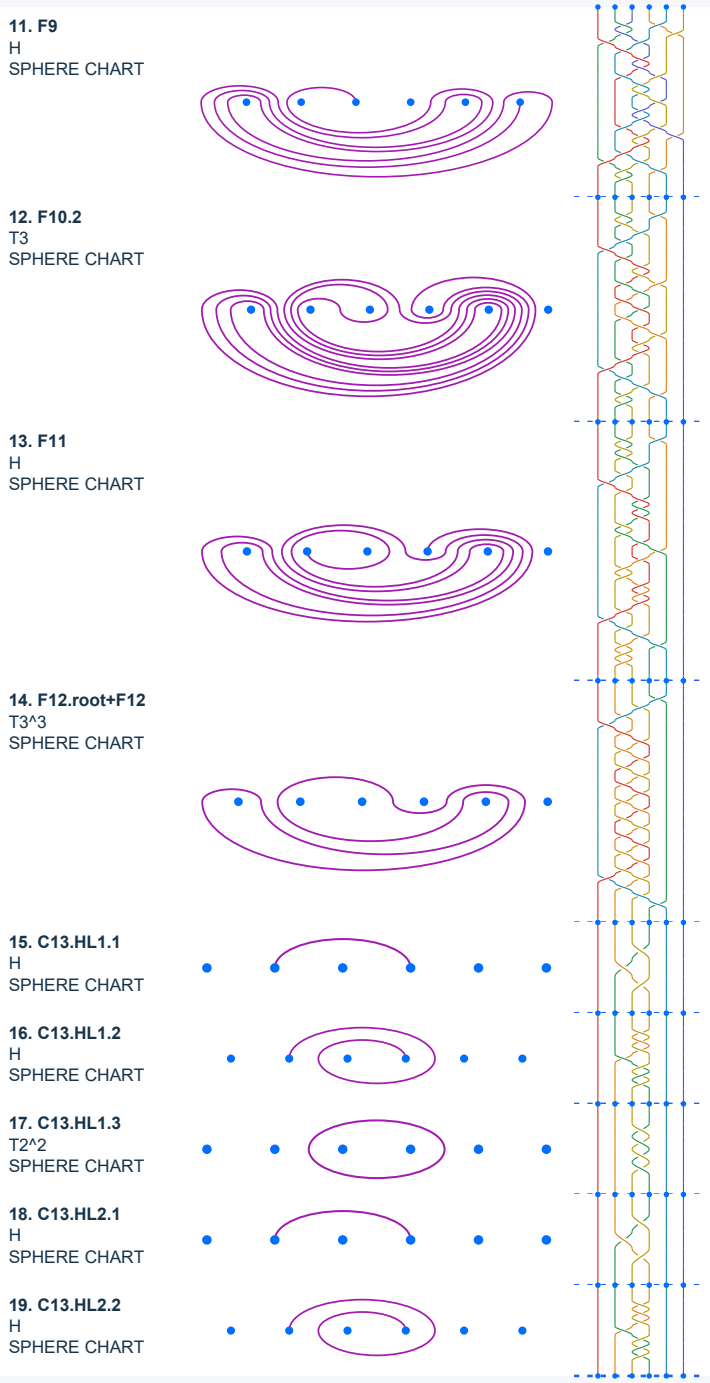
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -2 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



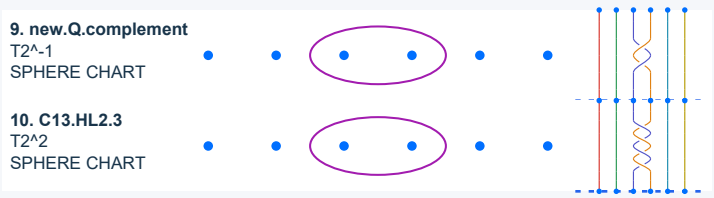
II - step 037

Cancel the inverse pair against one unit of the squared pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



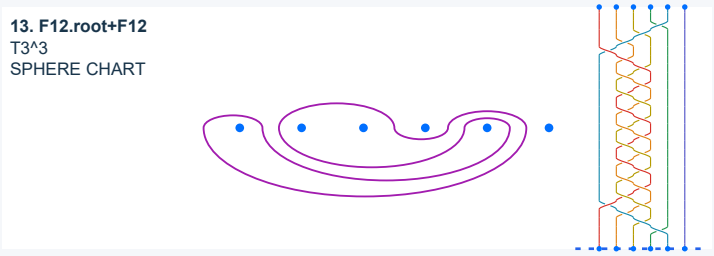
II - step 038

Split the C12 cube into three roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



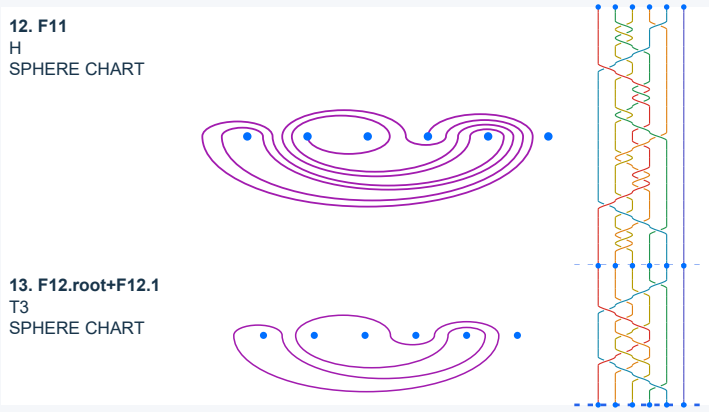
II - step 039

F12.root+F12.1 up past F11

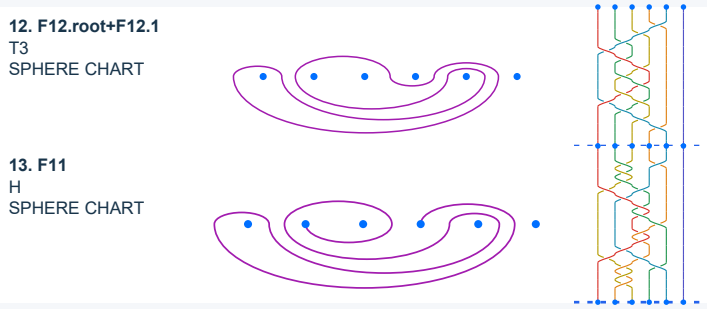
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



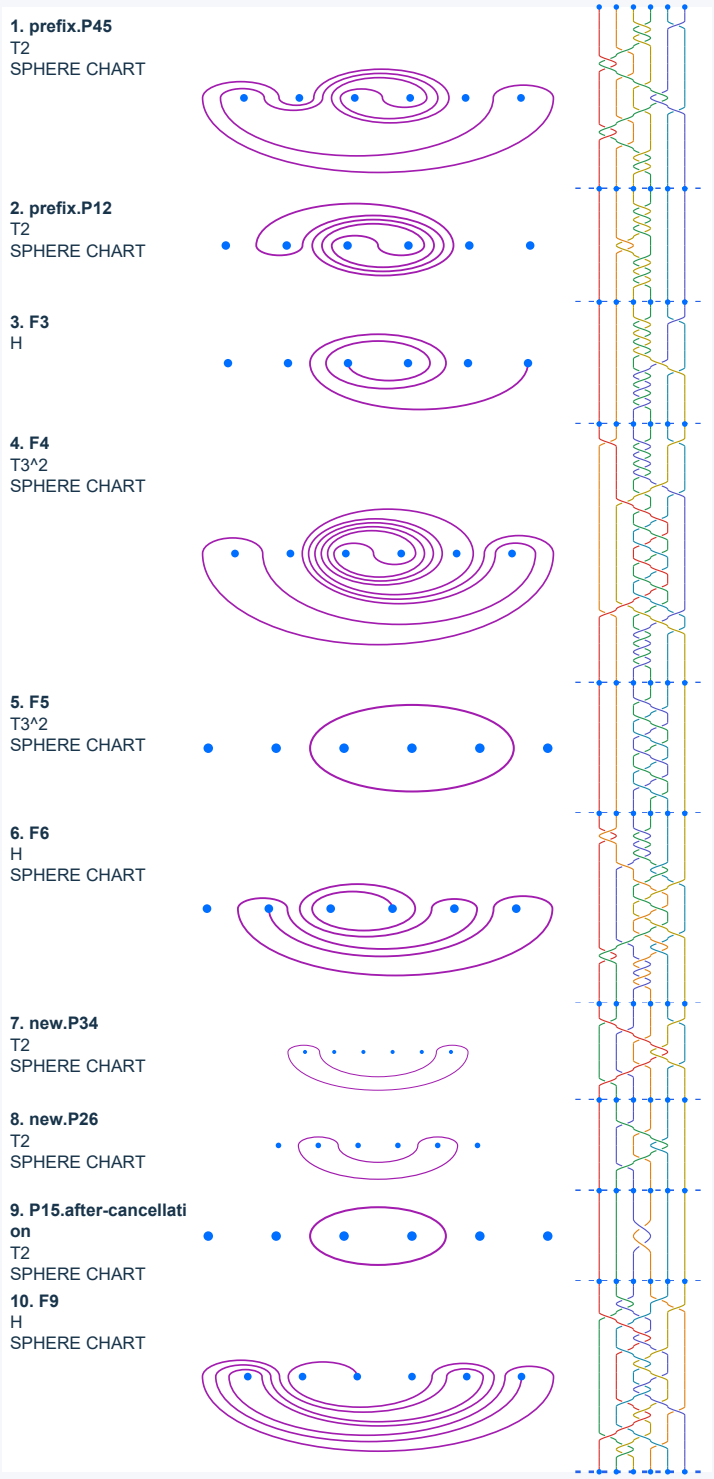
II - step 040

Use the residual lantern in the chart S C12, with the roots in their actual order

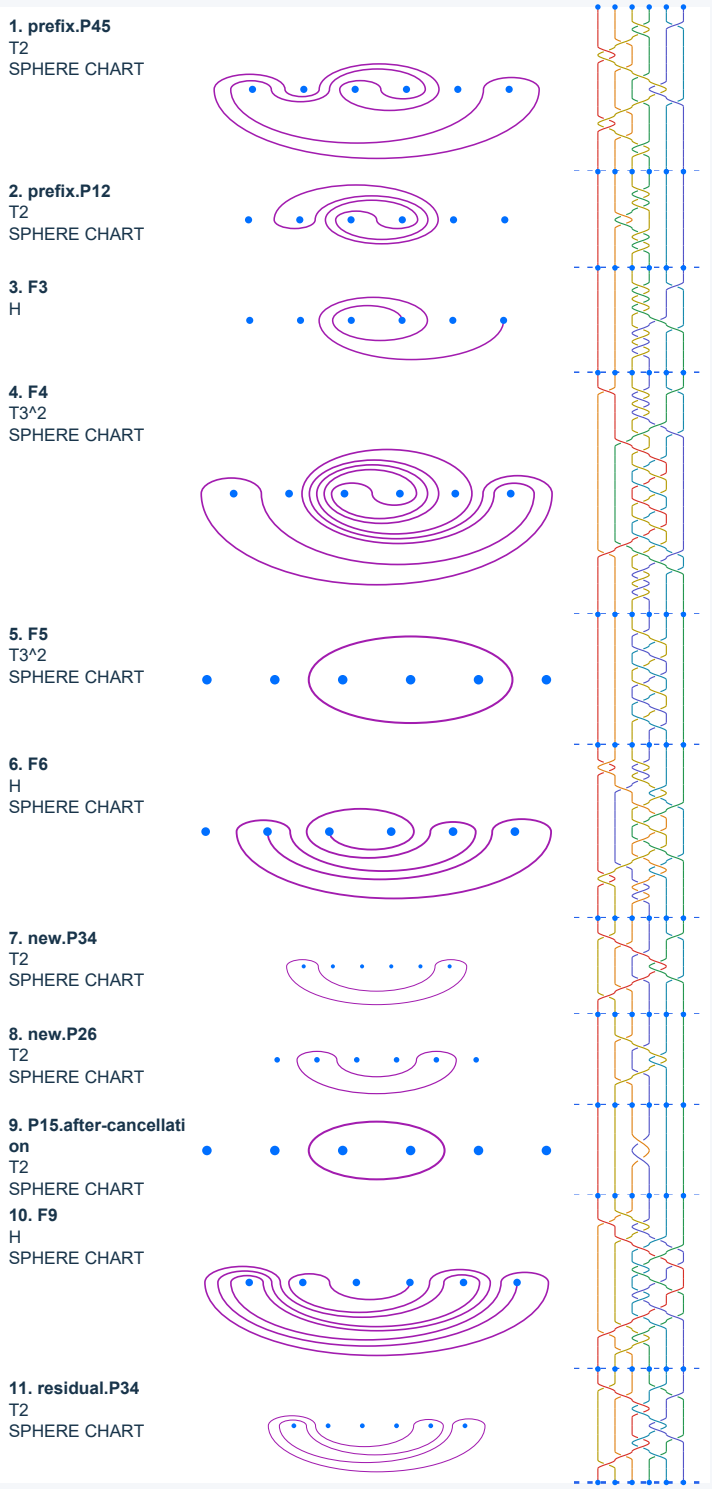
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



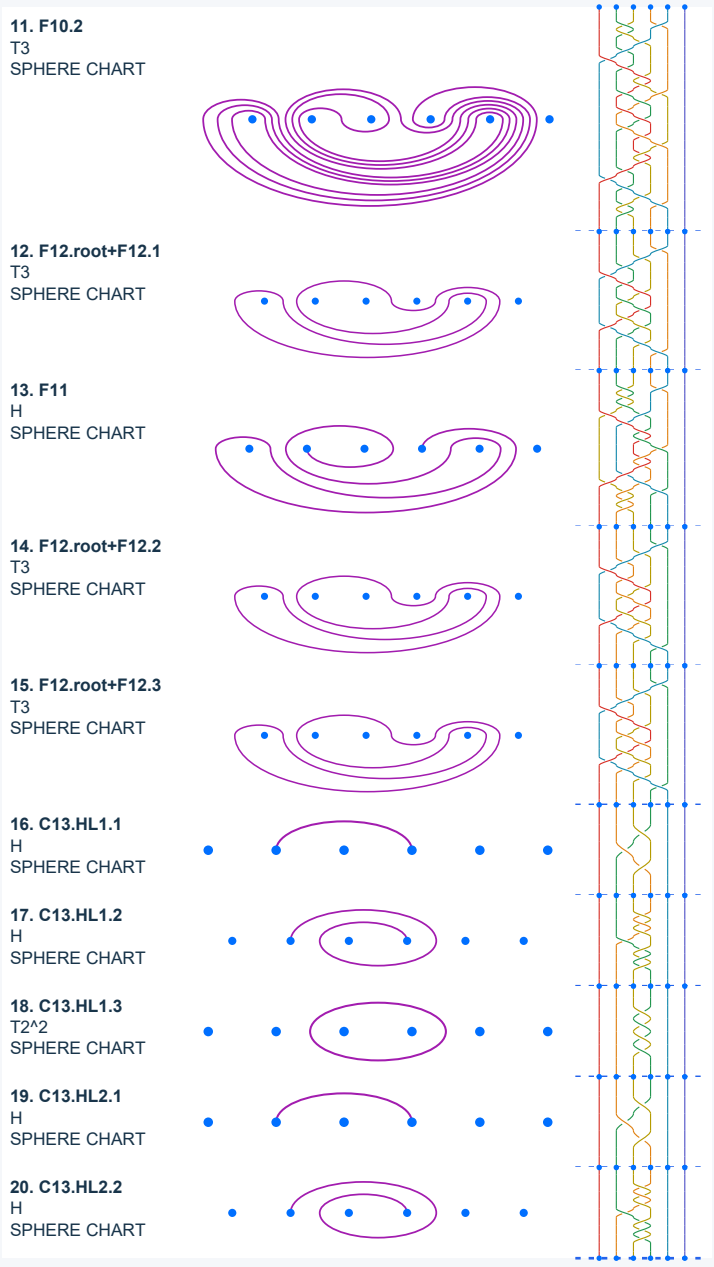
II - step 040 - continued

Use the residual lantern in the chart S C12, with the roots in their actual order

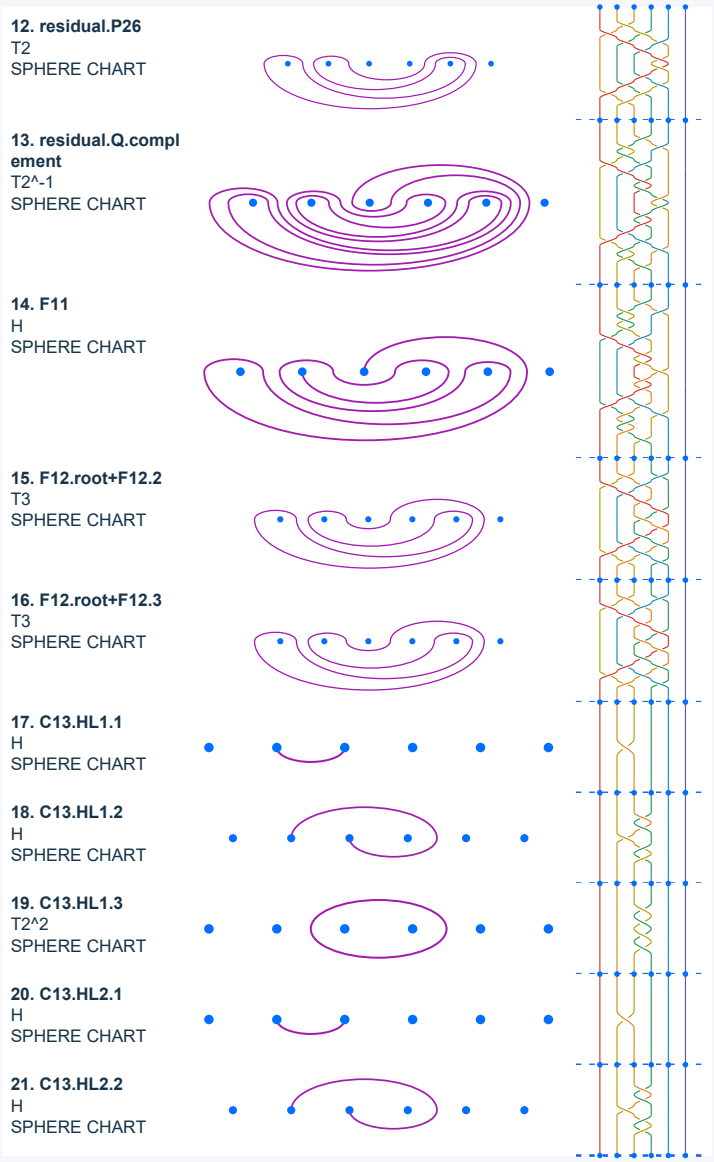
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



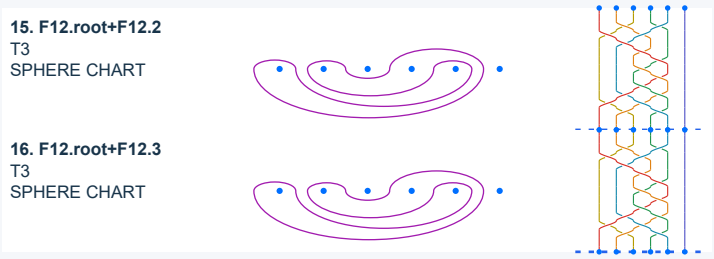
II - step 041

Combine the two unused C12 roots into a square

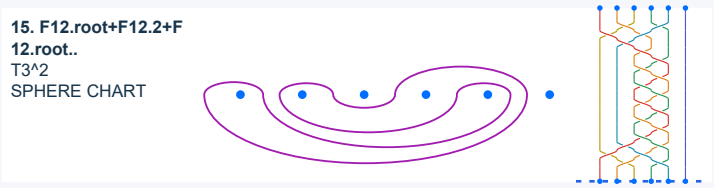
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



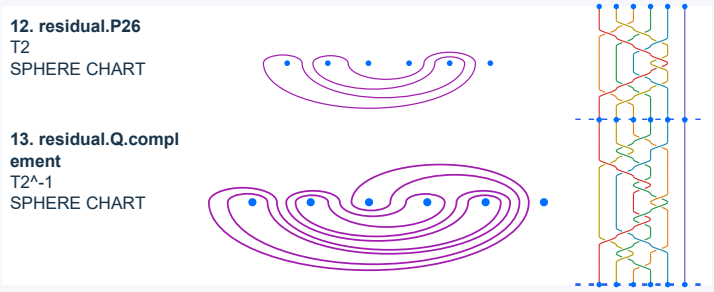
II - step 042

residual.P26 down past residual.Q.complement

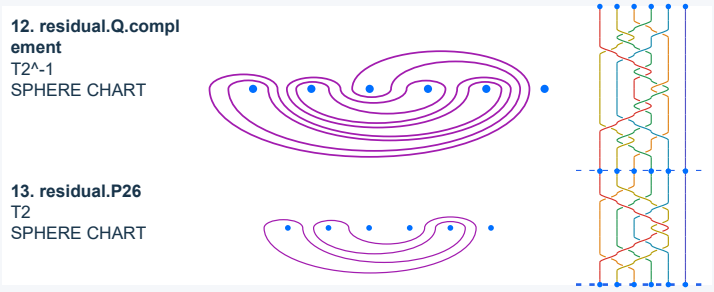
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



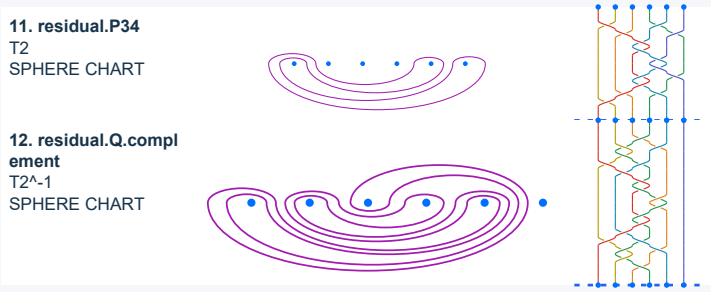
II - step 043

residual.P34 down past residual.Q.complement

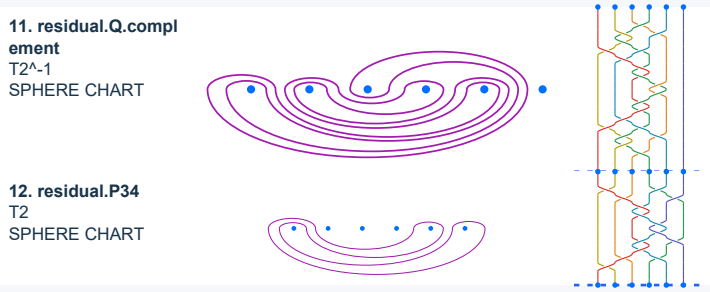
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



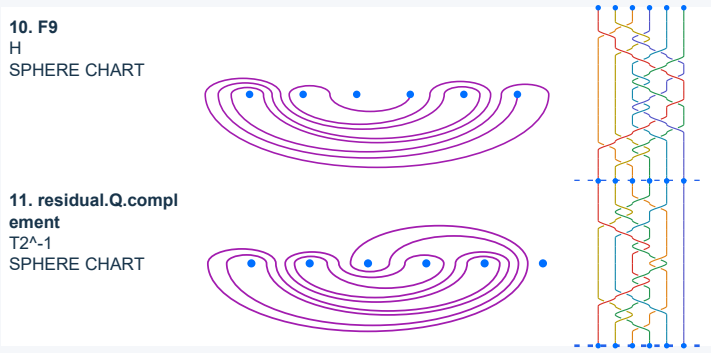
II - step 044

F9 down past residual.Q.complement

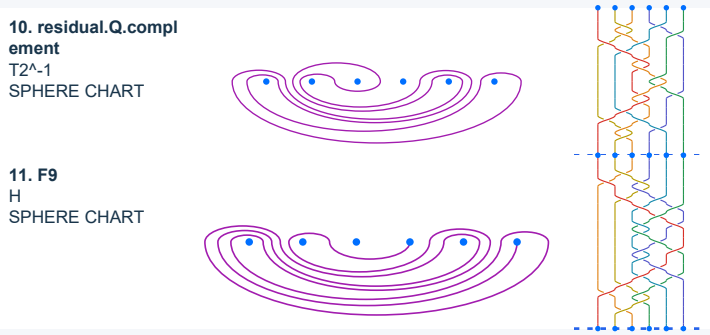
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



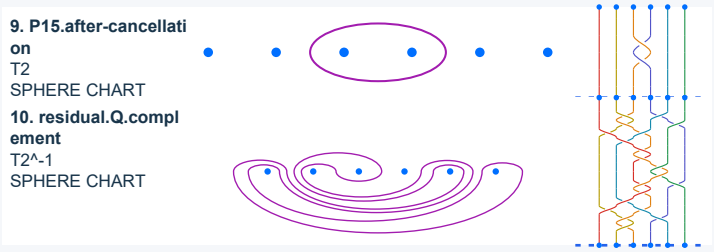
II - step 045

P15.after-cancellation down past residual.Q.complement

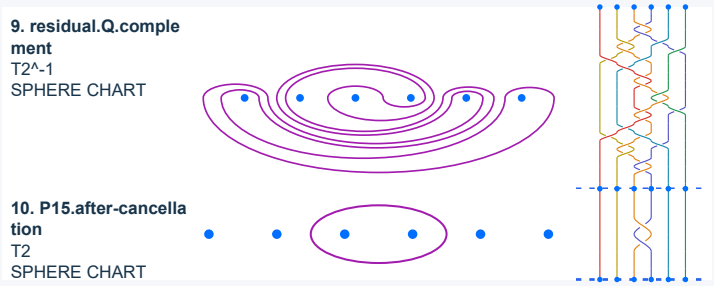
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



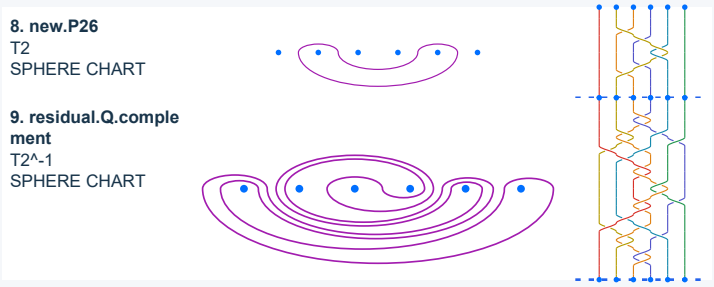
II - step 046

new.P26 down past residual.Q.complement

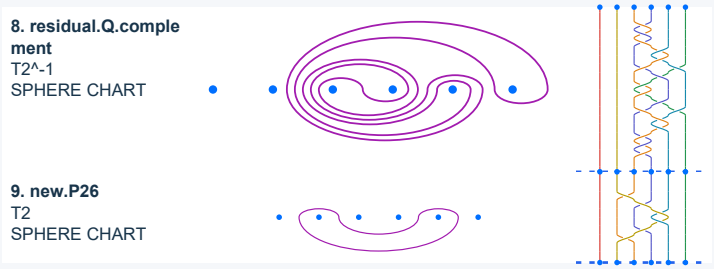
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



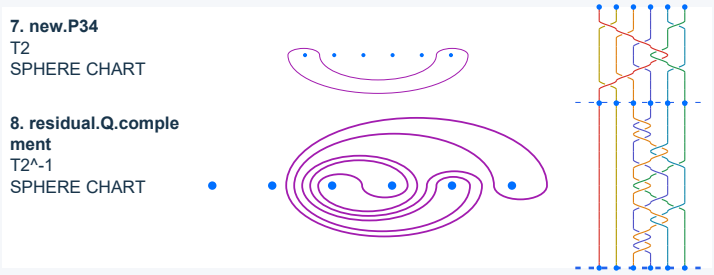
II - step 047

new.P34 down past residual.Q.complement

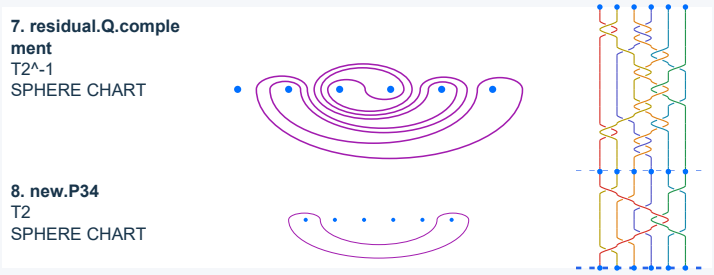
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



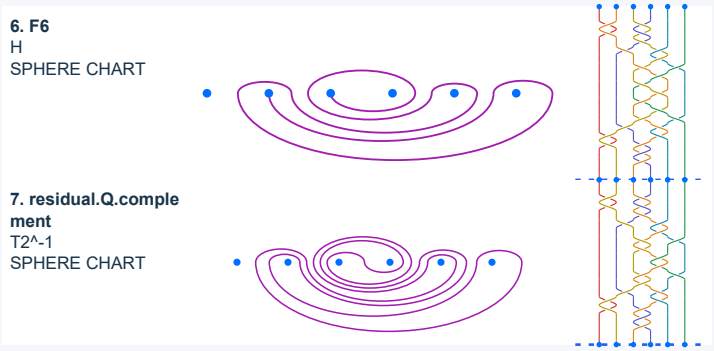
II - step 048

F6 down past residual.Q.complement

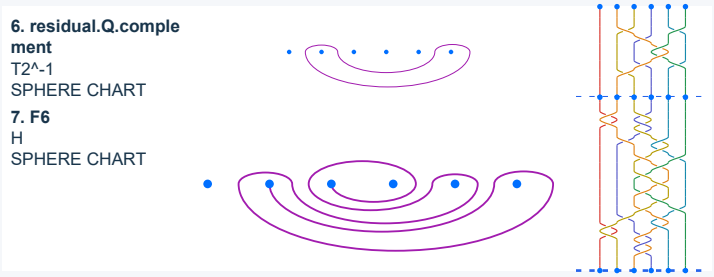
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



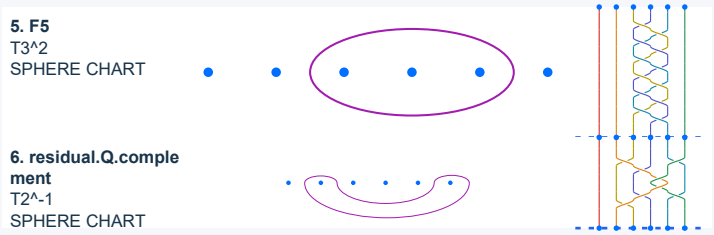
II - step 049

F5 down past residual.Q.complement

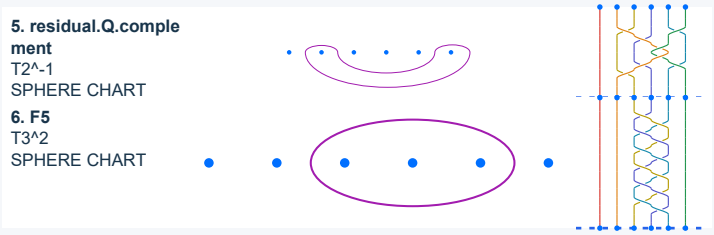
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



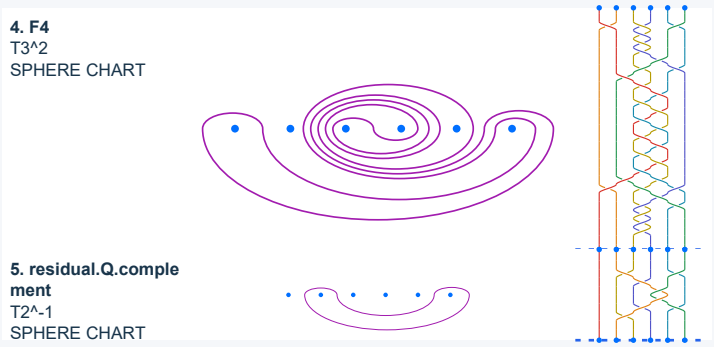
II - step 050

F4 down past residual.Q.complement

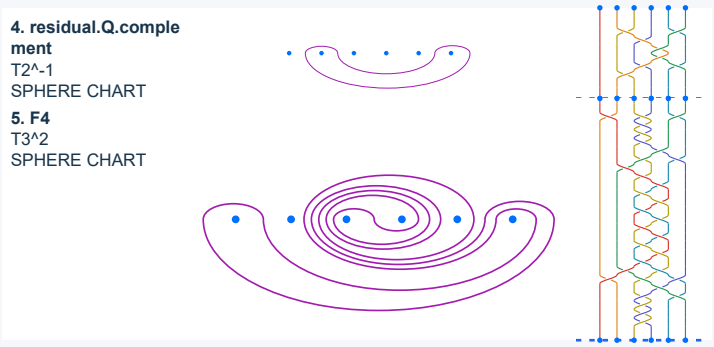
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 15 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



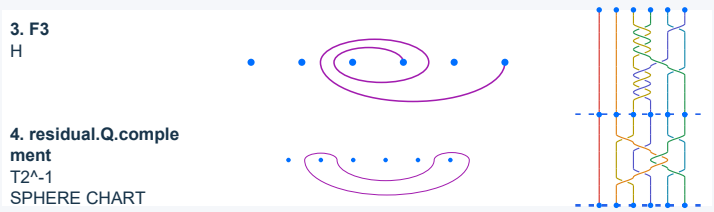
II - step 051

F3 down past residual.Q.complement

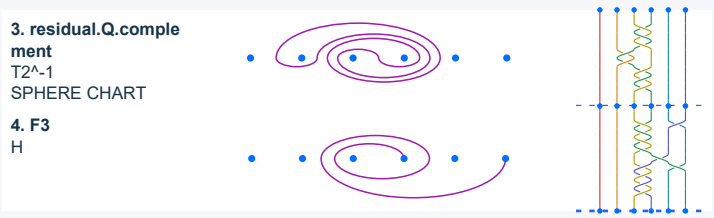
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 16 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 052

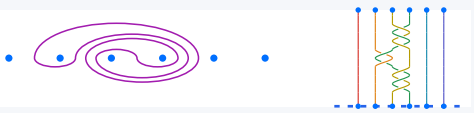
Rechart the transported inverse pair along the prefix pair on the sphere

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

2 unchanged displayed factors above; 17 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

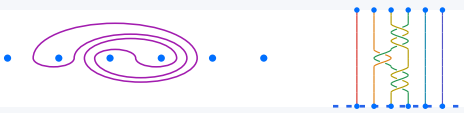
Before

3. residual.Q.complement
T2^-1
SPHERE CHART



After

3. residual.inverse.prefix..
T2^-1
SPHERE CHART



II - step 053

Cancel the equal positive and negative pair twists

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 17 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

2. prefix.P12
T2
SPHERE CHART

3. residual.inverse.prefix..
T2^-1
SPHERE CHART



After

Identity / no factors in this interval.

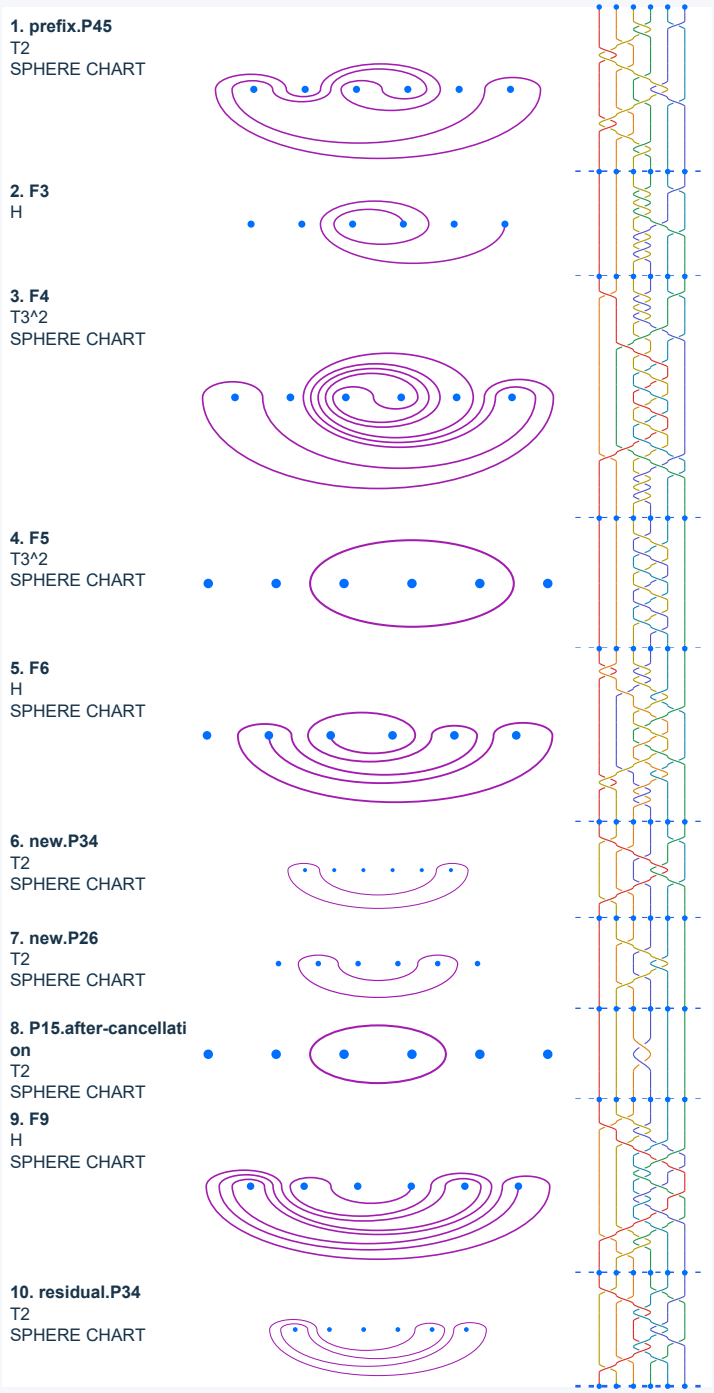
II - step 054

P15.after-cancellation down past F9

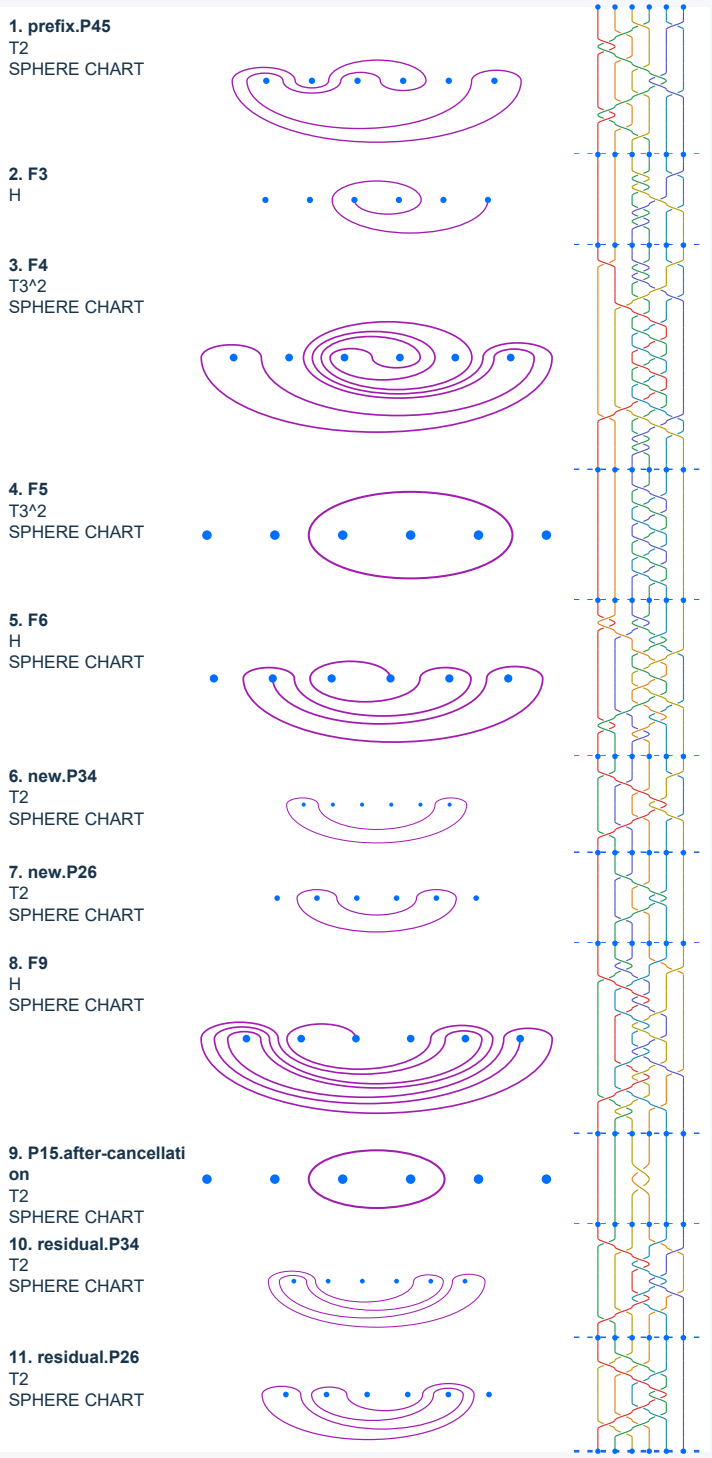
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



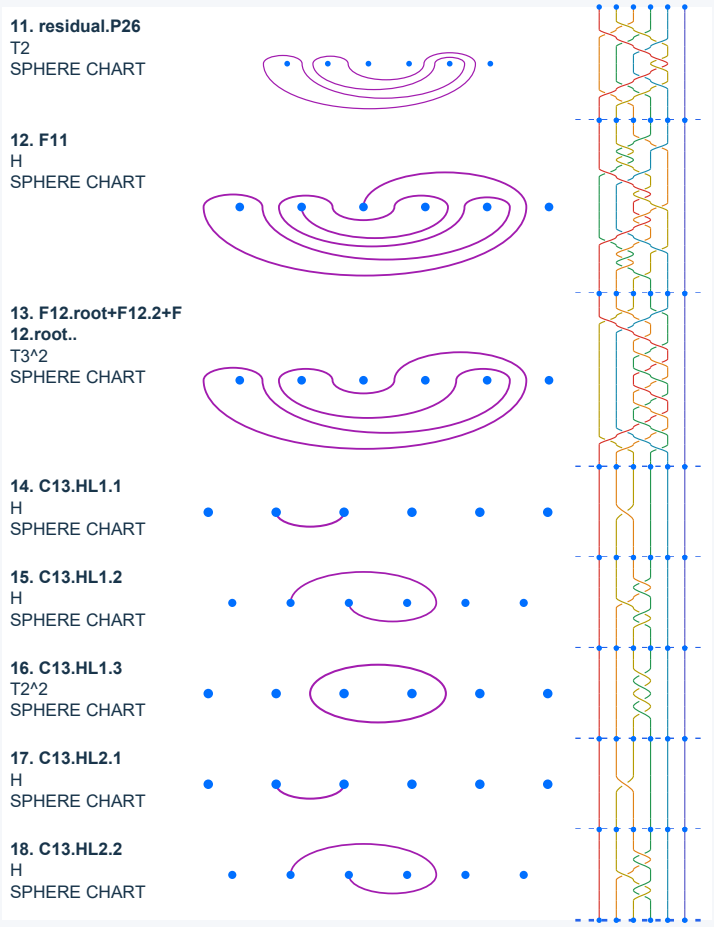
II - step 054 - continued

P15.after-cancellation down past F9

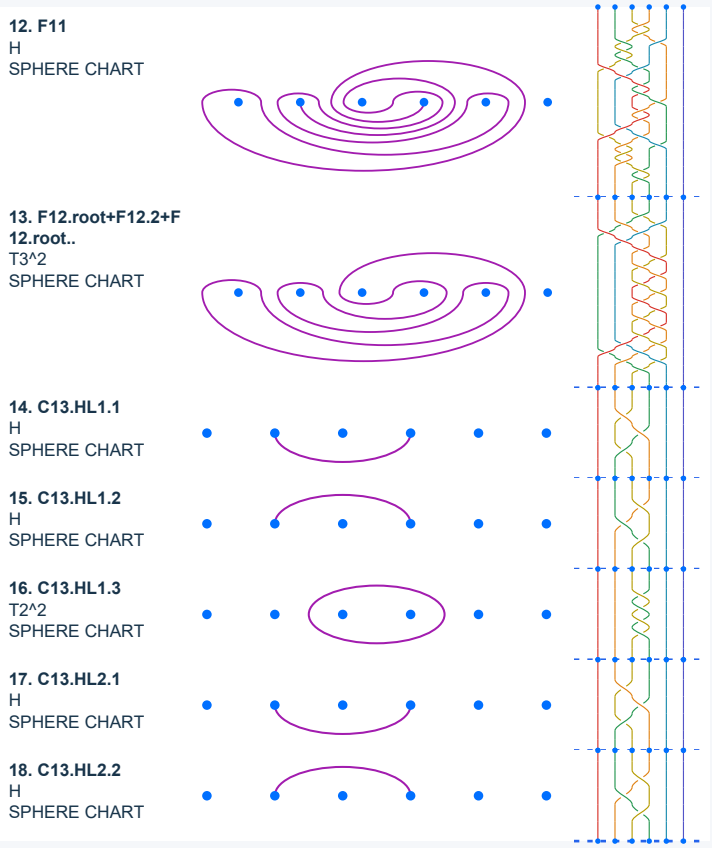
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



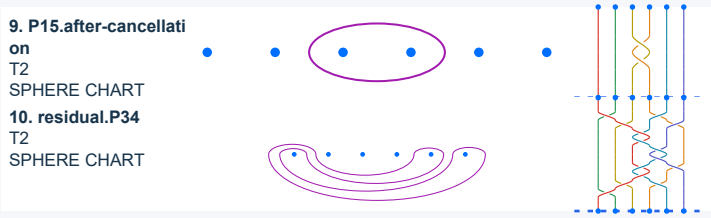
II - step 055

P15.after-cancellation down past residual.P34

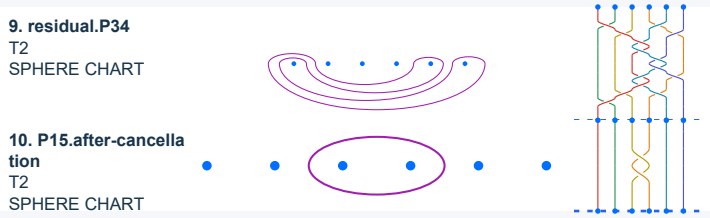
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



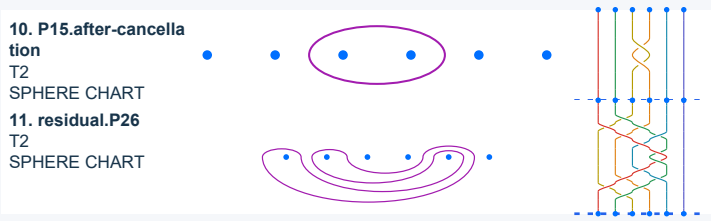
II - step 056

P15.after-cancellation down past residual.P26

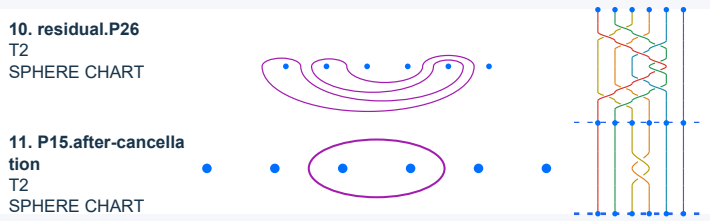
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



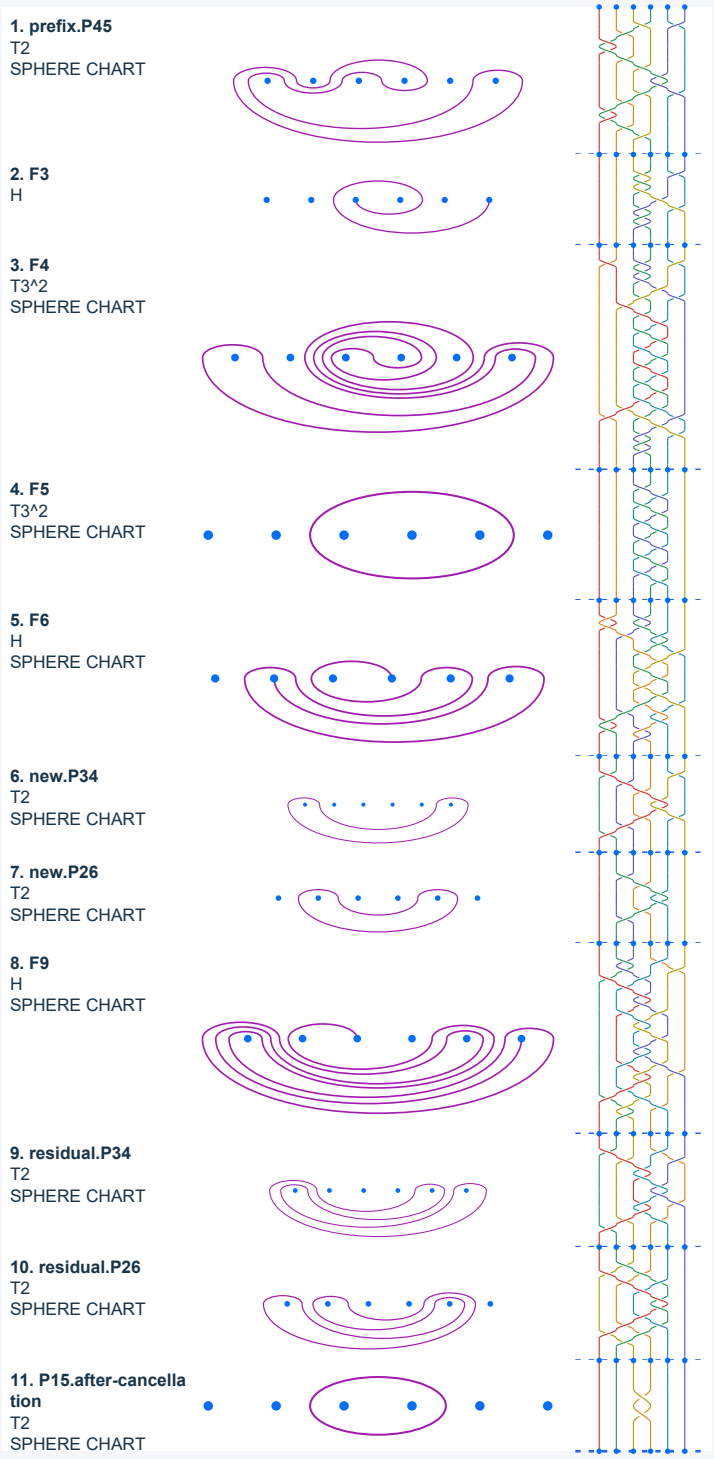
II - step 057

P15.after-cancellation down past F11

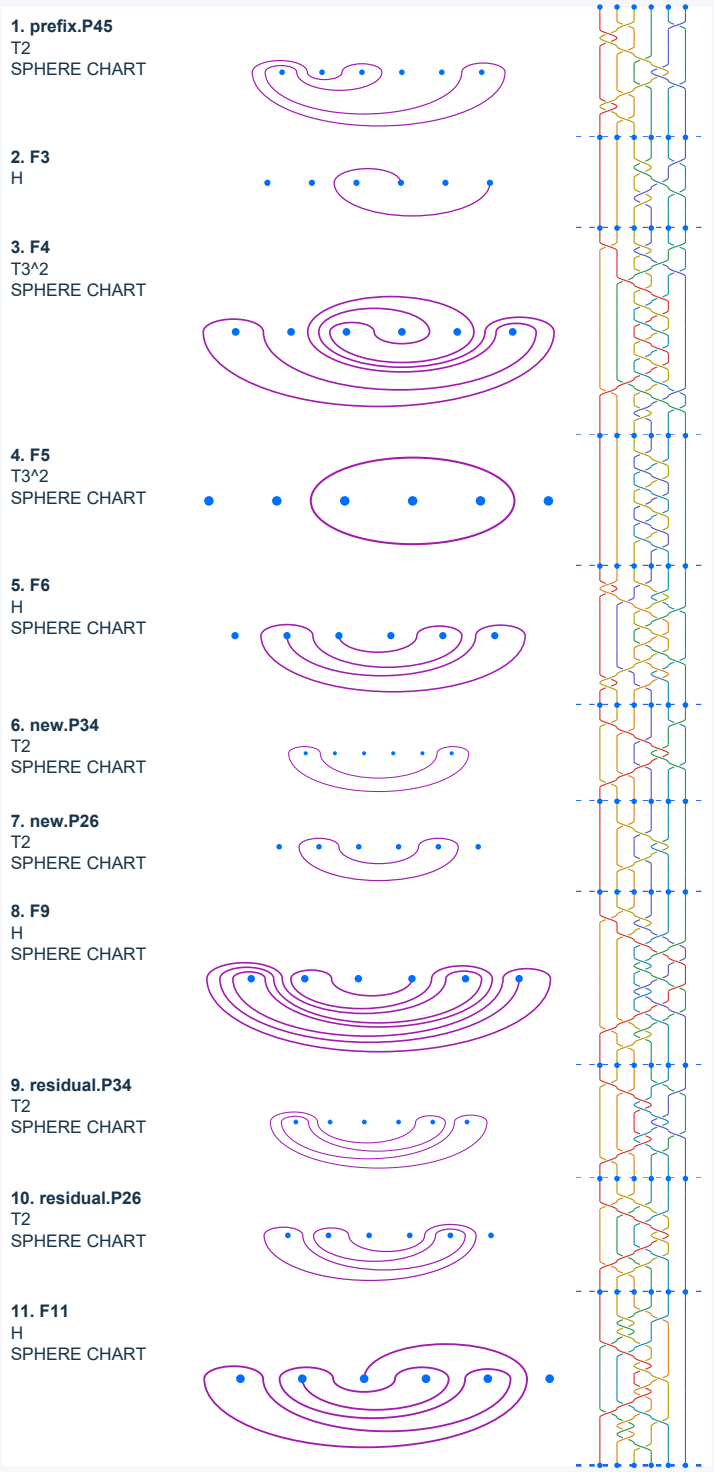
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



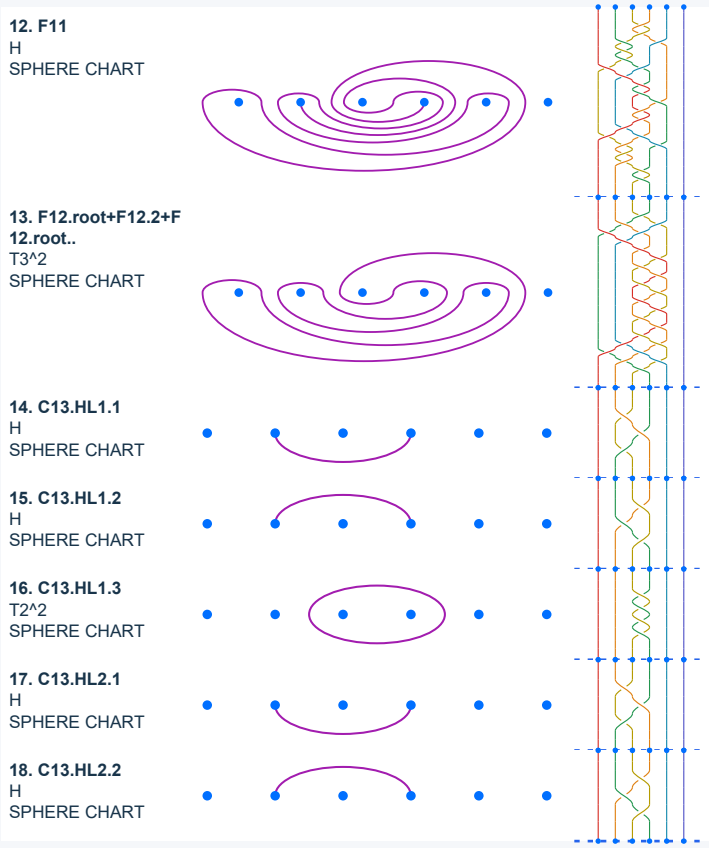
II - step 057 - continued

P15.after-cancellation down past F11

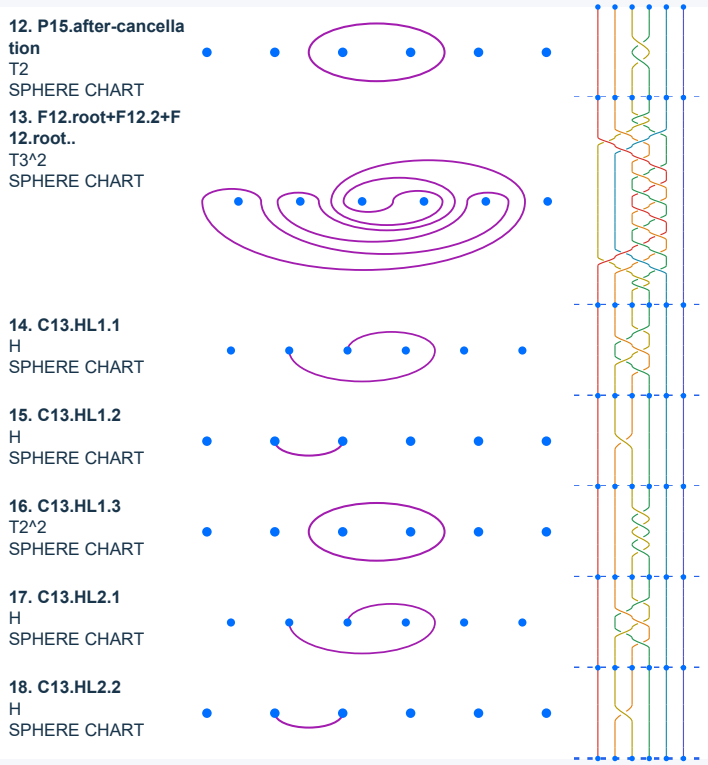
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



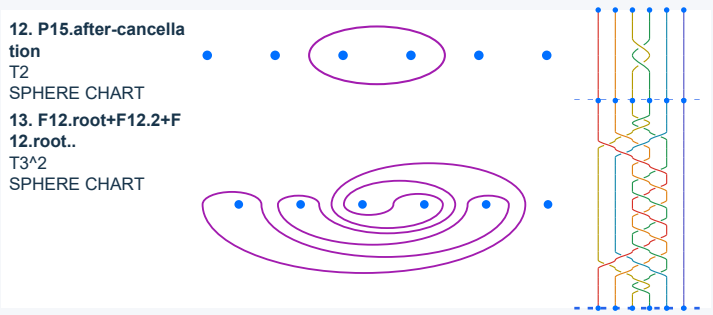
II - step 058

P15.after-cancellation down past F12.root+F12.2+F12.root+F12.3

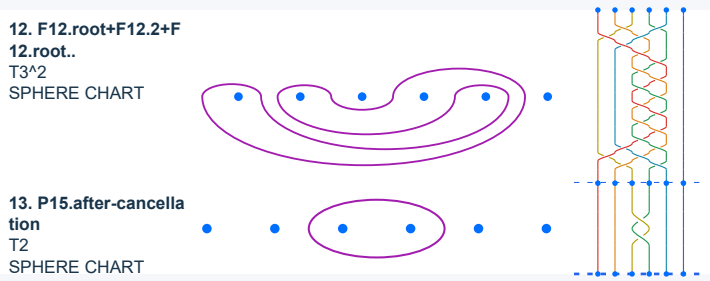
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



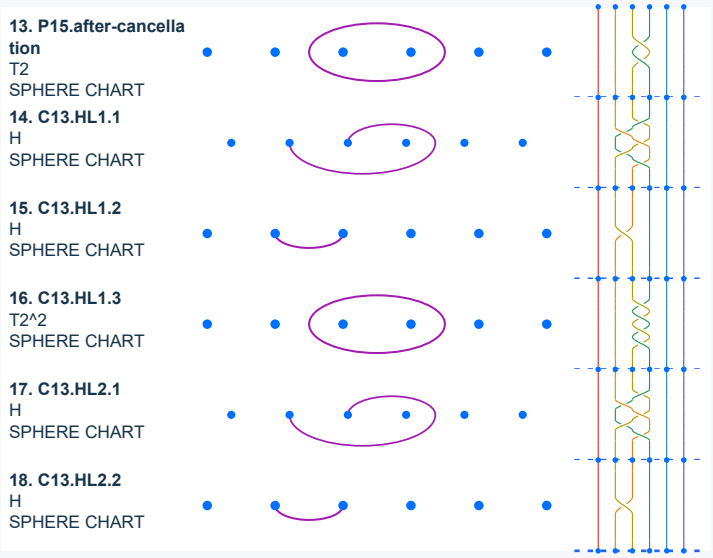
II - step 059

Reverse the two PC half lanterns and retain the inverse pair

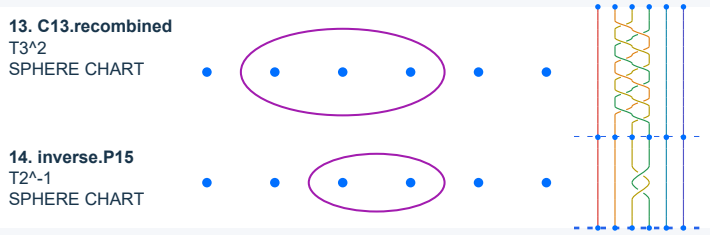
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



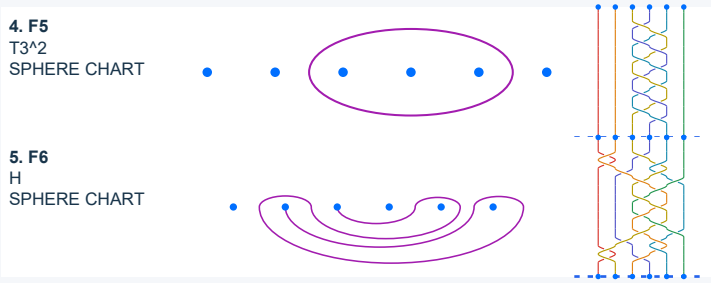
II - step 060

F5 down past F6

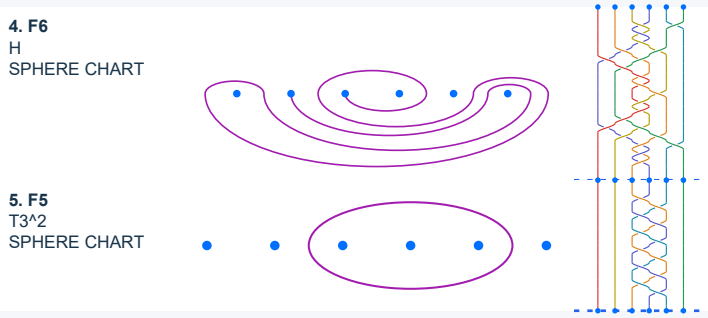
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



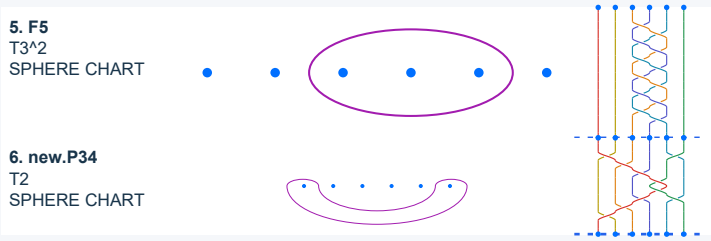
II - step 061

F5 down past new.P34

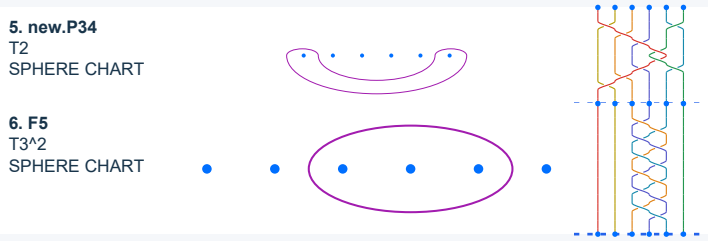
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



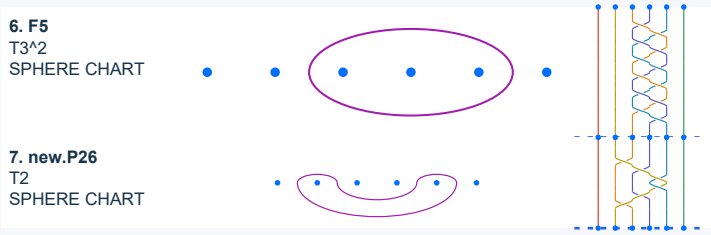
II - step 062

F5 down past new.P26

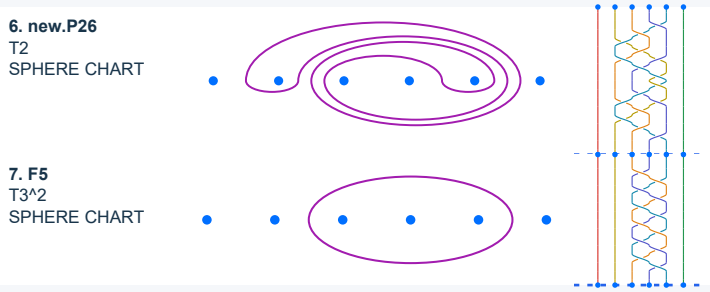
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



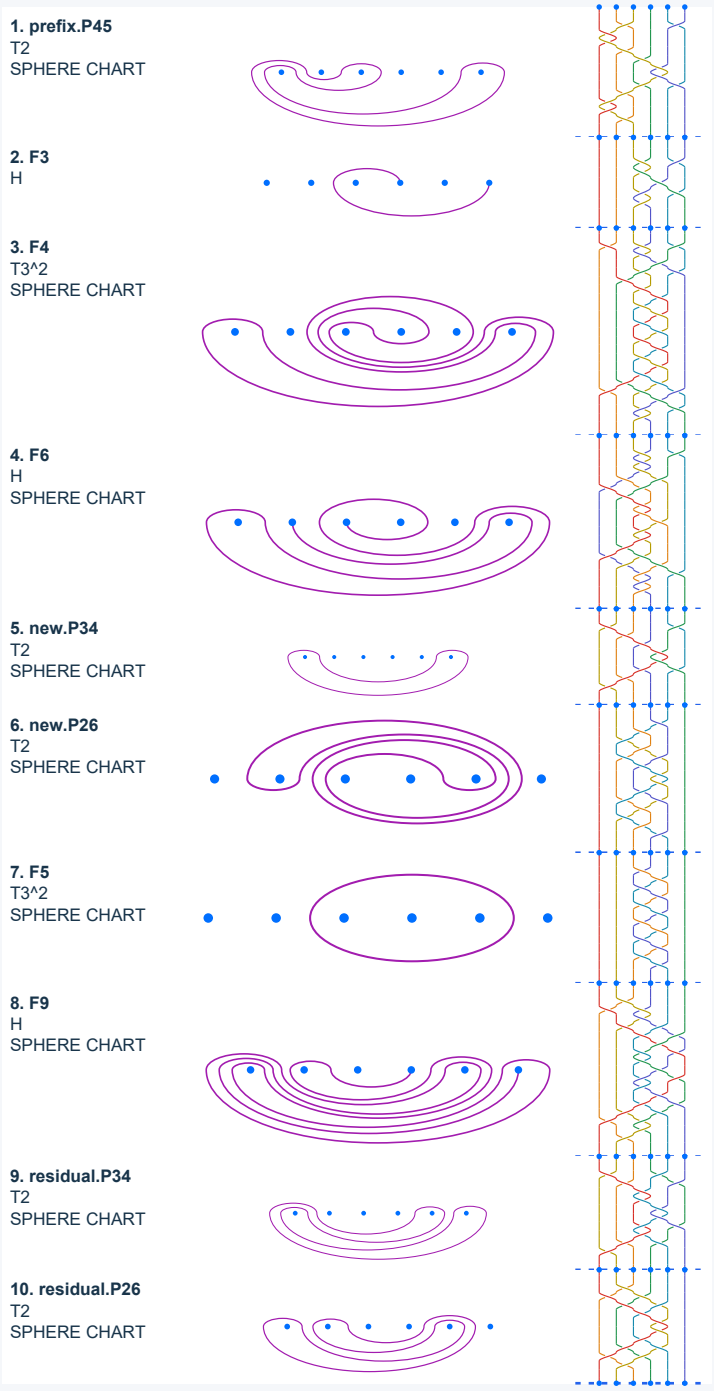
II - step 063

F5 down past F9

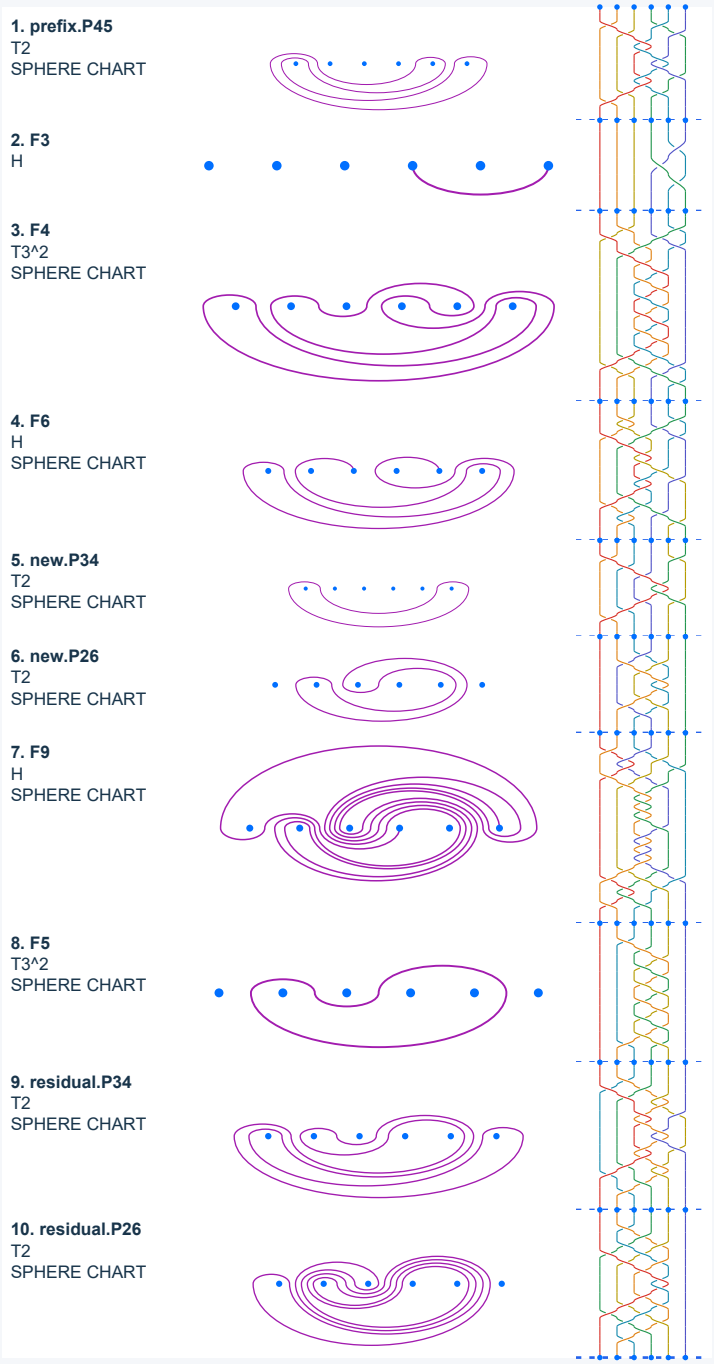
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



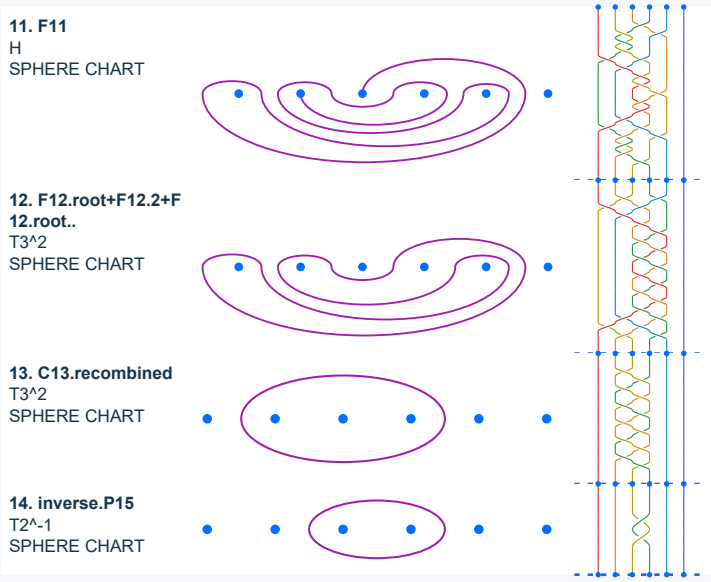
II - step 063 - continued

F5 down past F9

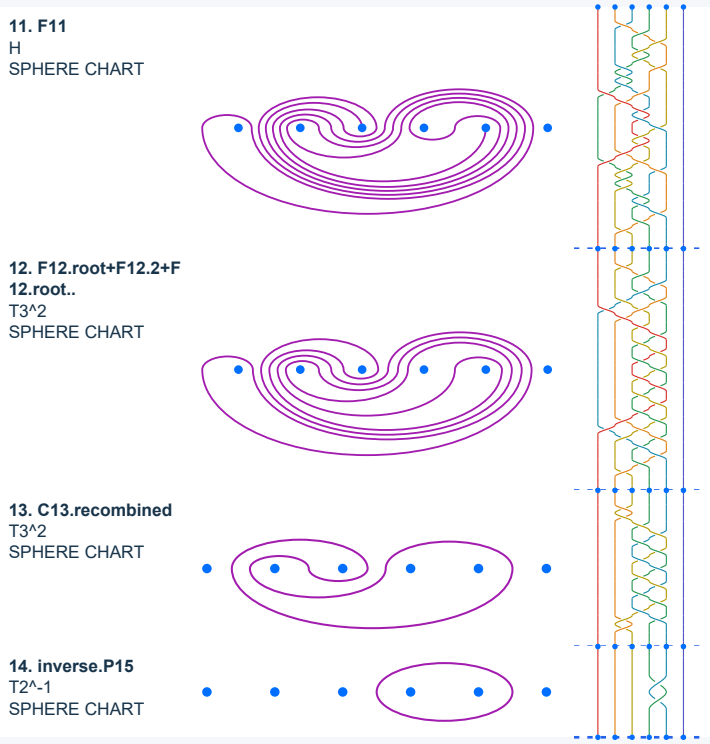
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 -2 4 -3 2 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



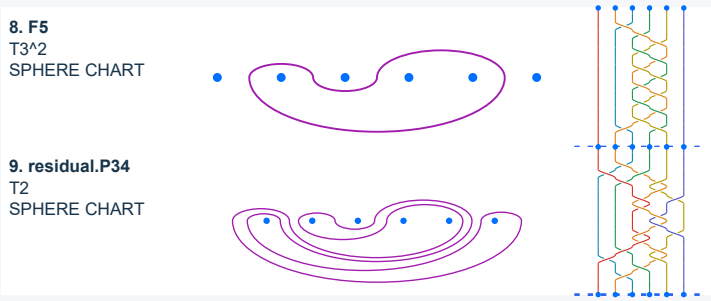
II - step 064

F5 down past residual.P34

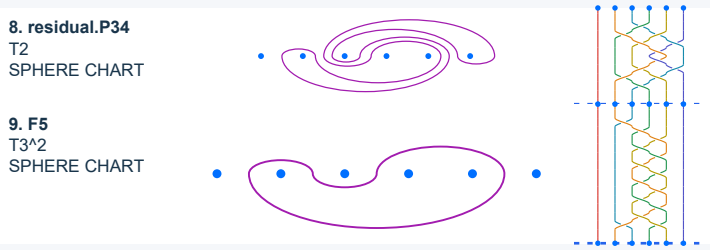
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



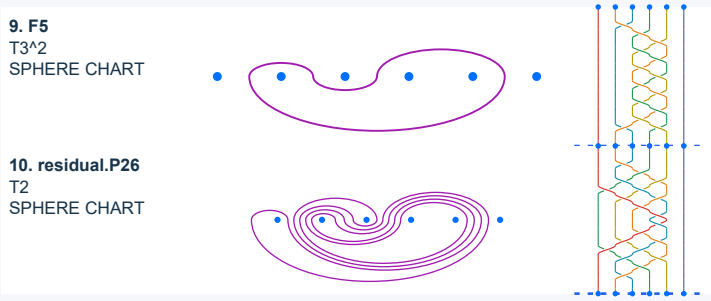
II - step 065

F5 down past residual.P26

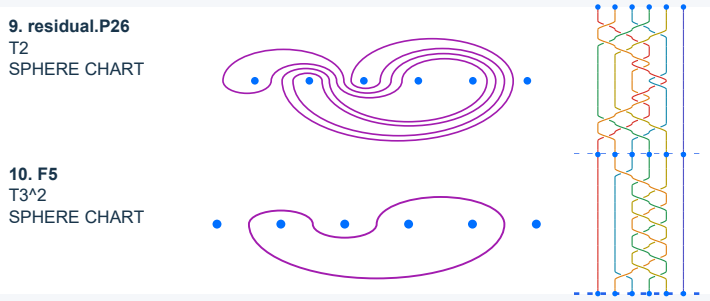
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



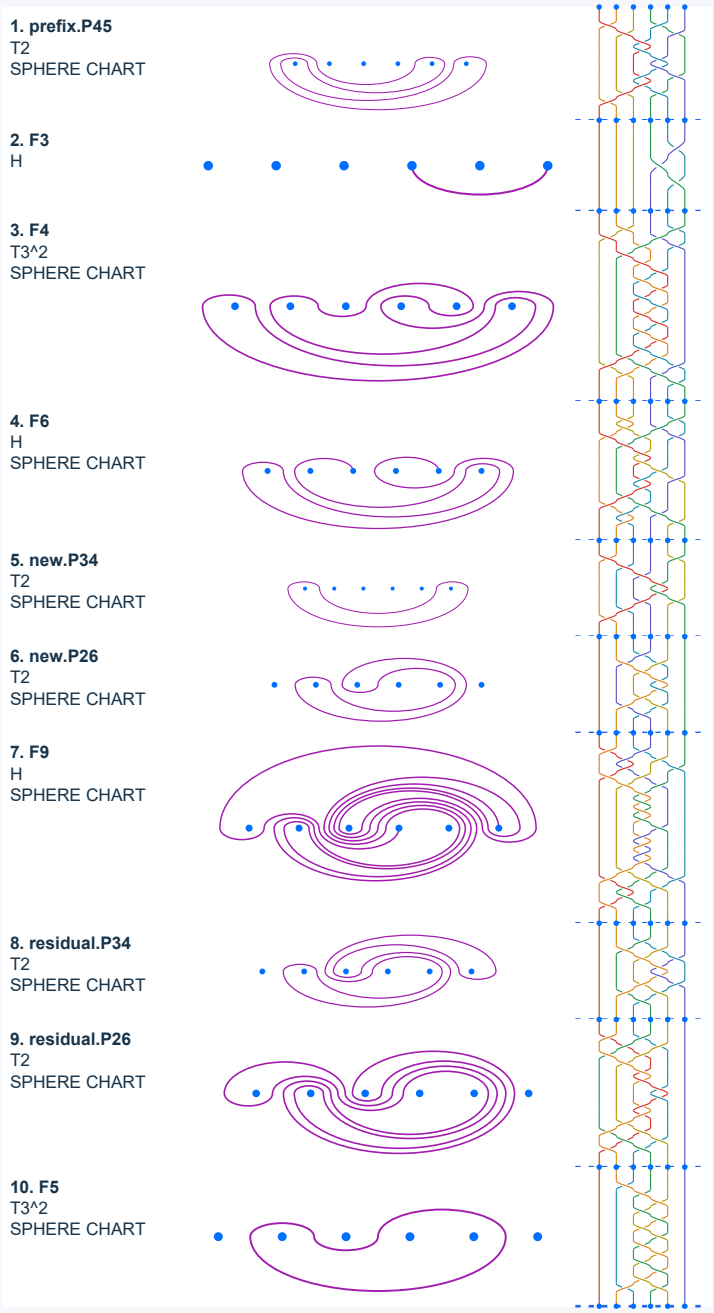
II - step 066

F5 down past F11

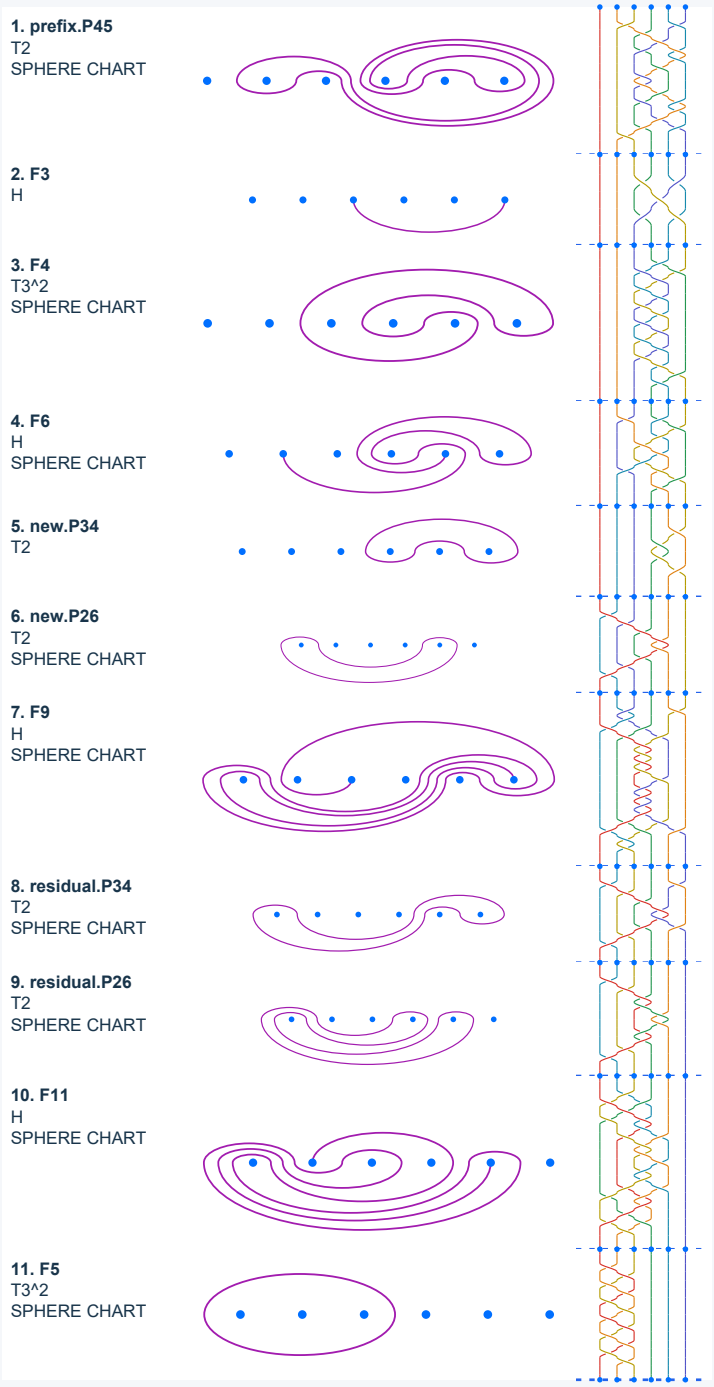
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



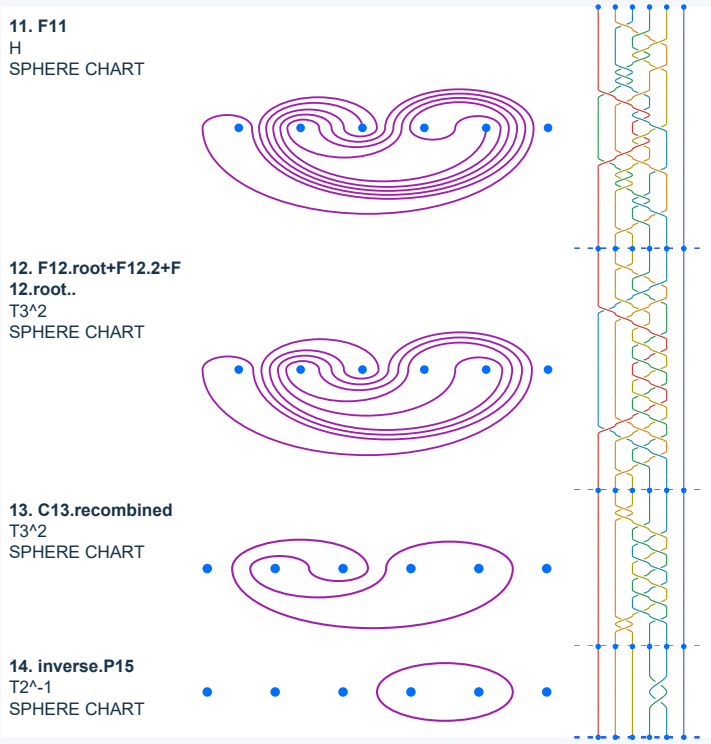
II - step 066 - continued

F5 down past F11

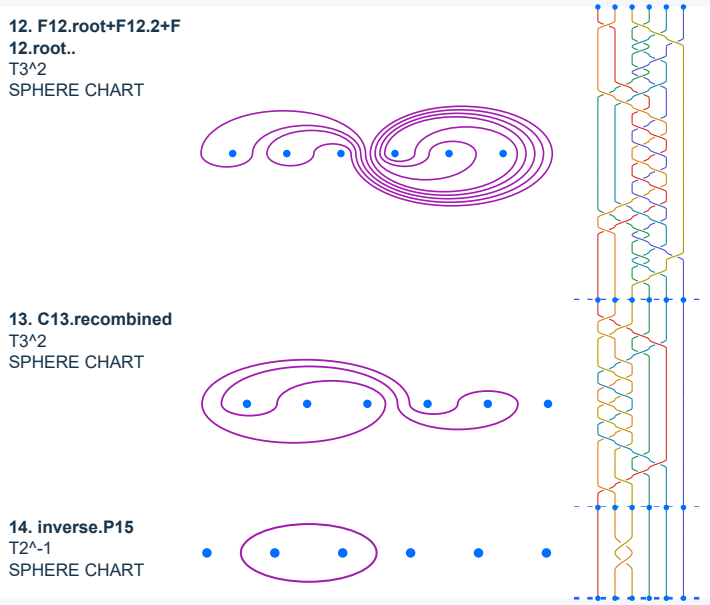
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-3 -1 3 4 -3 -2 -4 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



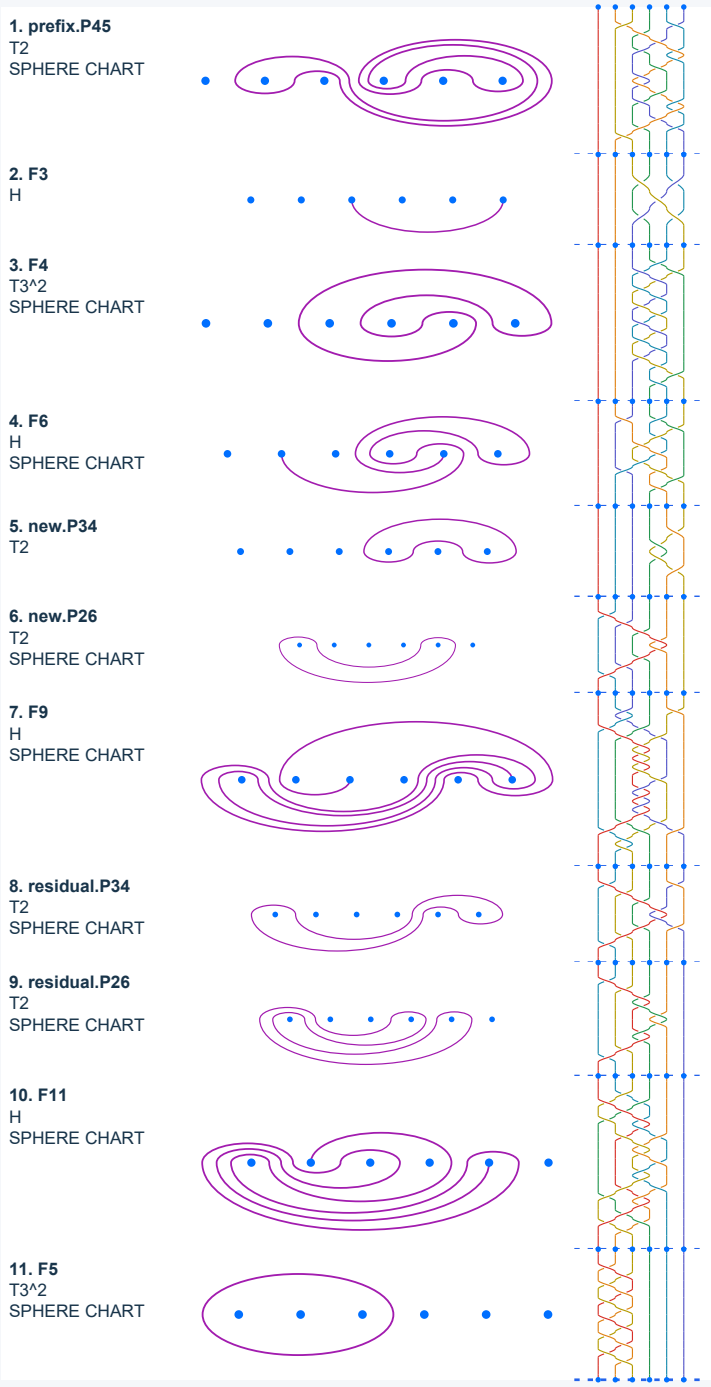
II - step 067

F5 down past F12.root+F12.2+F12.root+F12.3

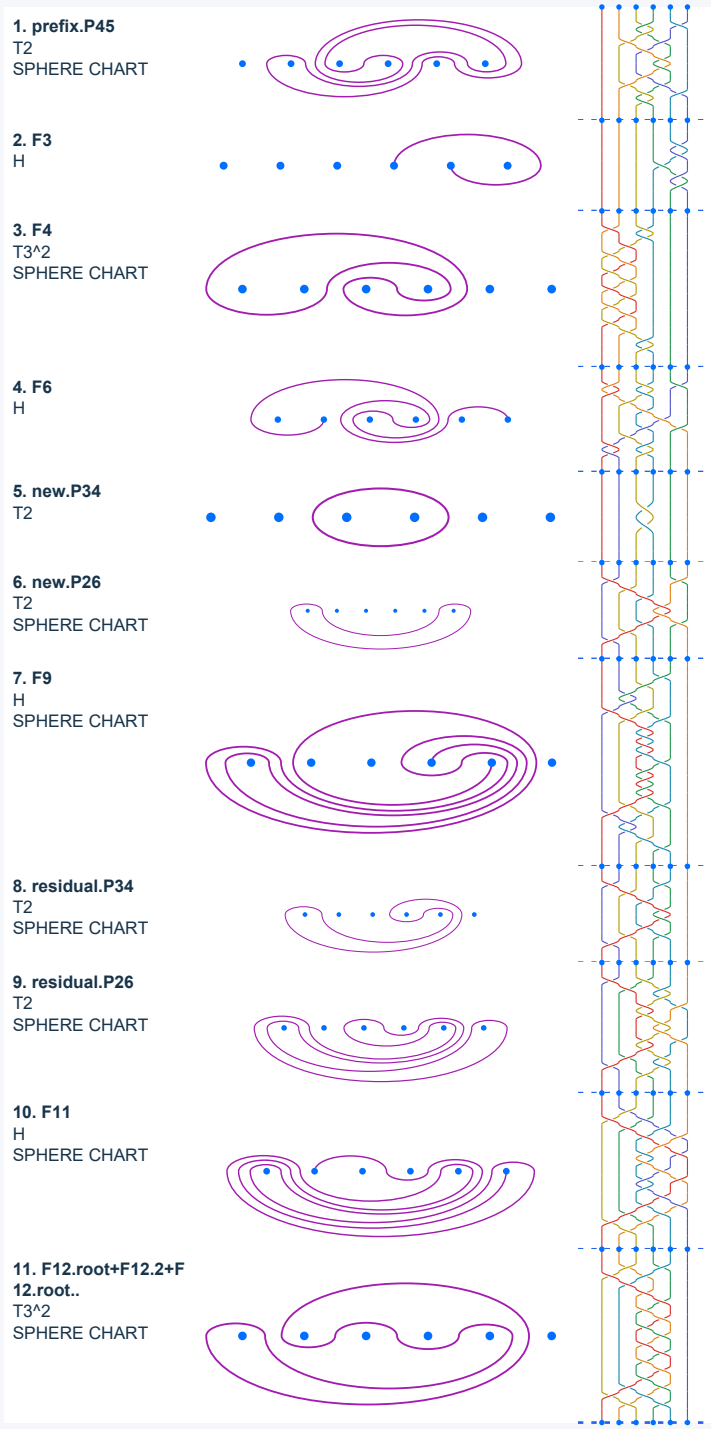
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



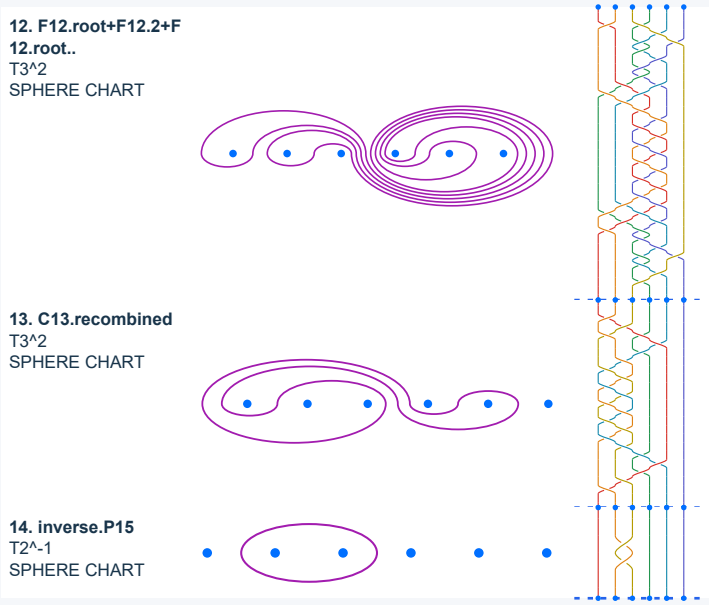
II - step 067 - continued

F5 down past F12.root+F12.2+F12.root+F12.3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=3 5 4 -1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



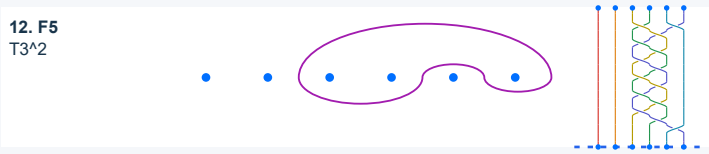
II - step 068

Rechart the whole C5 square along its complementary {1,3,5} triple

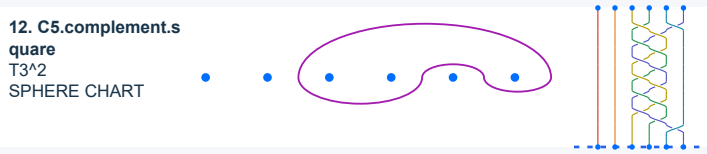
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



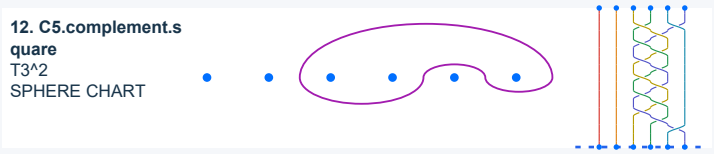
II - step 069

Use two cyclic PC roses, each ending in the P15 petal

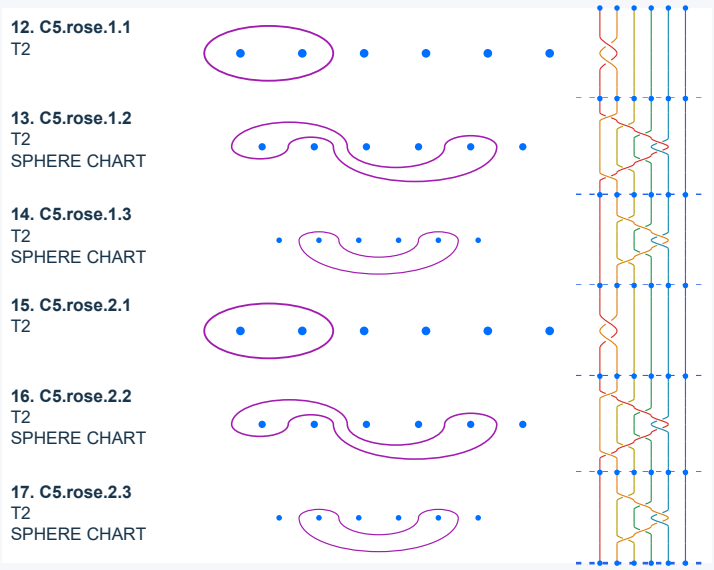
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 070

Rechart the last petal along the retained inverse pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



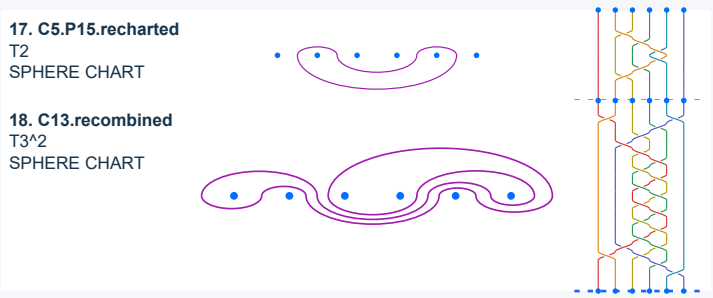
II - step 071

C5.P15.recharted down past C13.recombined

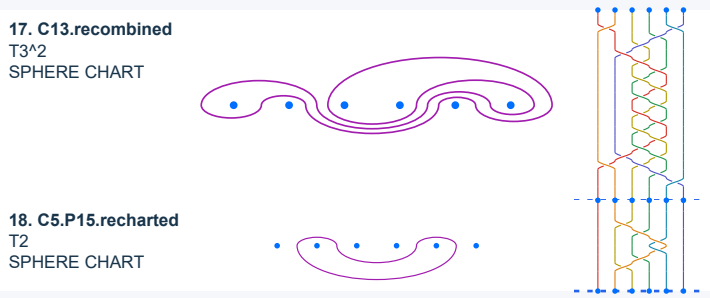
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



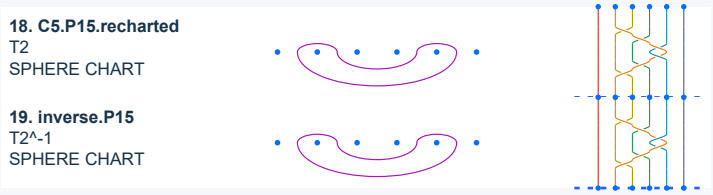
II - step 072

Cancel the last petal against the inverse pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

17 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After

Identity / no factors in this interval.

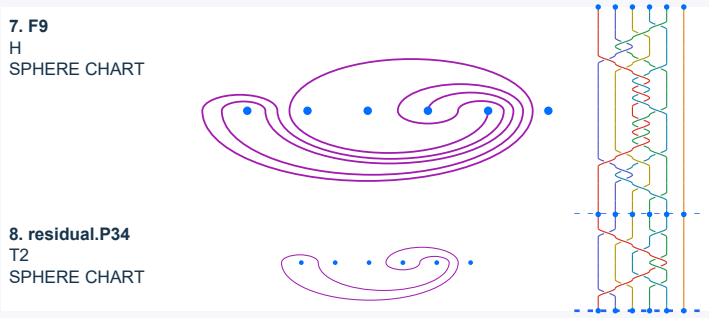
II - step 073

F9 down past residual.P34

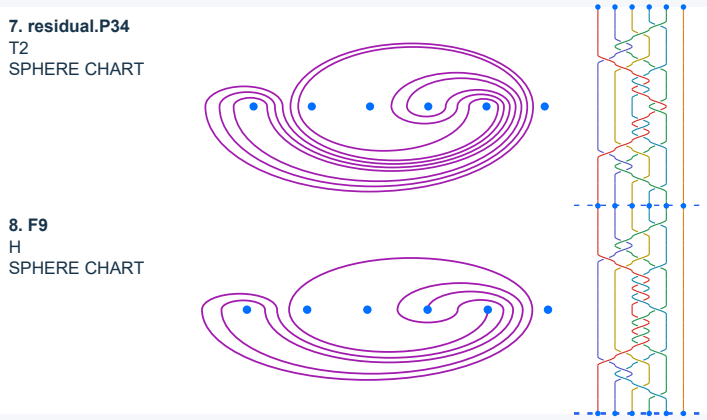
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



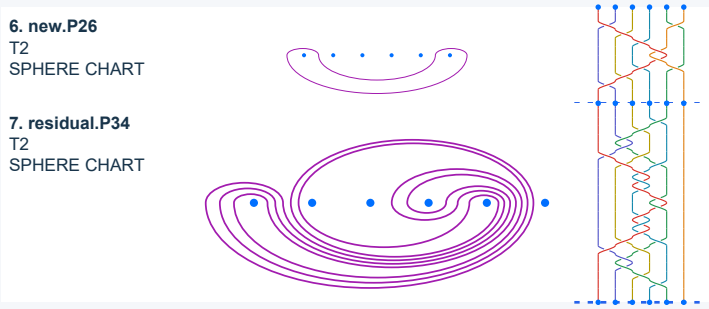
II - step 074

new.P26 down past residual.P34

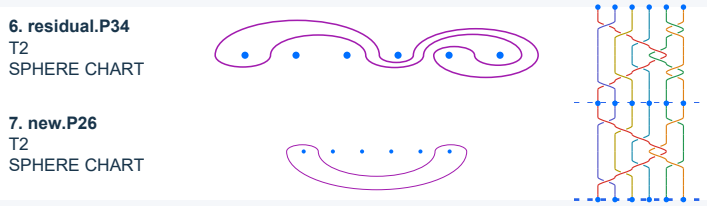
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



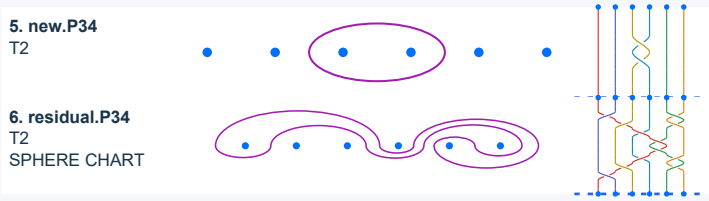
II - step 075

new.P34 down past residual.P34

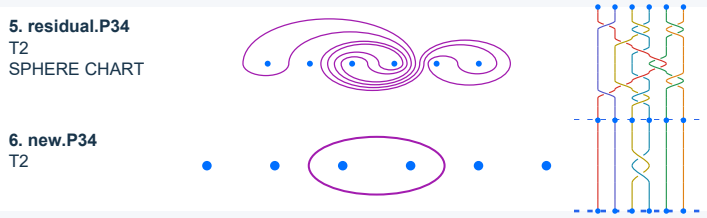
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



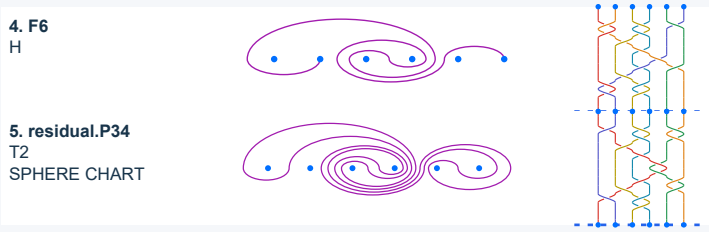
II - step 076

F6 down past residual.P34

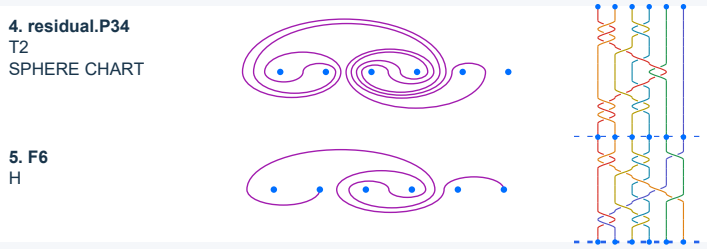
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



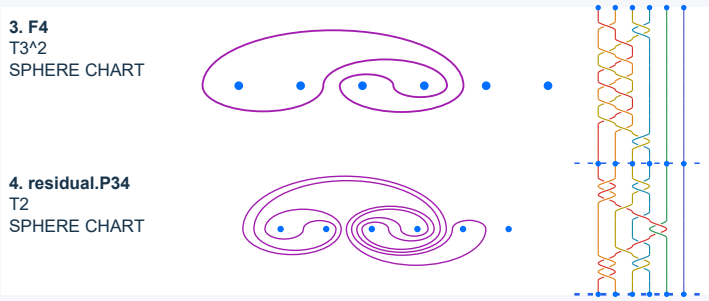
II - step 077

F4 down past residual.P34

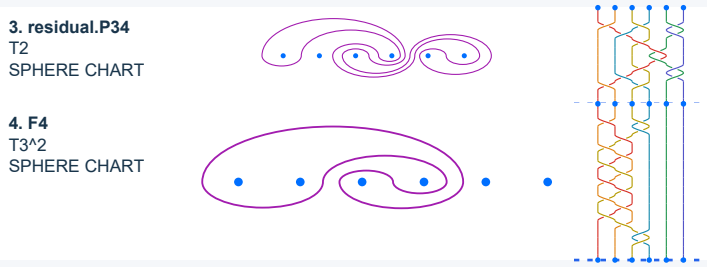
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



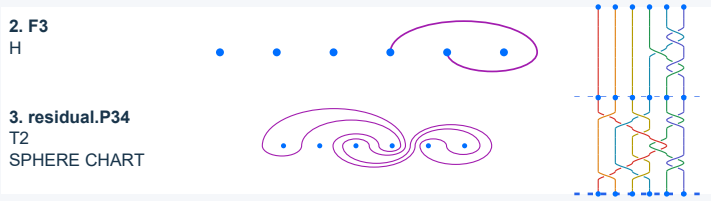
II - step 078

F3 down past residual.P34

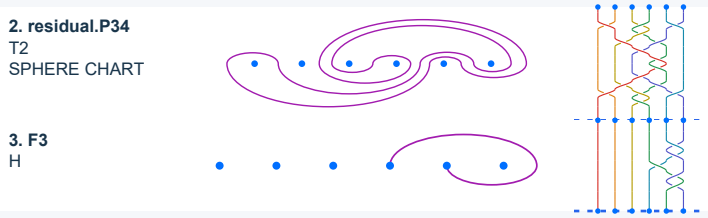
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



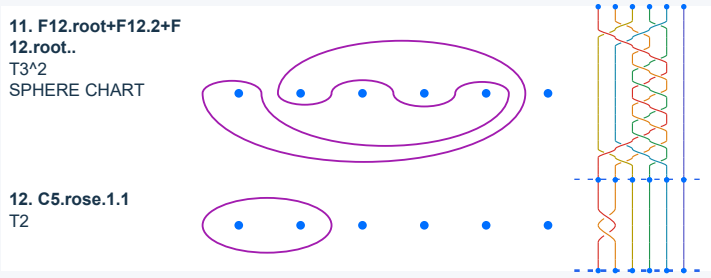
II - step 079

C5.rose.1.1 up past F12.root+F12.2+F12.root+F12.3

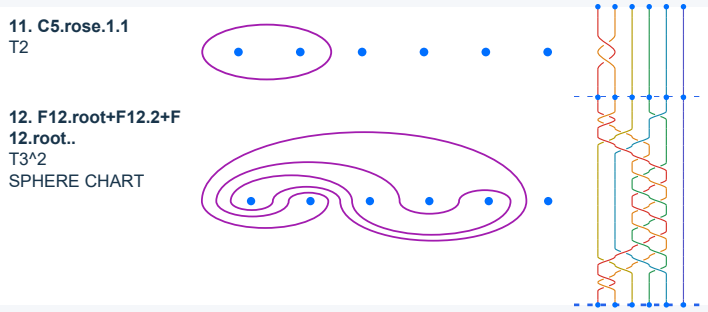
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



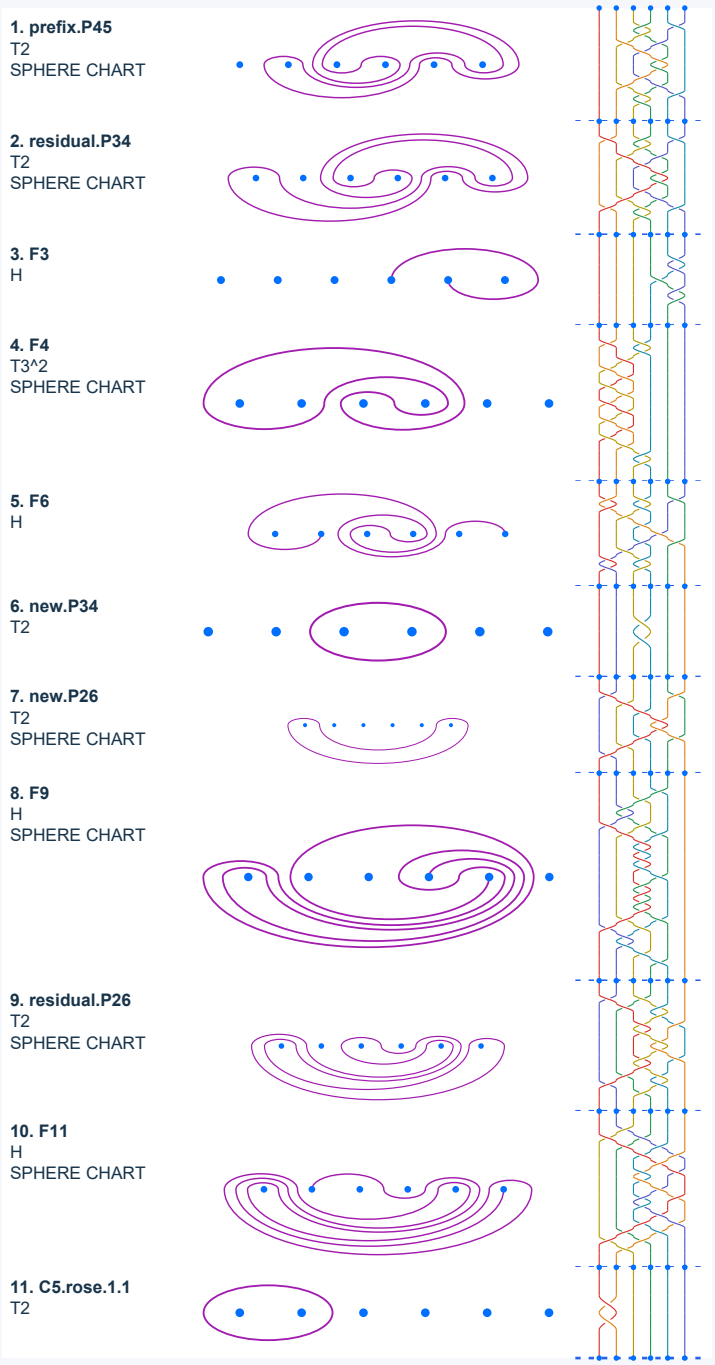
II - step 080

C5.rose.1.1 up past F11

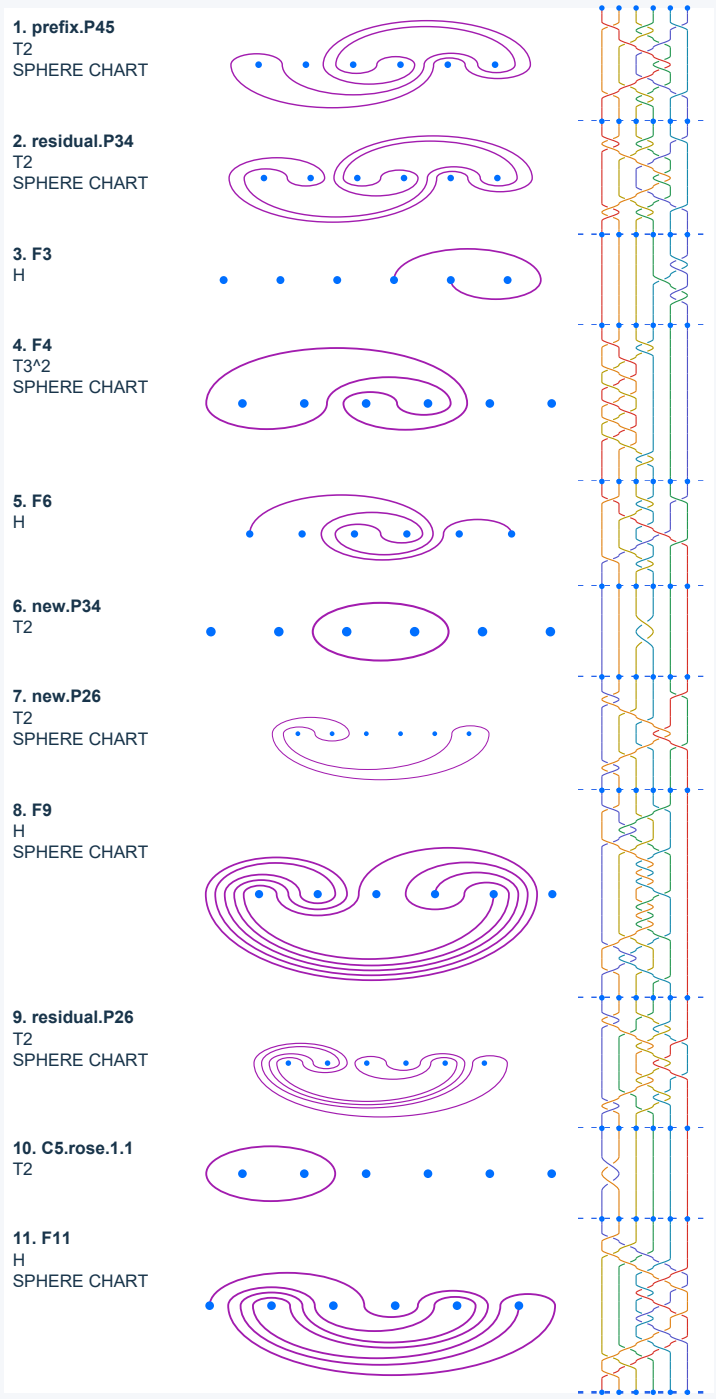
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



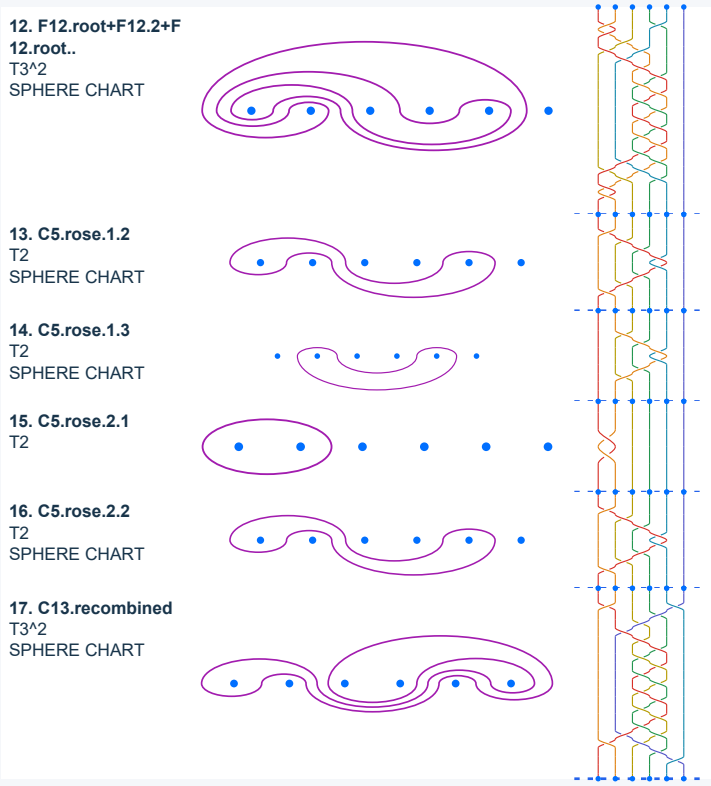
II - step 080 - continued

C5.rose.1.1 up past F11

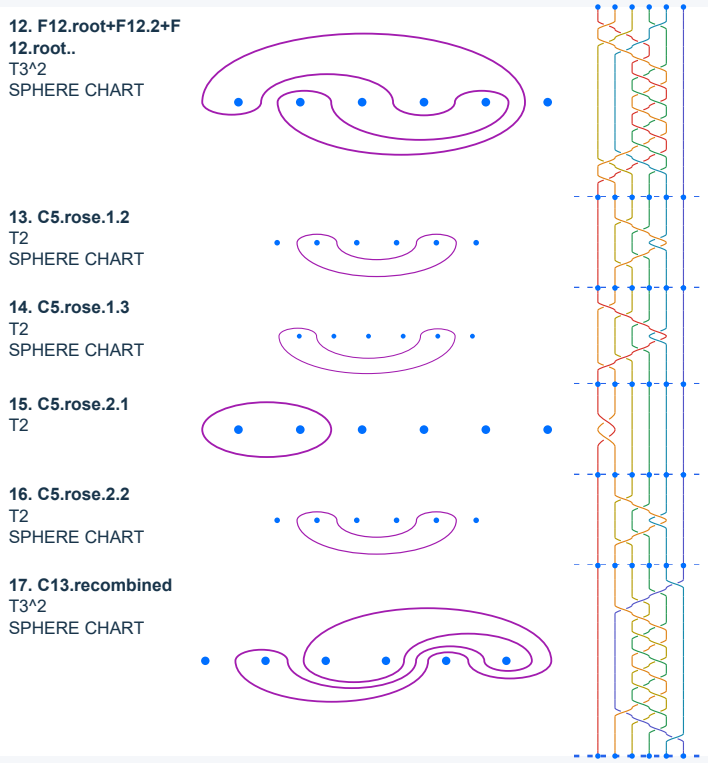
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 -1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



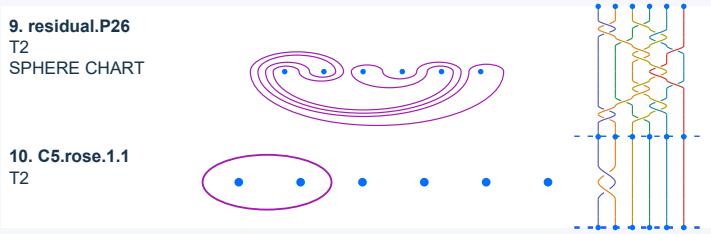
II - step 081

C5.rose.1.1 up past residual.P26

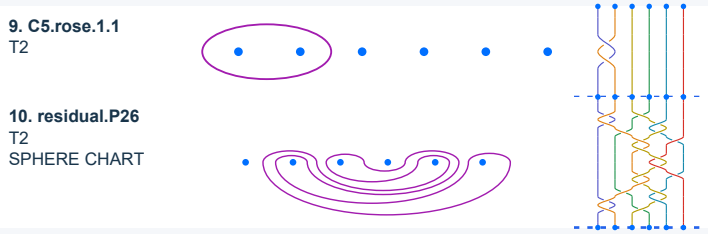
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



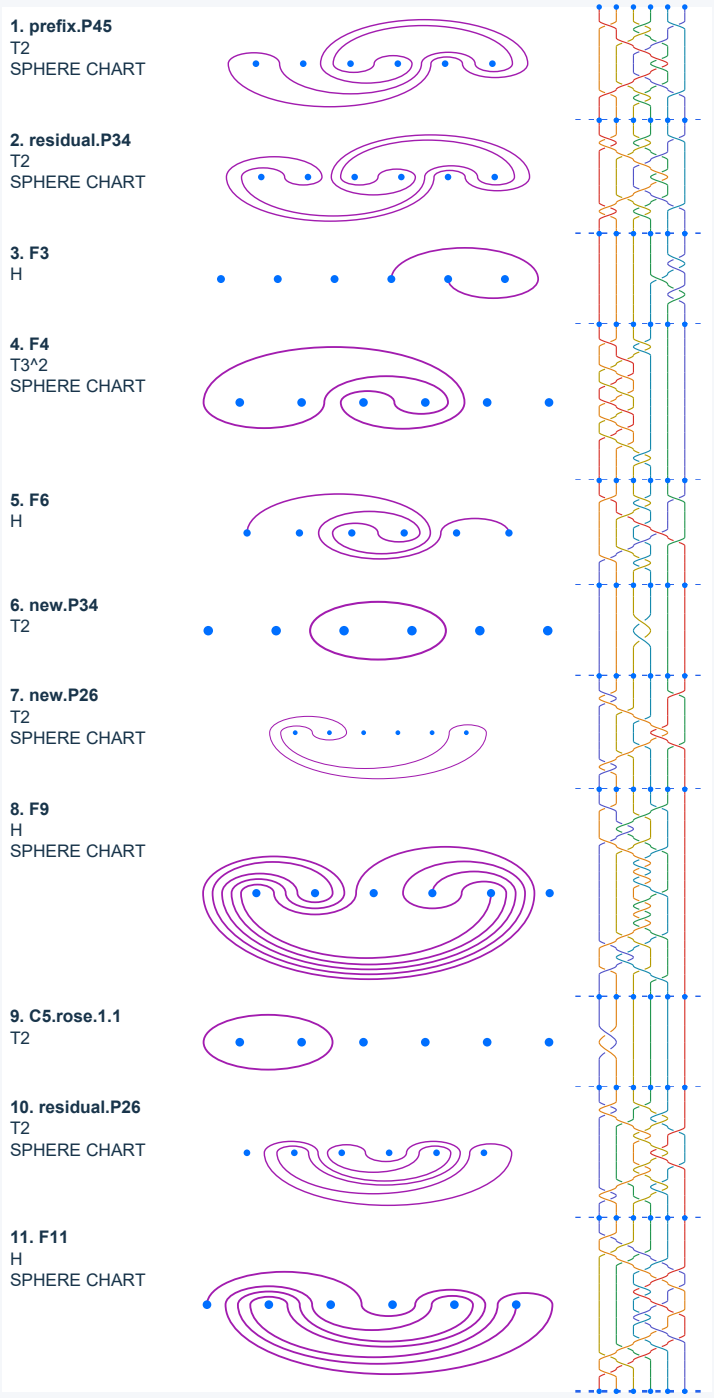
II - step 082

C5.rose.1.1 up past F9

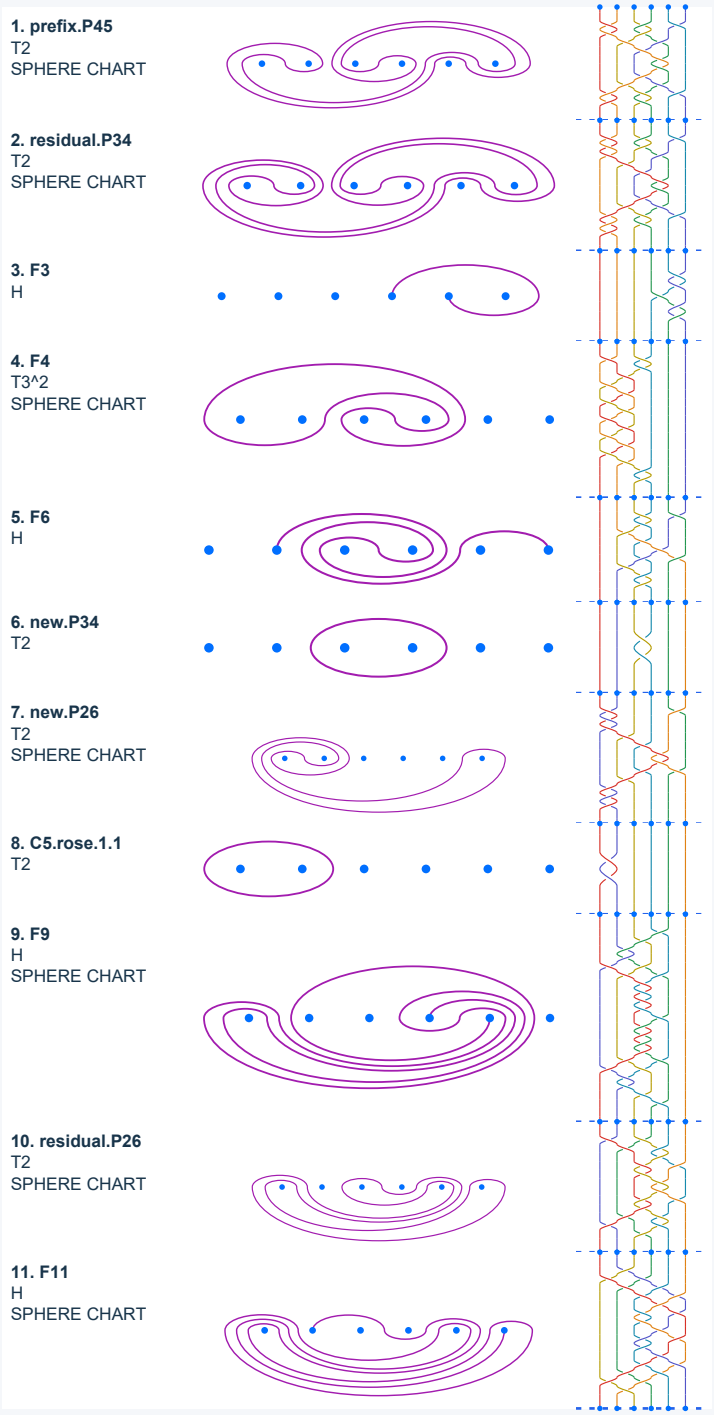
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



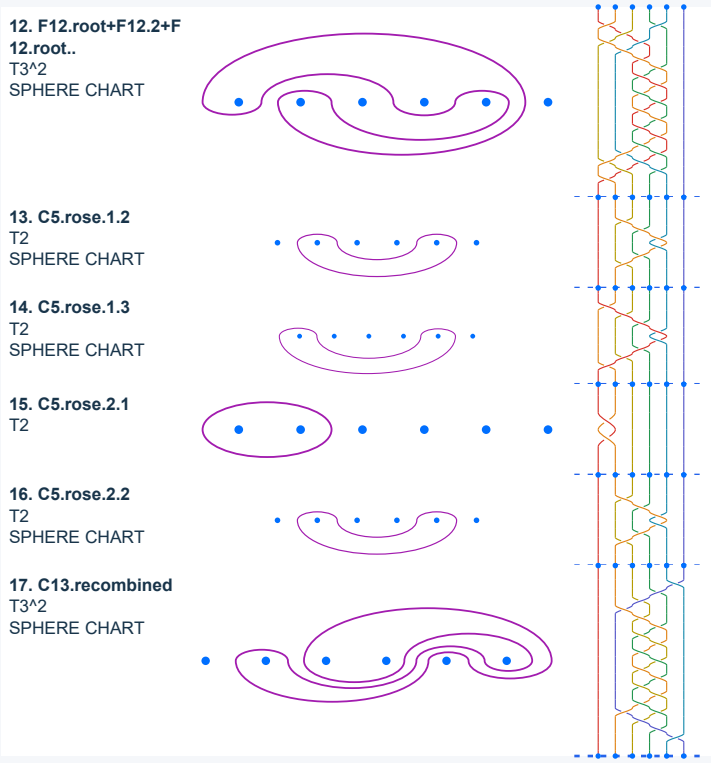
II - step 082 - continued

C5.rose.1.1 up past F9

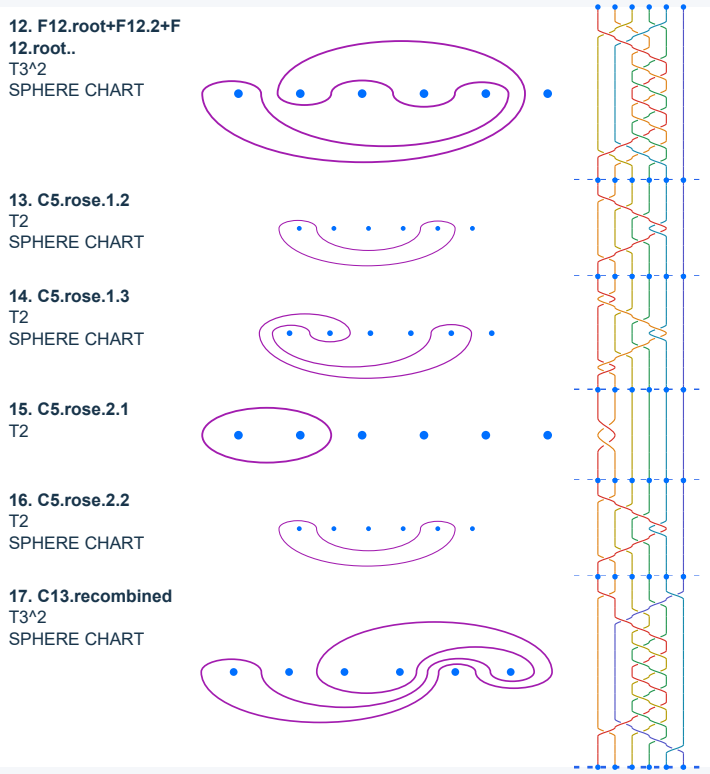
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-1 2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



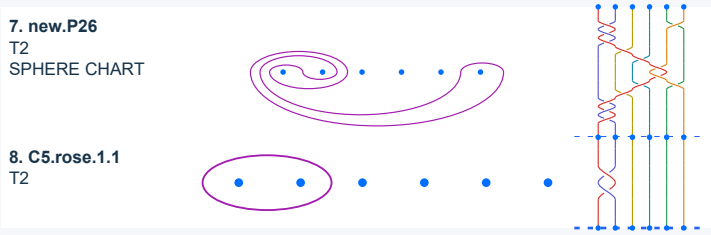
II - step 083

C5.rose.1.1 up past new.P26

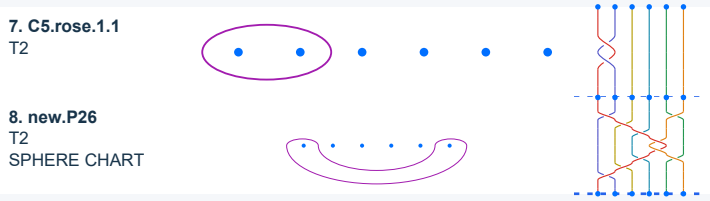
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



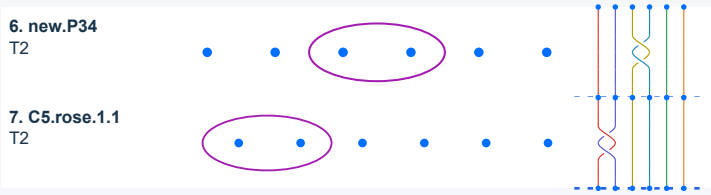
II - step 084

C5.rose.1.1 up past new.P34

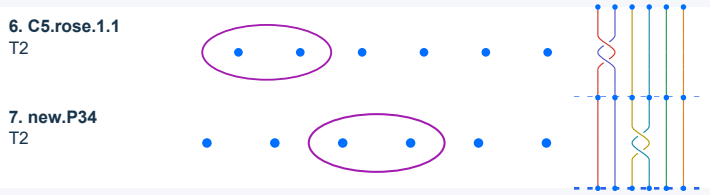
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



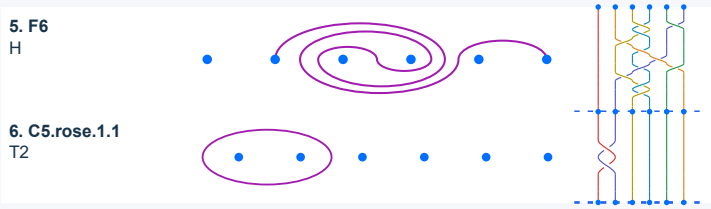
II - step 085

C5.rose.1.1 up past F6

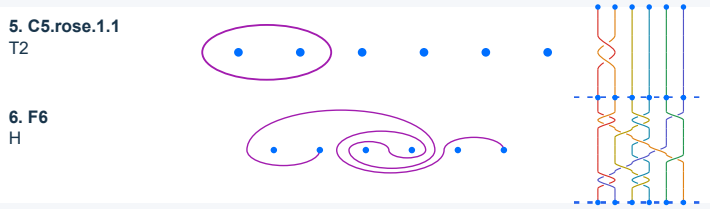
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



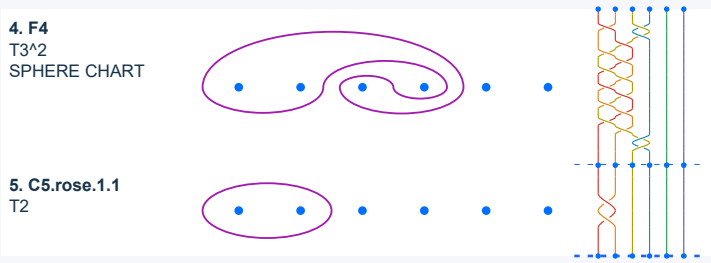
II - step 086

C5.rose.1.1 up past F4

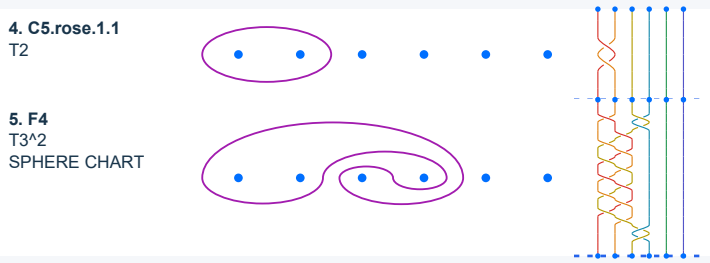
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



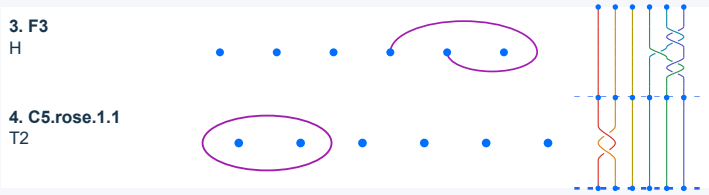
II - step 087

C5.rose.1.1 up past F3

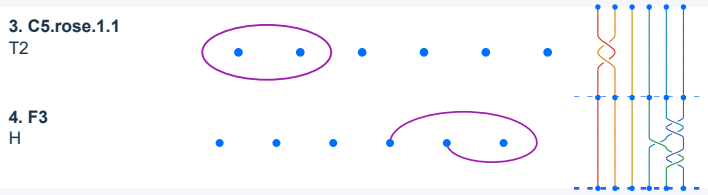
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



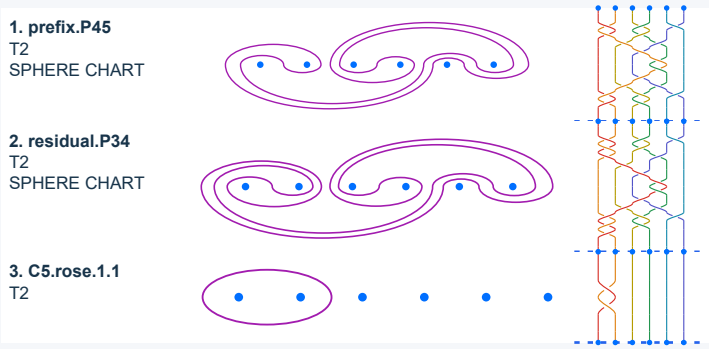
II - step 088

Reverse the transported PC lantern on {3,4,5}

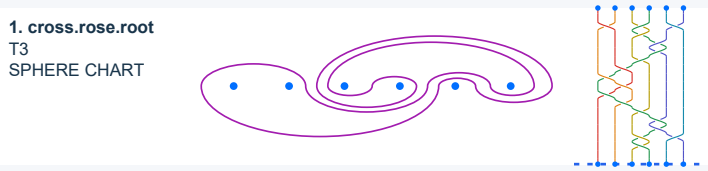
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



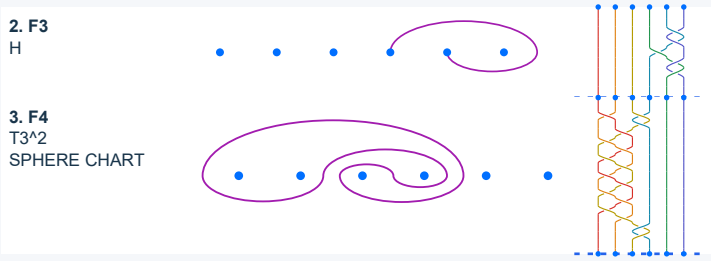
II - step 089

F3 down past F4

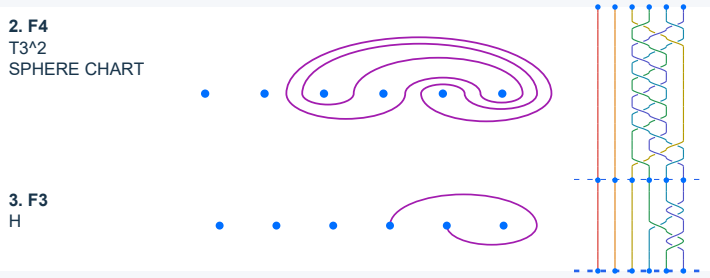
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



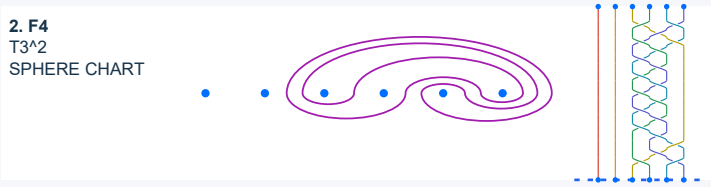
II - step 090

Put the transported C4 square in the common {35},4,6 chart

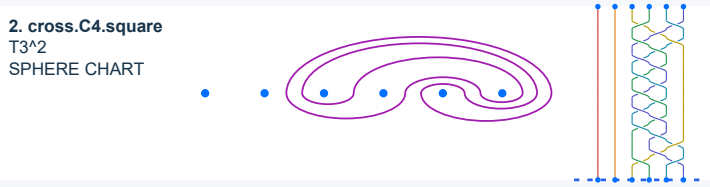
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



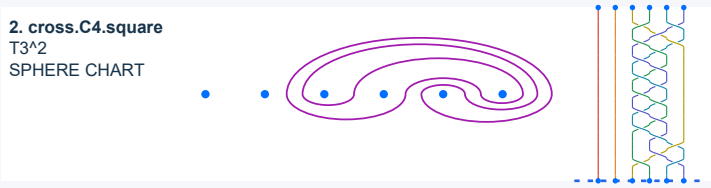
II - step 091

Split the common-chart C4 square

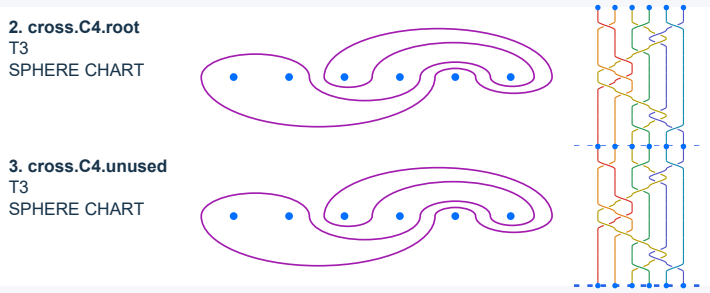
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



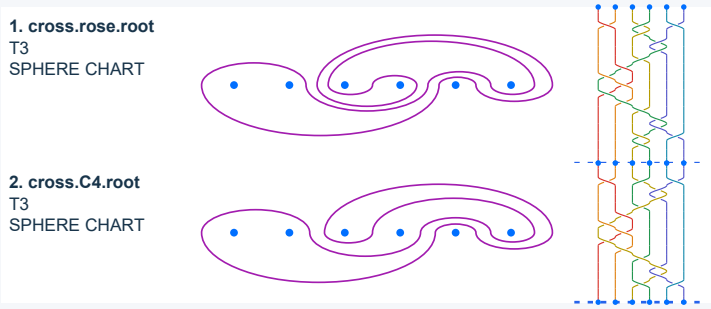
II - step 092

Apply the transported cluster lantern AB=O P35 Q46^-1

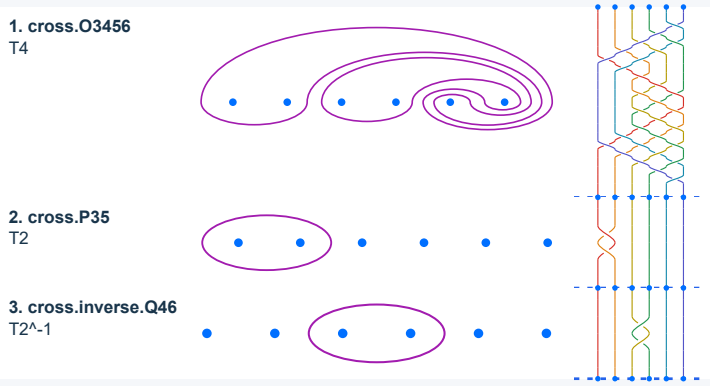
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



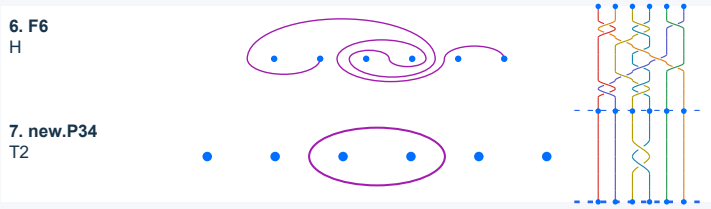
II - step 093

new.P34 up past F6

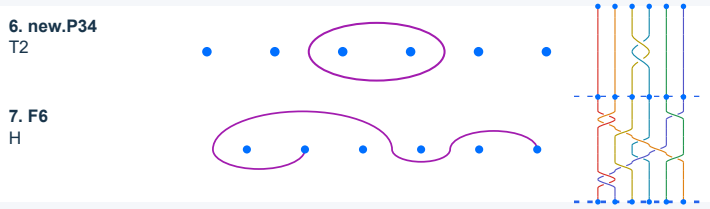
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



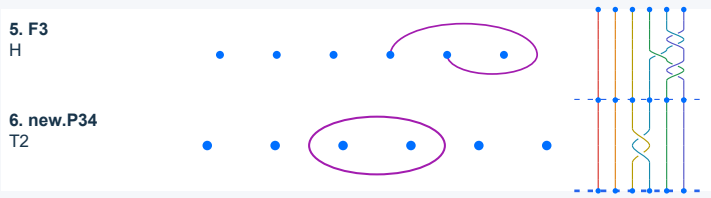
II - step 094

new.P34 up past F3

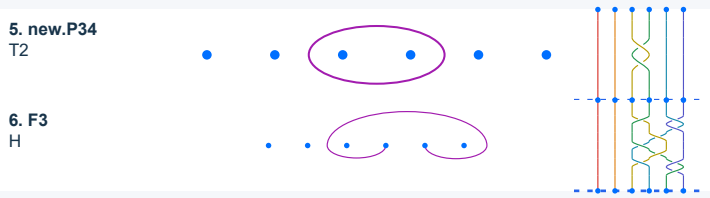
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



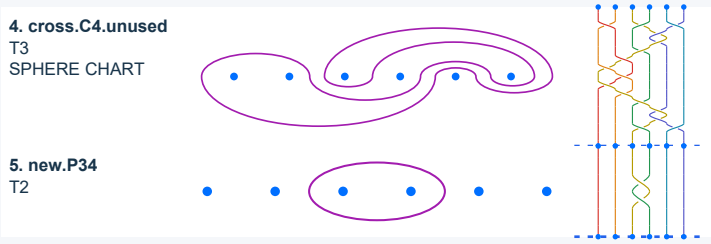
II - step 095

new.P34 up past cross.C4.unused

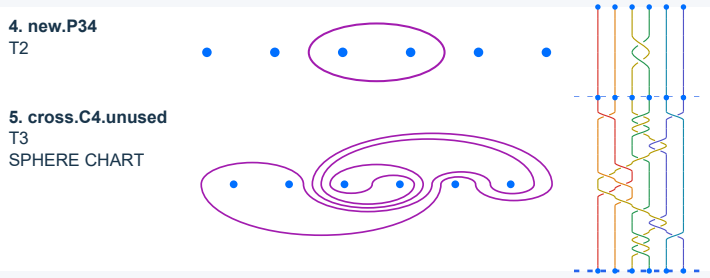
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



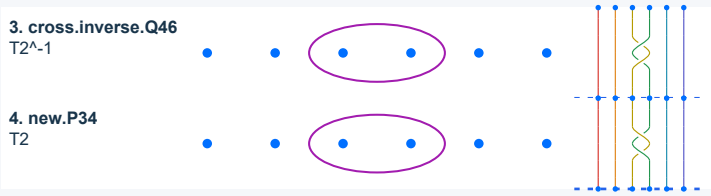
II - step 096

Cancel the equal inverse and positive Q46 pairs in the disk

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 13 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After

Identity / no factors in this interval.

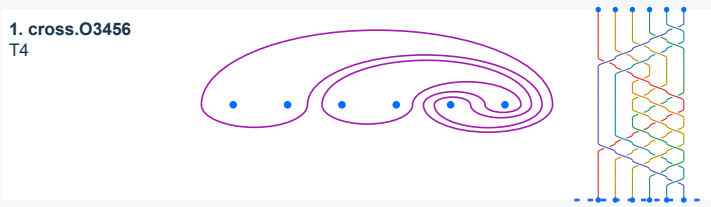
II - step 097

Draw O3456 along its complementary P12 sphere curve

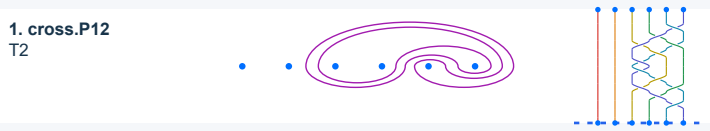
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



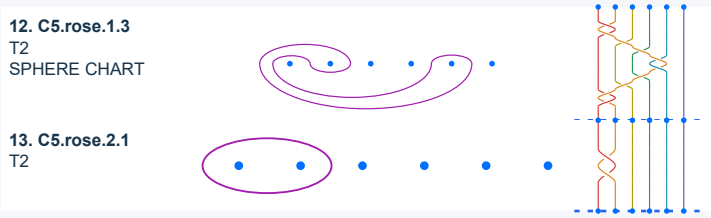
II - step 098

C5.rose.2.1 up past C5.rose.1.3

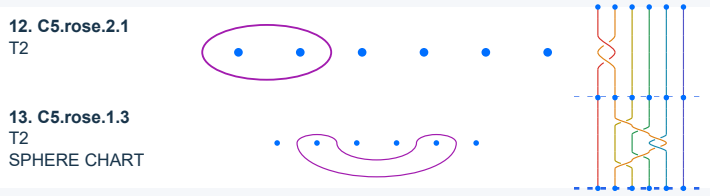
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



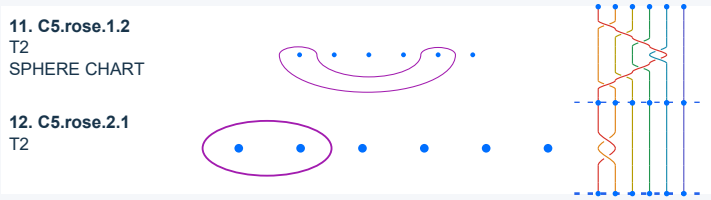
II - step 099

C5.rose.2.1 up past C5.rose.1.2

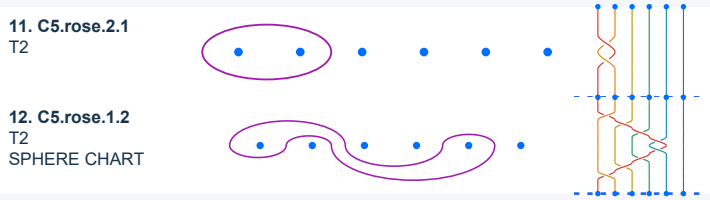
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



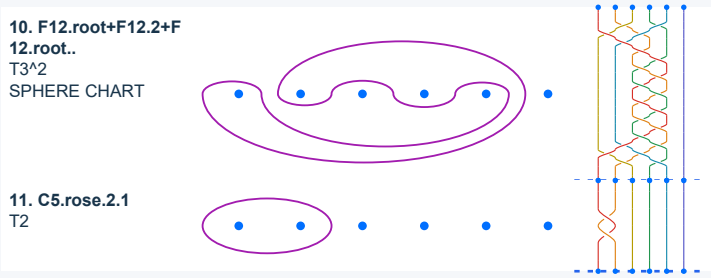
II - step 100

C5.rose.2.1 up past F12.root+F12.2+F12.root+F12.3

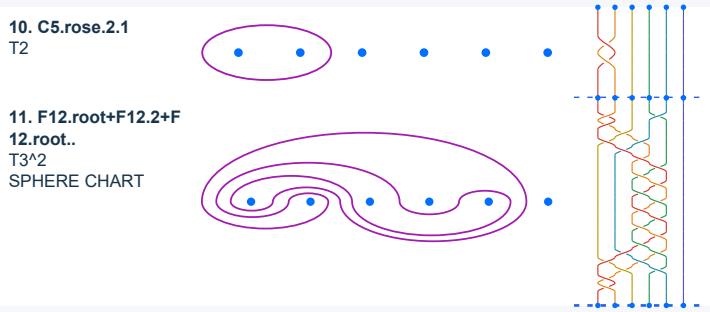
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



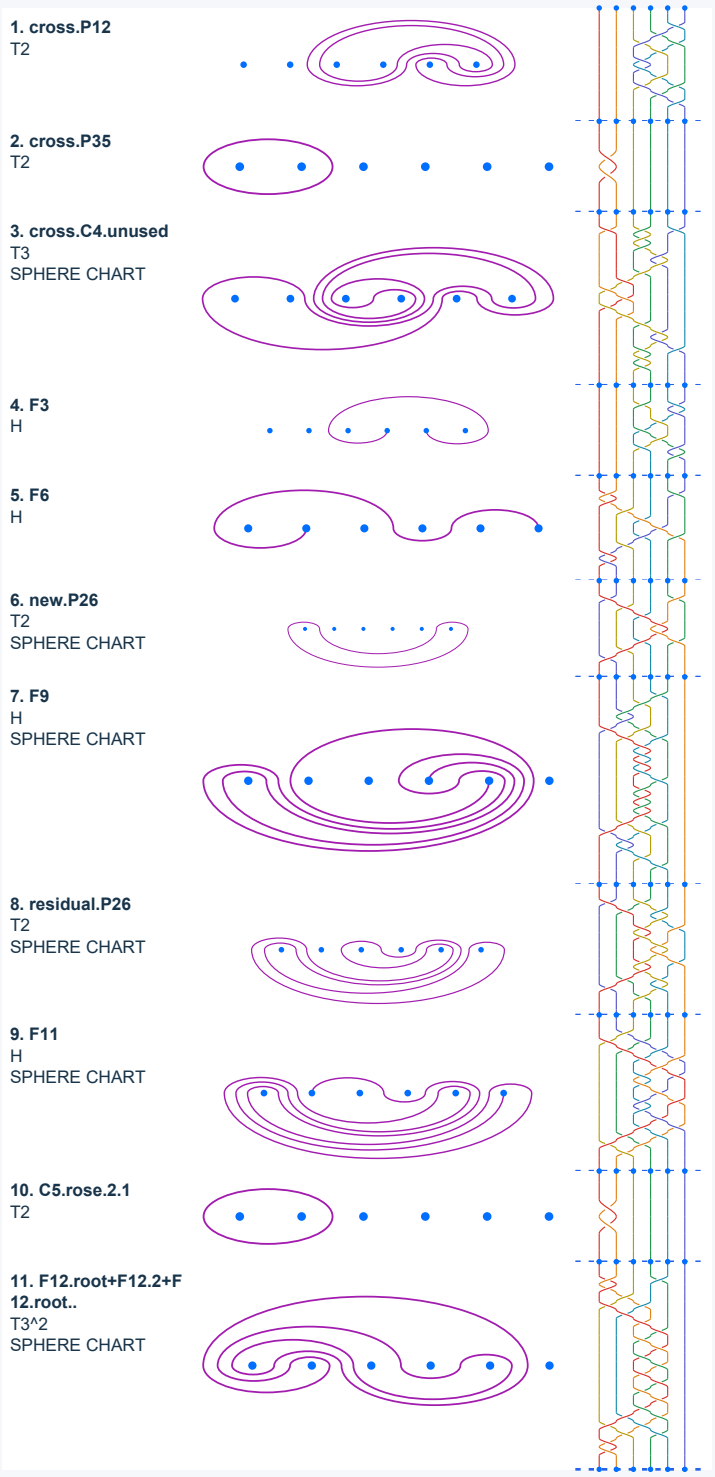
II - step 101

C5.rose.2.1 up past F11

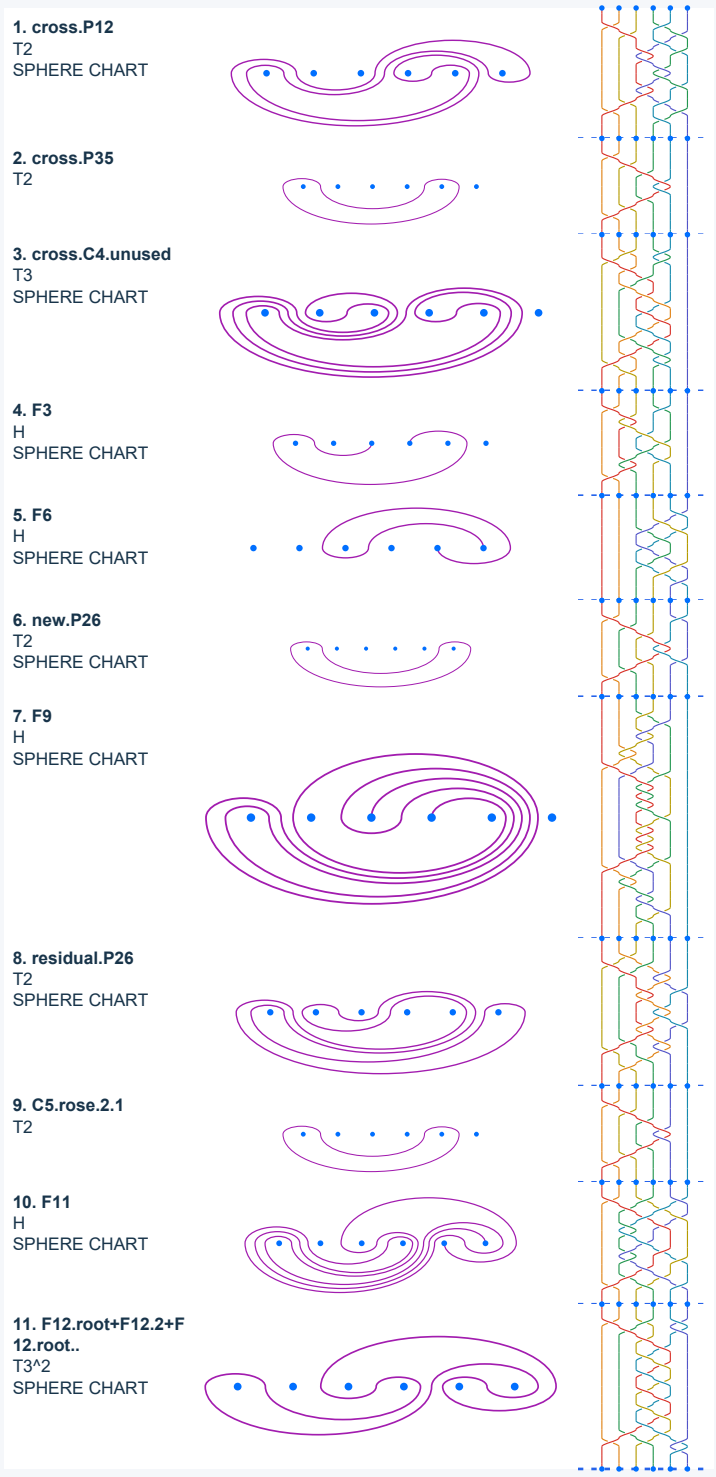
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



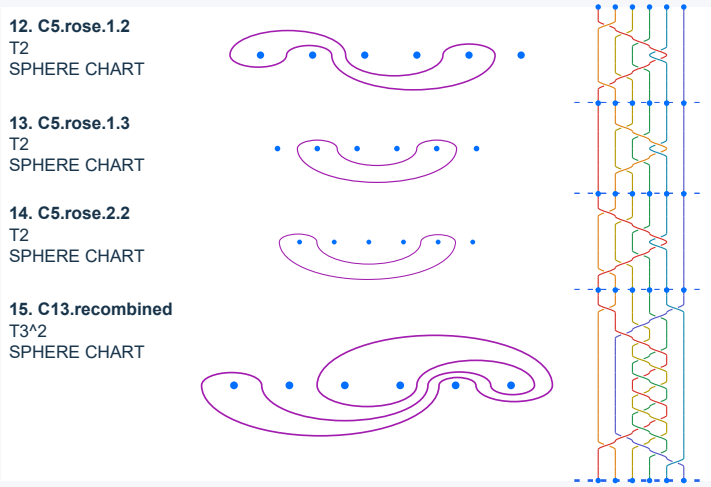
II - step 101 - continued

C5.rose.2.1 up past F11

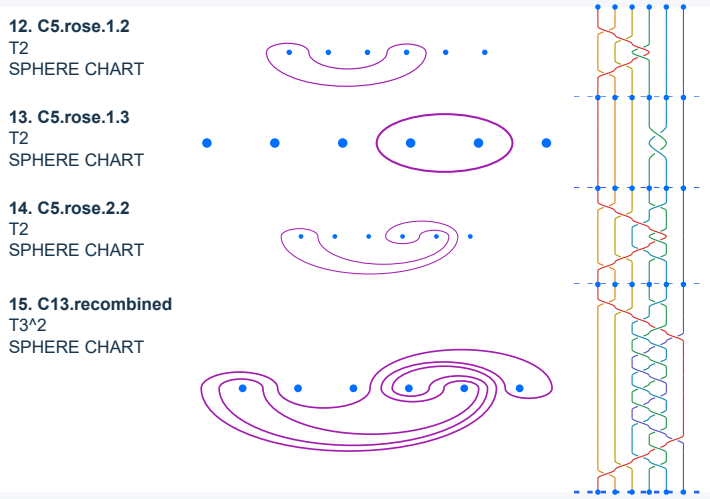
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 3 4 5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



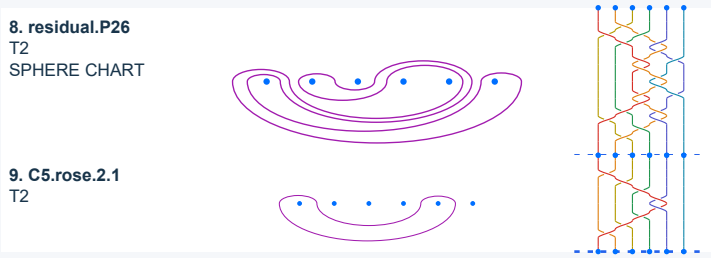
II - step 102

C5.rose.2.1 up past residual.P26

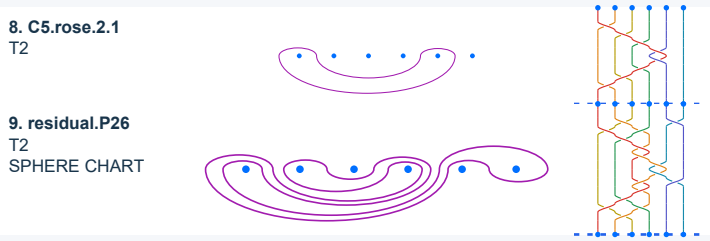
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



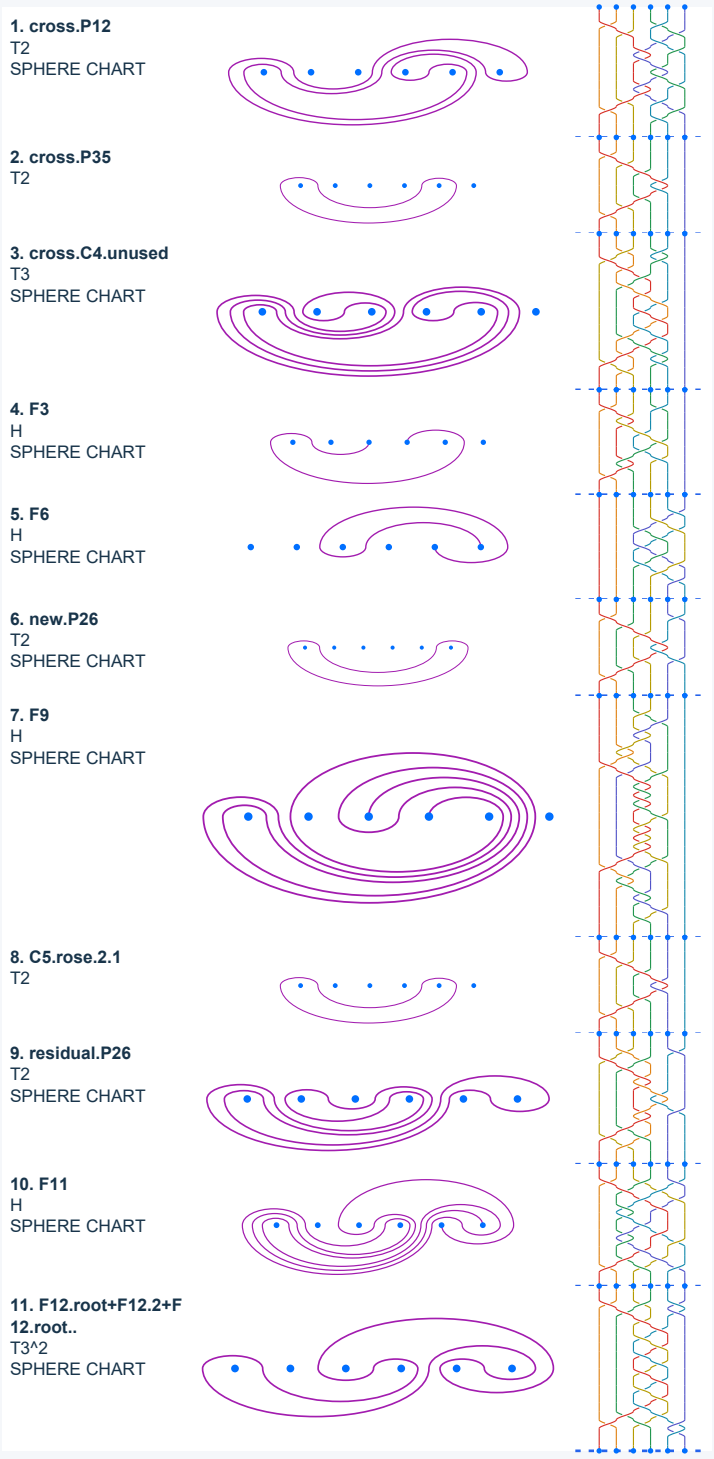
II - step 103

C5.rose.2.1 up past F9

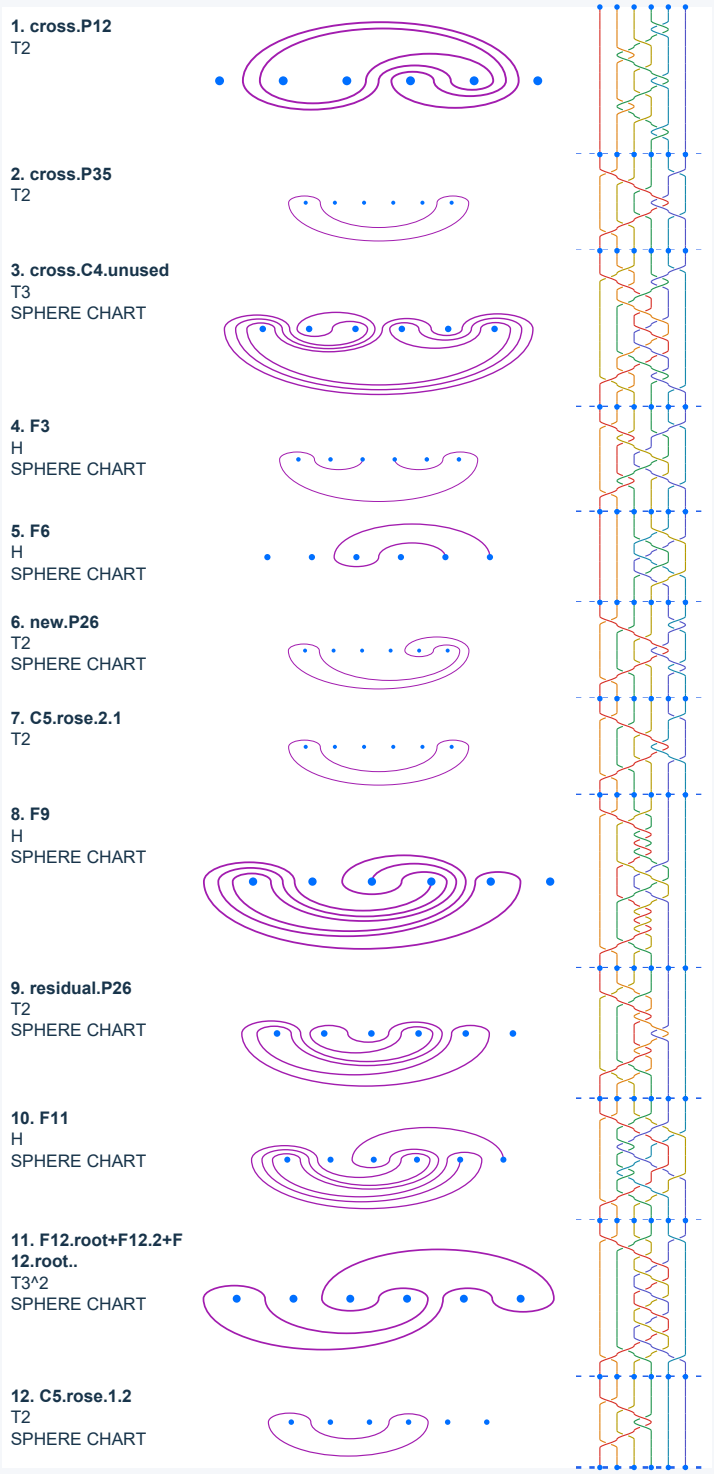
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



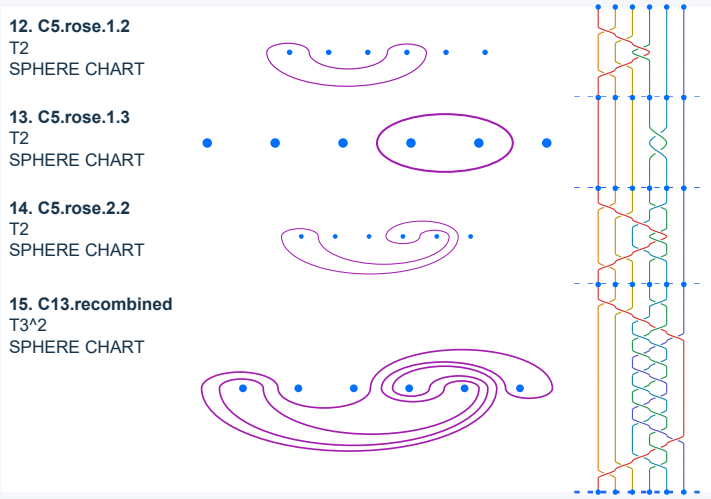
II - step 103 - continued

C5.rose.2.1 up past F9

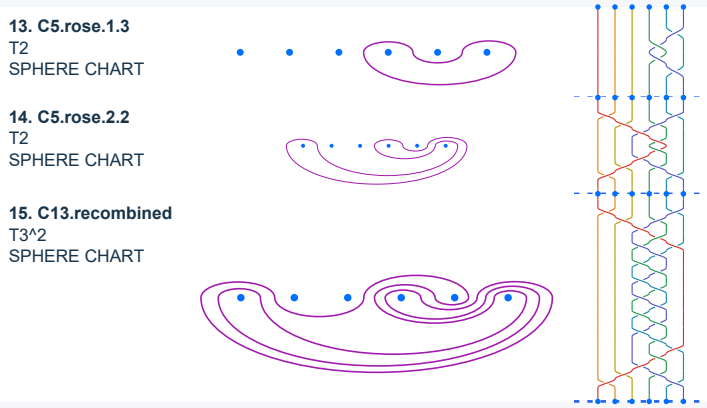
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=5 4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



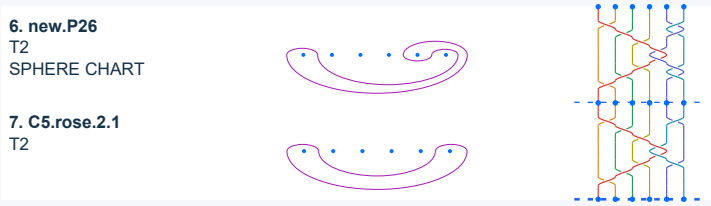
II - step 104

C5.rose.2.1 up past new.P26

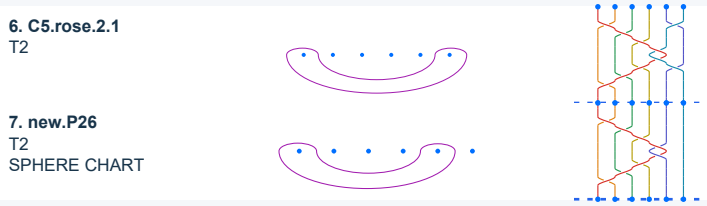
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 105

C5.rose.2.1 up past F6

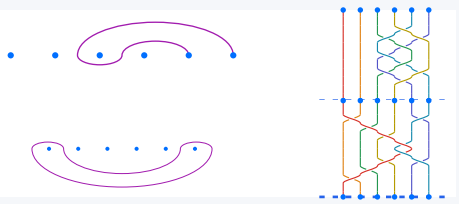
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

5. F6
H
SPHERE CHART

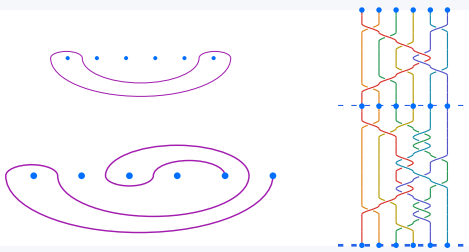
6. C5.rose.2.1
T2



After

5. C5.rose.2.1
T2

6. F6
H
SPHERE CHART



II - step 106

C5.rose.2.1 up past F3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

4. F3
H
SPHERE CHART

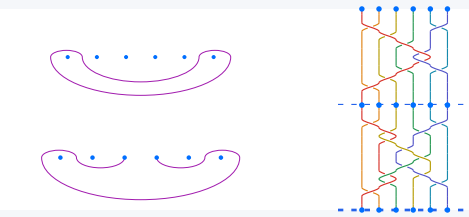
5. C5.rose.2.1
T2



After

4. C5.rose.2.1
T2

5. F3
H
SPHERE CHART



II - step 107

C5.rose.2.1 up past cross.C4.unused

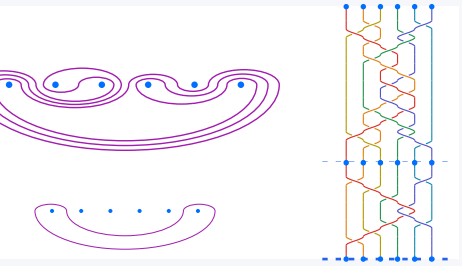
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before

3. cross.C4.unused
T3
SPHERE CHART

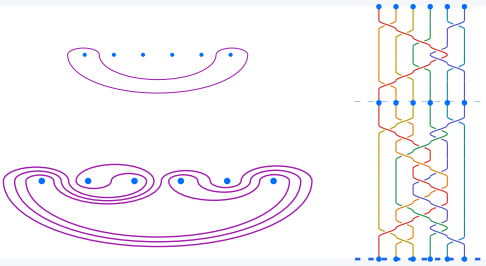
4. C5.rose.2.1
T2



After

3. C5.rose.2.1
T2

4. cross.C4.unused
T3
SPHERE CHART



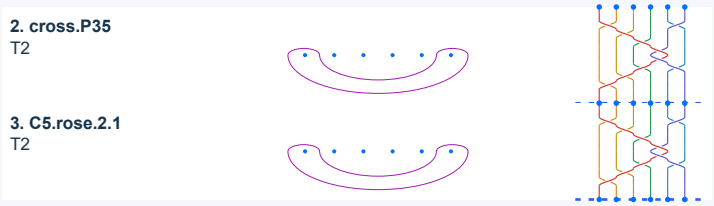
II - step 108

Combine the two equal P35 twists in the disk

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 12 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



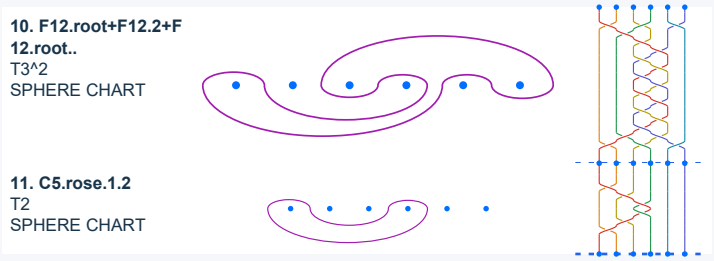
II - step 109

C5.rose.1.2 up past F12.root+F12.2+F12.root+F12.3

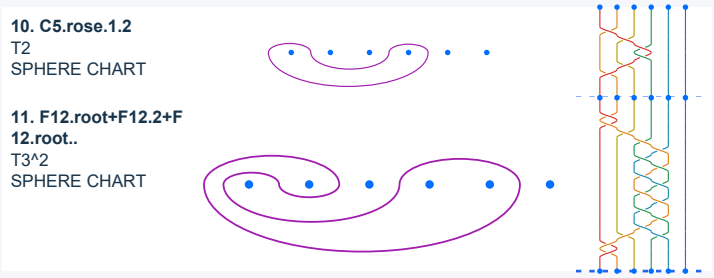
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



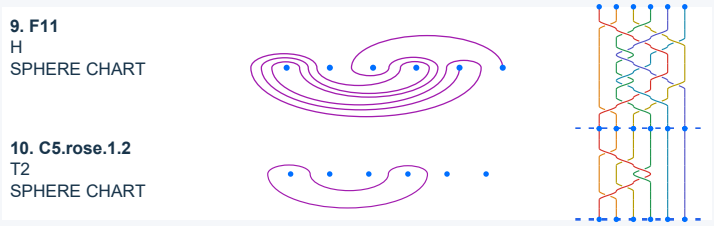
II - step 110

C5.rose.1.2 up past F11

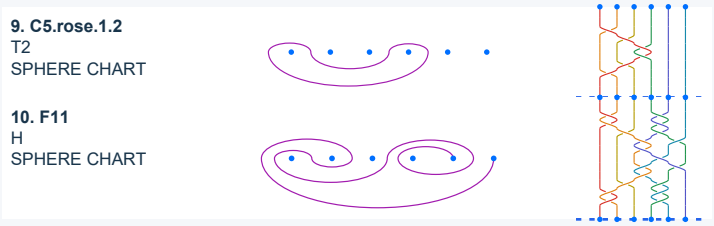
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



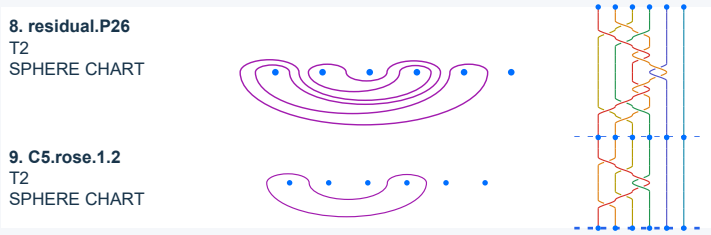
II - step 111

C5.rose.1.2 up past residual.P26

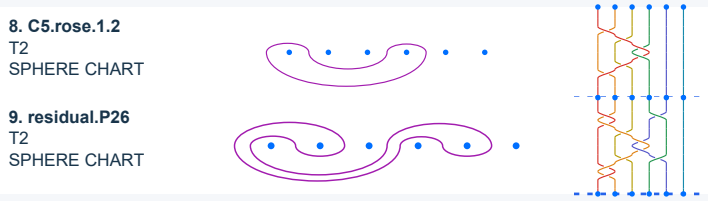
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



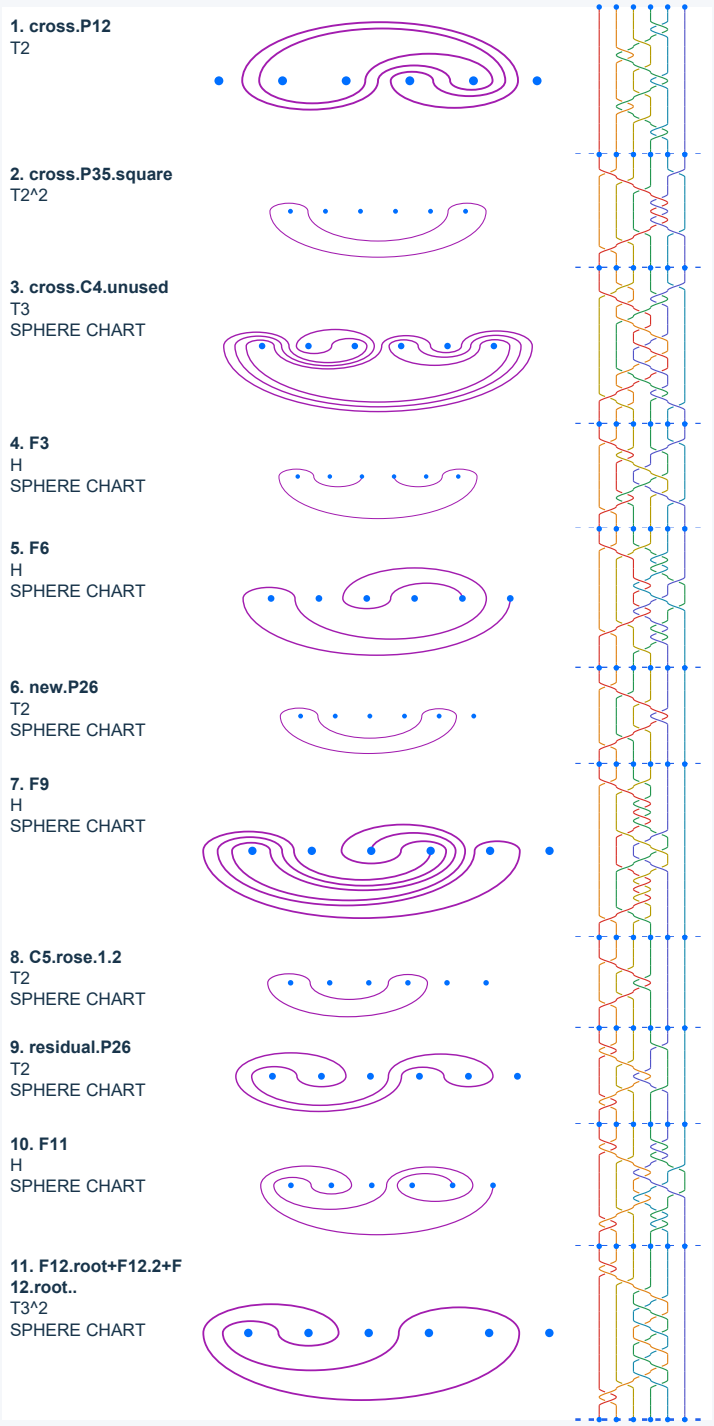
II - step 112

C5.rose.1.2 up past F9

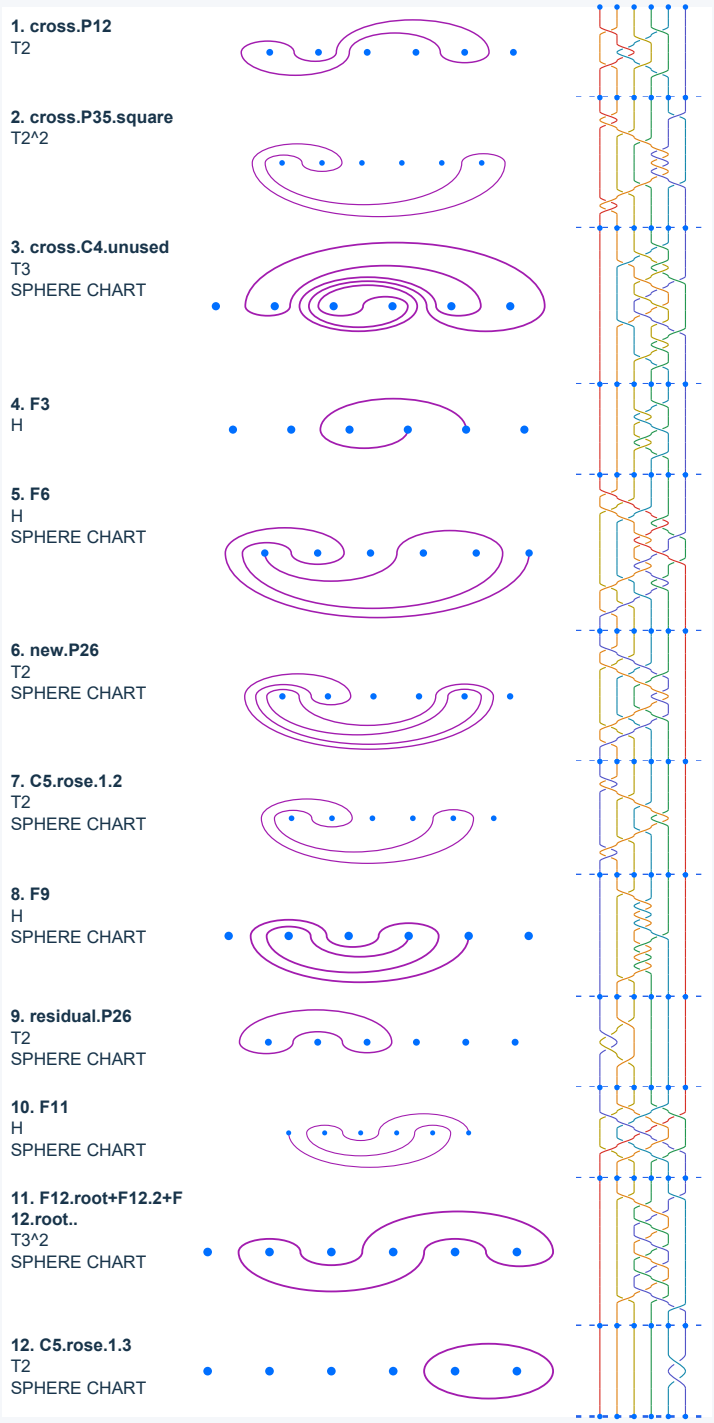
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



II - step 112 - continued

C5.rose.1.2 up past F9

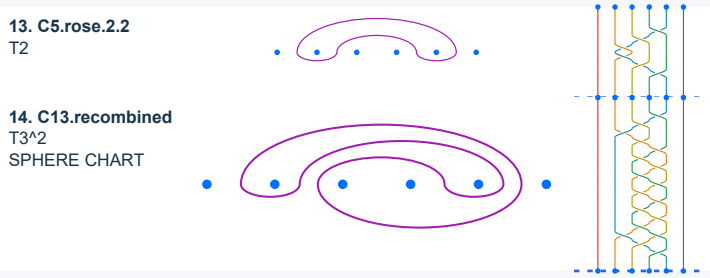
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -3 -2 -1 -3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



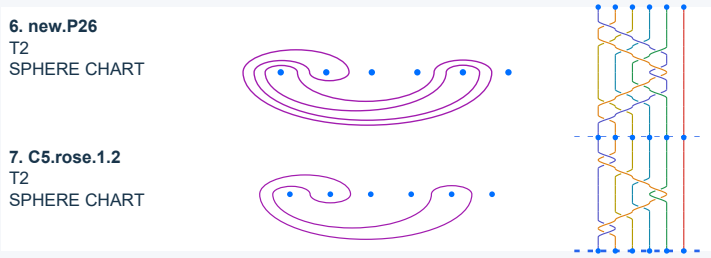
II - step 113

C5.rose.1.2 up past new.P26

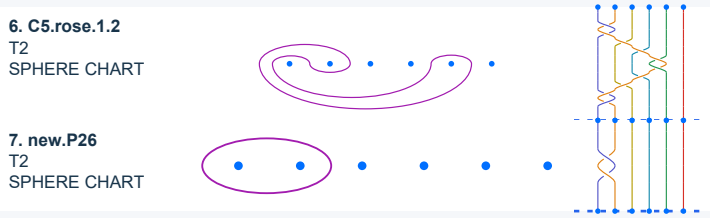
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



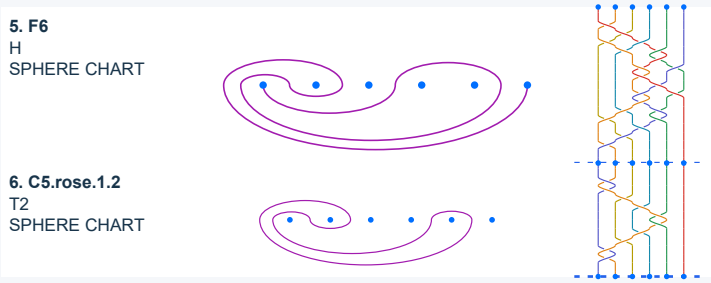
II - step 114

C5.rose.1.2 up past F6

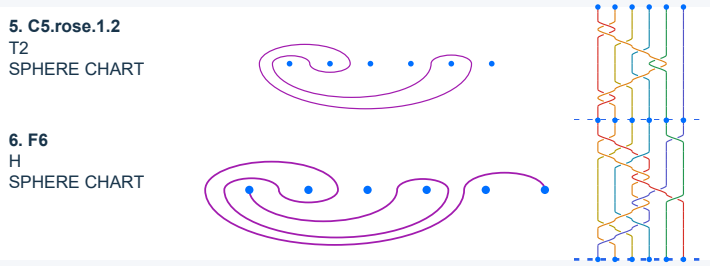
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



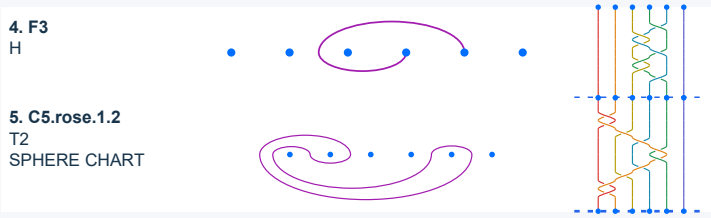
II - step 115

C5.rose.1.2 up past F3

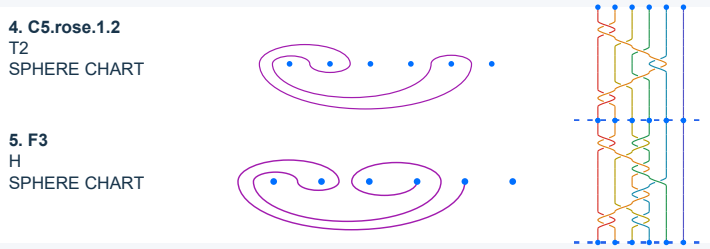
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



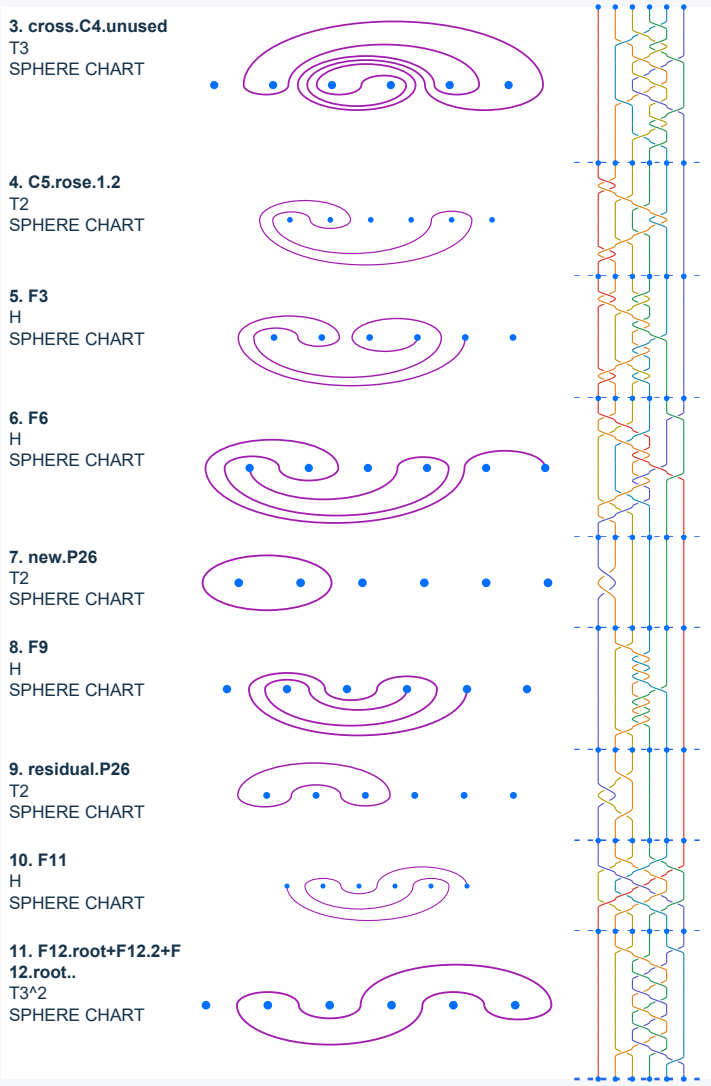
II - step 116

C5.rose.1.2 up past cross.C4.unused

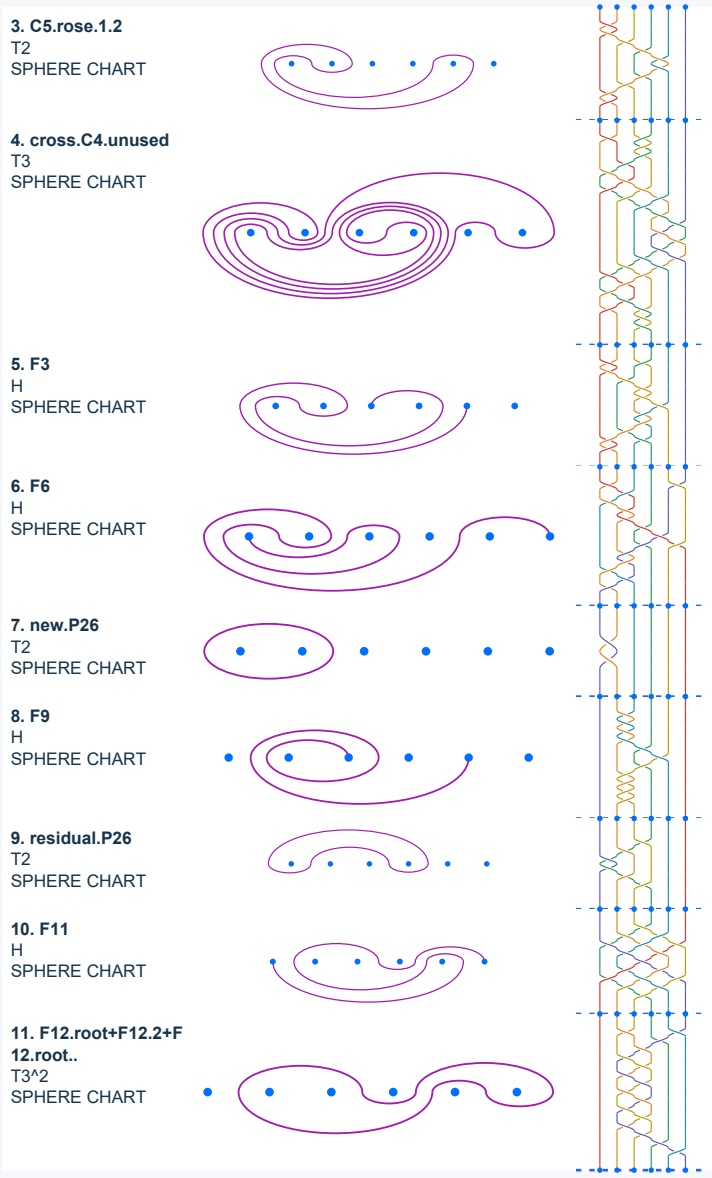
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-3 4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



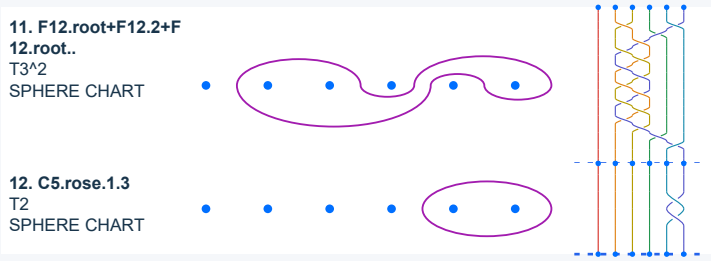
II - step 117

C5.rose.1.3 up past F12.root+F12.2+F12.root+F12.3

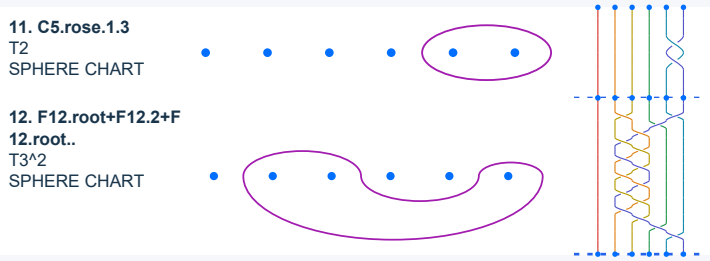
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=4 3 2 4 -2 -2 1 2 -4 -2 5; after g=4 3 2 4 -2 -2 1 2 -4 -2 5.

Before



After



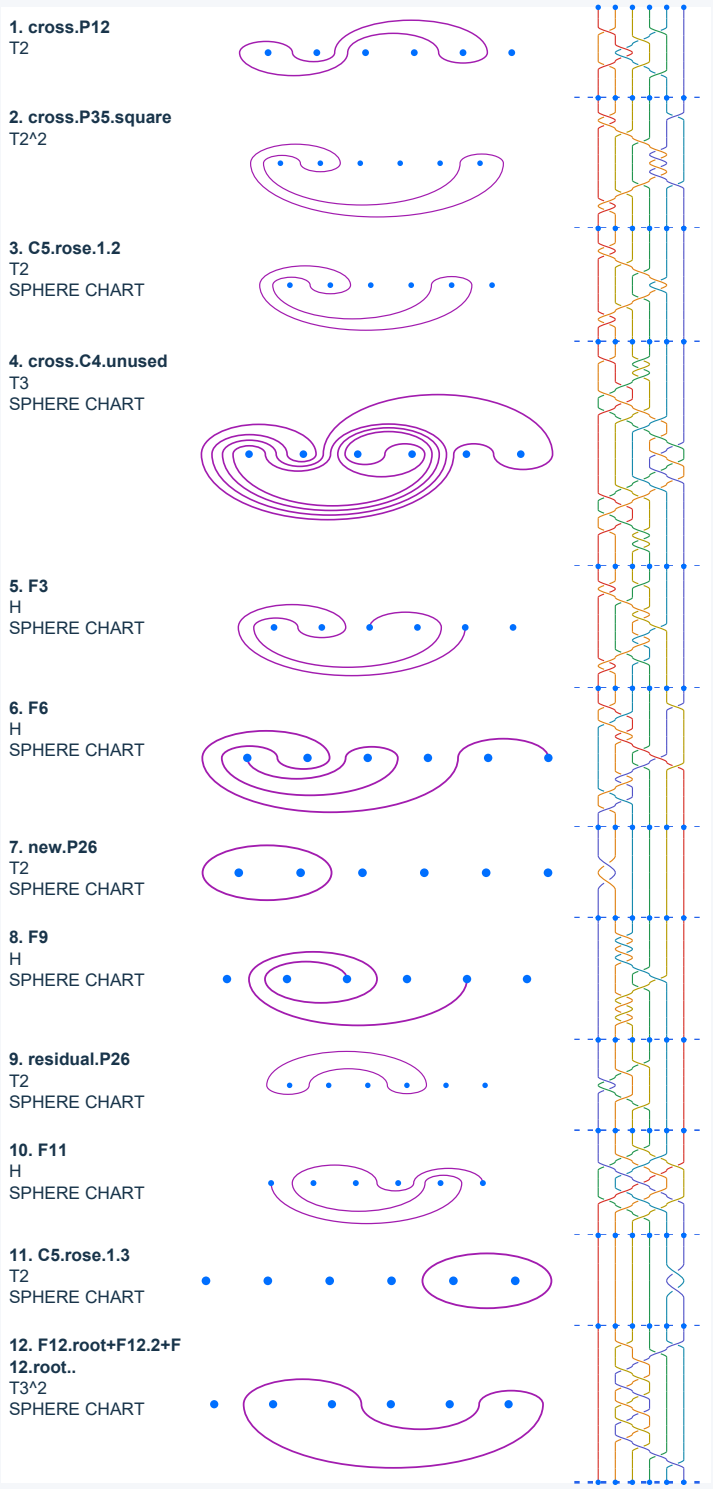
II - step 118

C5.rose.1.3 up past F11

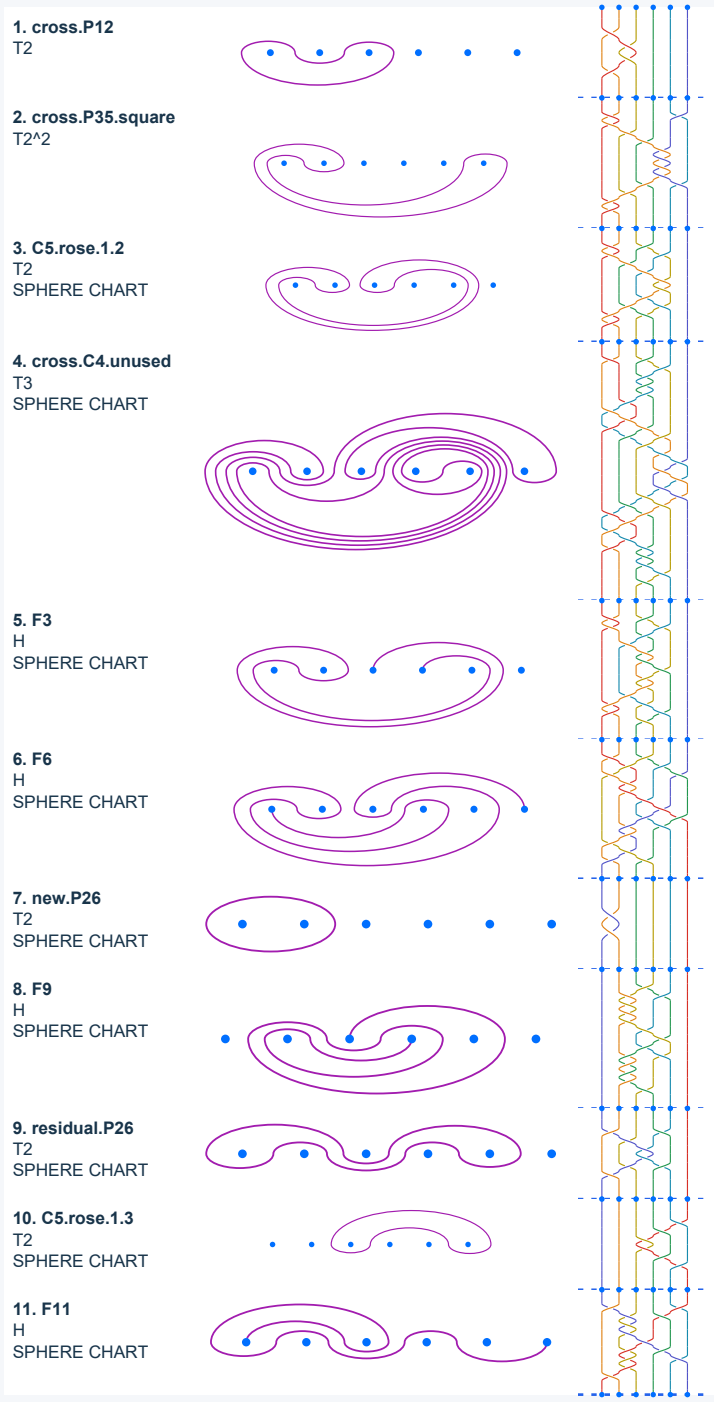
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



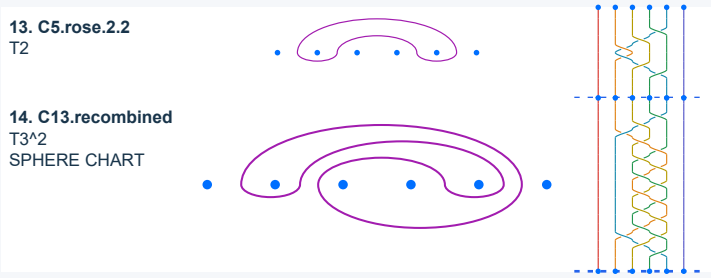
II - step 118 - continued

C5.rose.1.3 up past F11

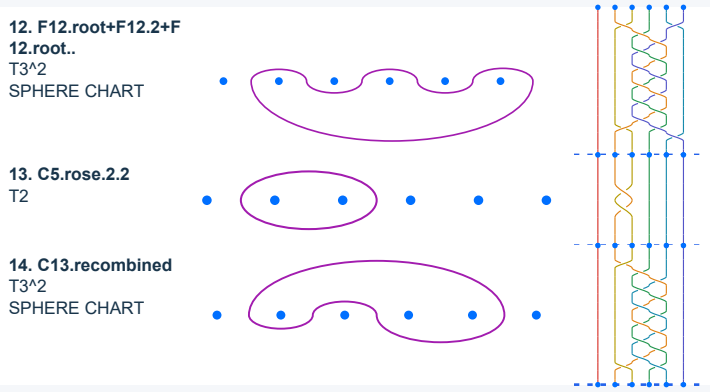
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 3 2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



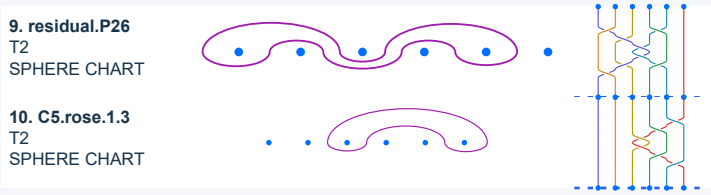
II - step 119

C5.rose.1.3 up past residual.P26

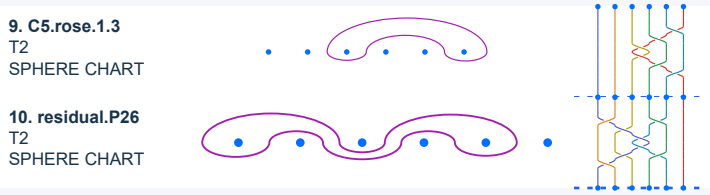
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



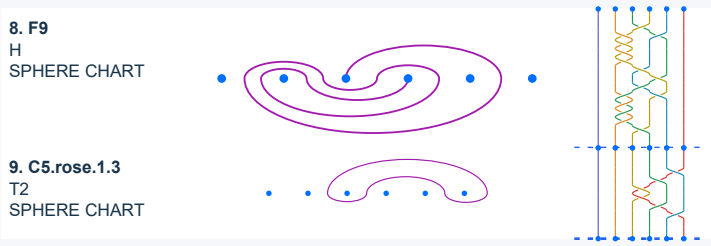
II - step 120

C5.rose.1.3 up past F9

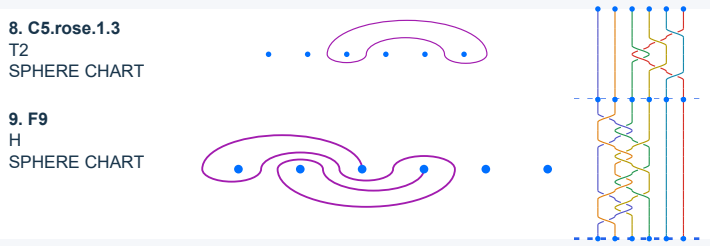
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



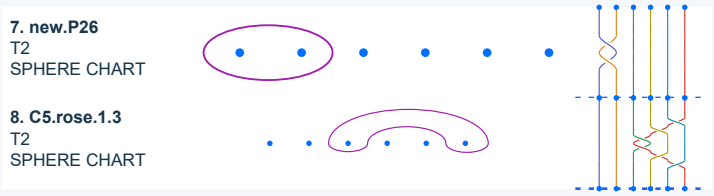
II - step 121

C5.rose.1.3 up past new.P26

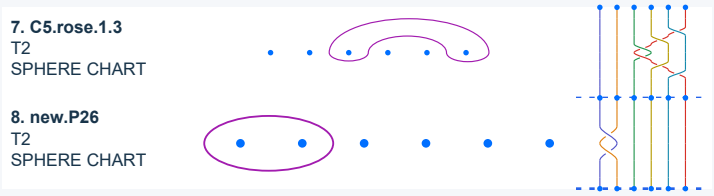
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 6 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



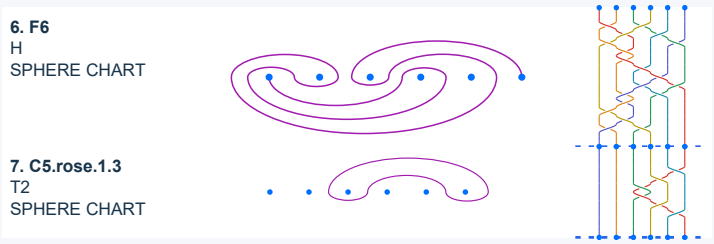
II - step 122

C5.rose.1.3 up past F6

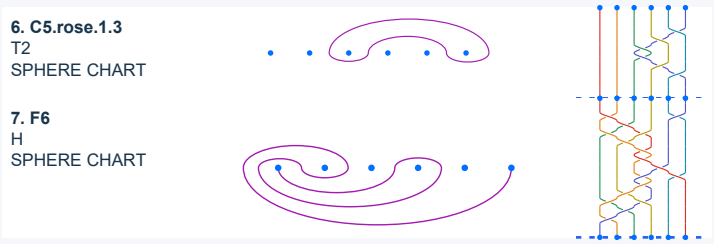
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



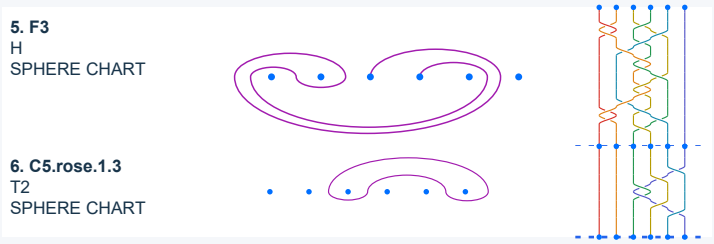
II - step 123

C5.rose.1.3 up past F3

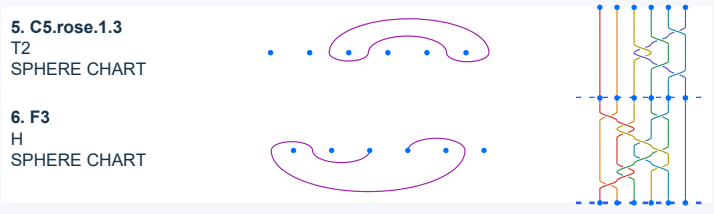
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 8 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=2 4 -2 -2 1 2 -4 -2 5.

Before



After



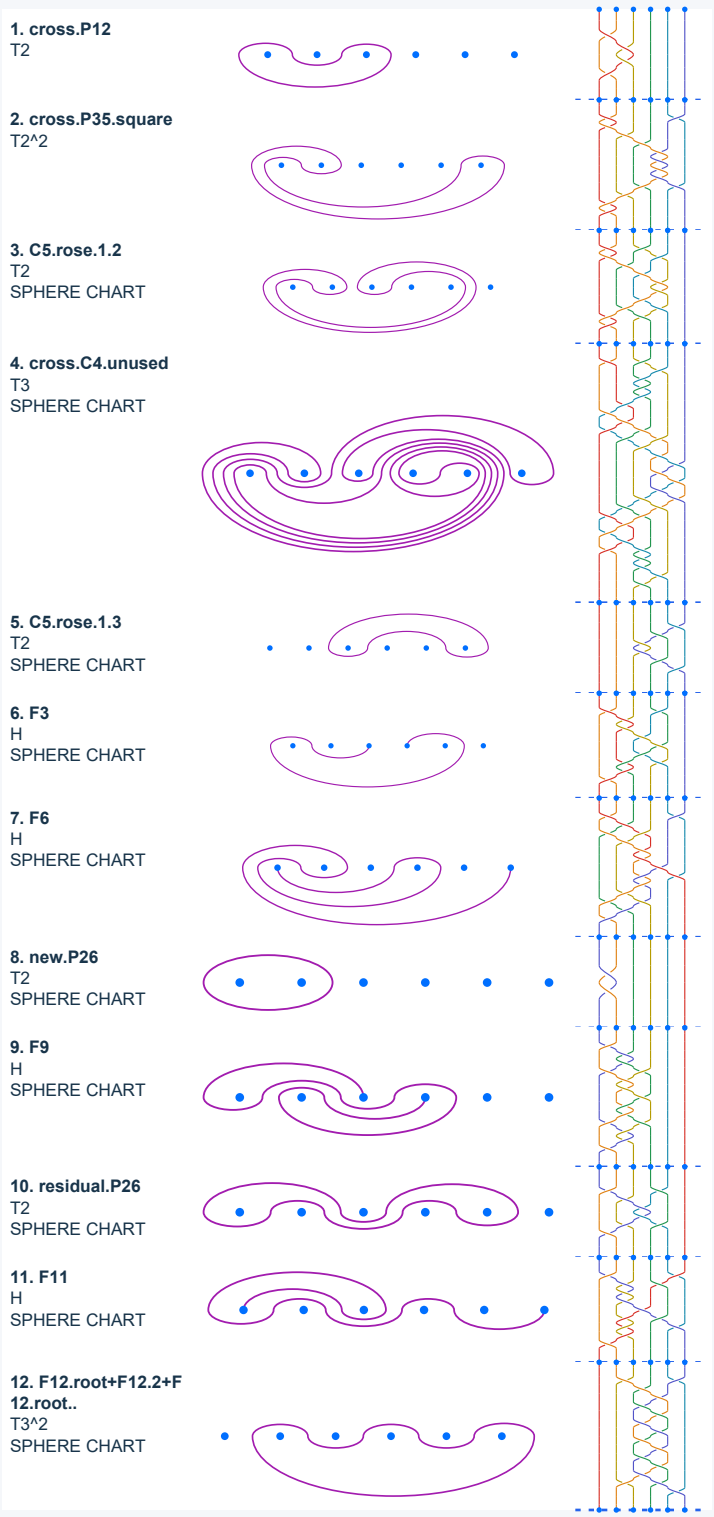
II - step 124

C5.rose.1.3 up past cross.C4.unused

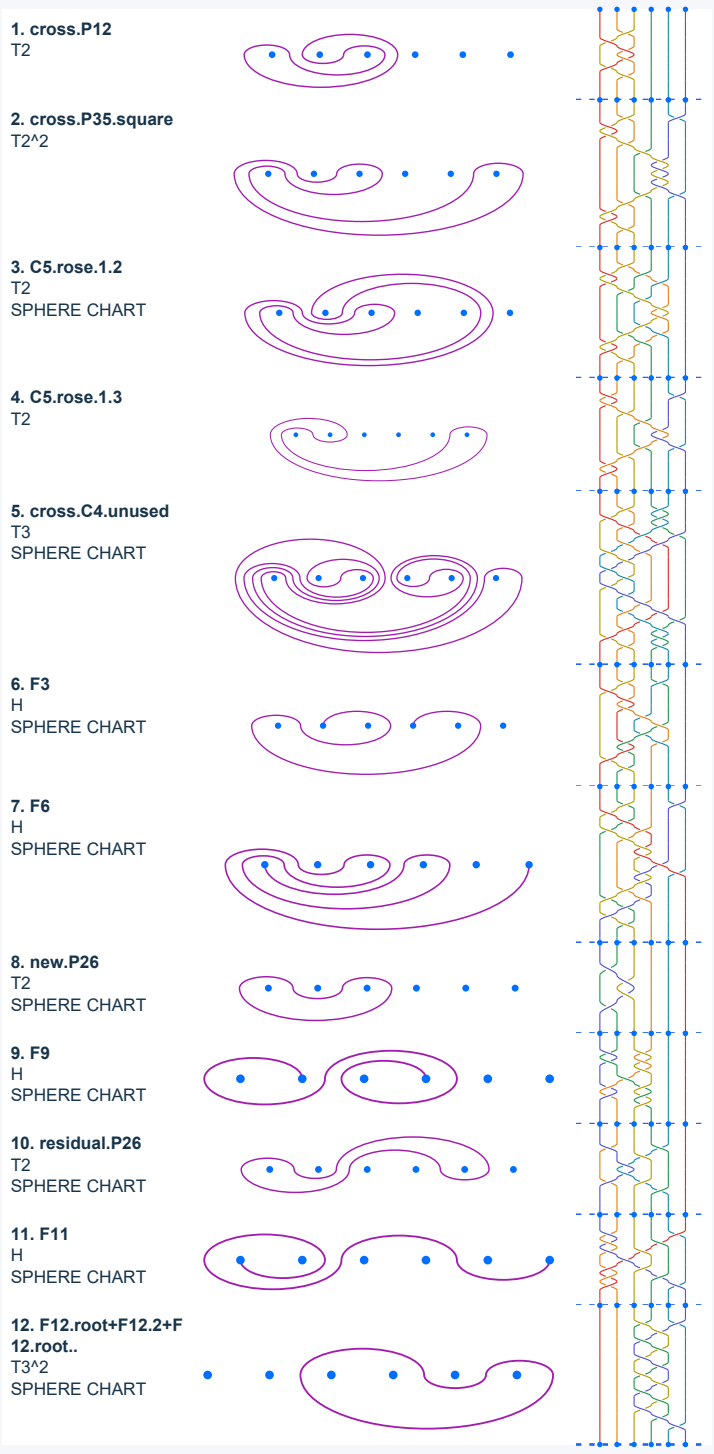
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=4 -2 -2 1 2 -4 -2 5.

Before



After



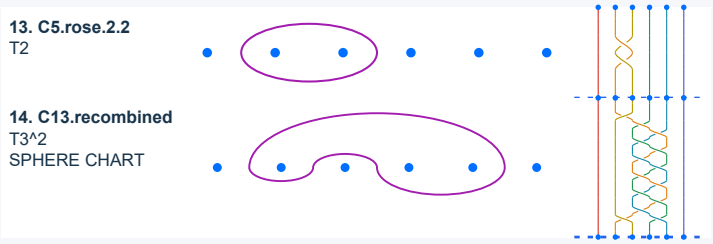
II - step 124 - continued

C5.rose.1.3 up past cross.C4.unused

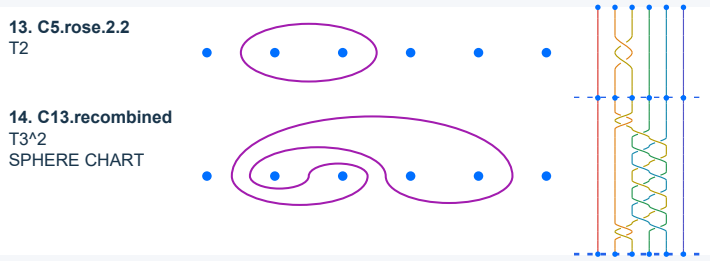
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=2 4 -2 -2 1 2 -4 -2 5; after g=4 -2 -2 1 2 -4 -2 5.

Before



After



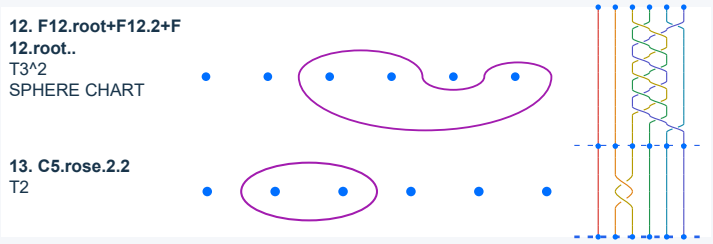
II - step 125

C5.rose.2.2 up past F12.root+F12.2+F12.root+F12.3

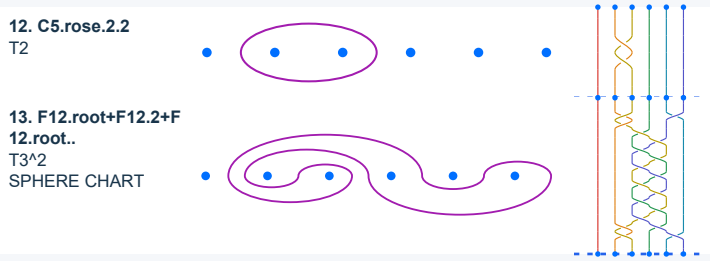
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=4 -2 -2 1 2 -4 -2 5; after g=4 -2 -2 1 2 -4 -2 5.

Before



After



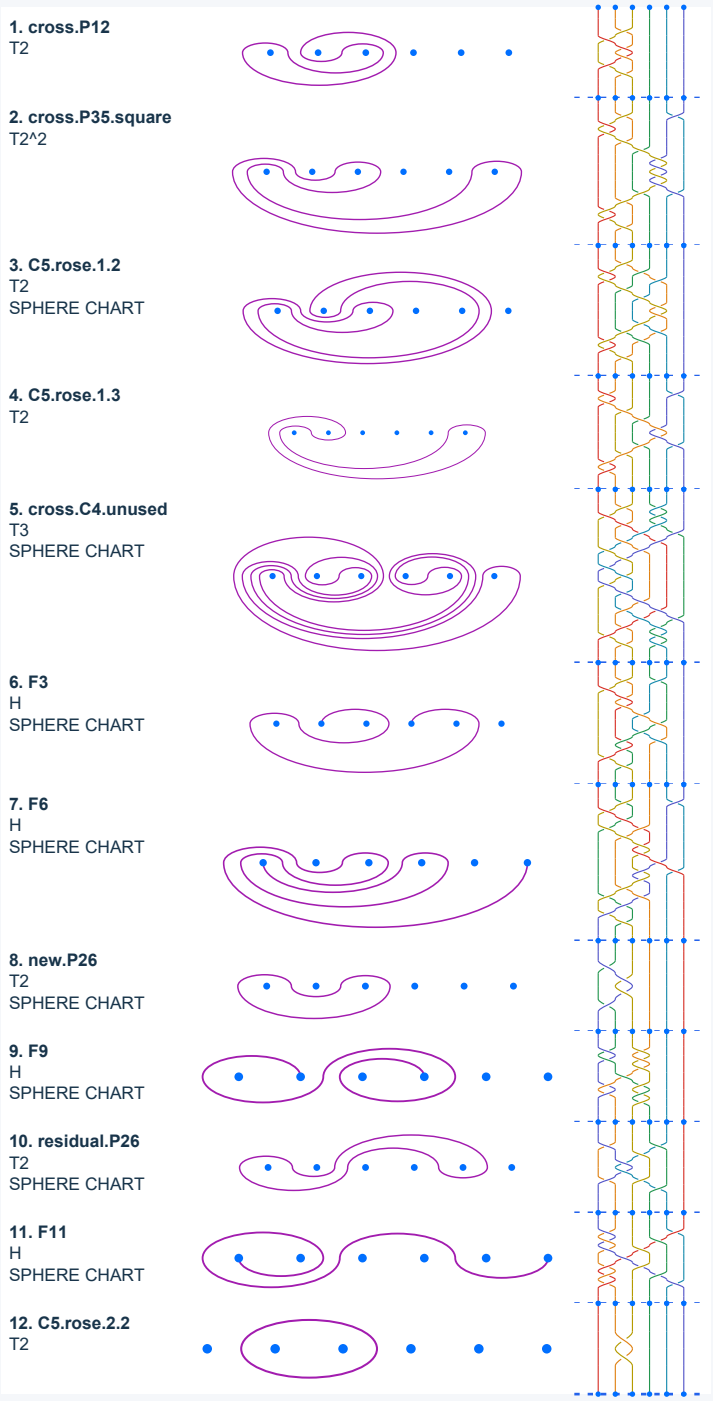
II - step 126

C5.rose.2.2 up past F11

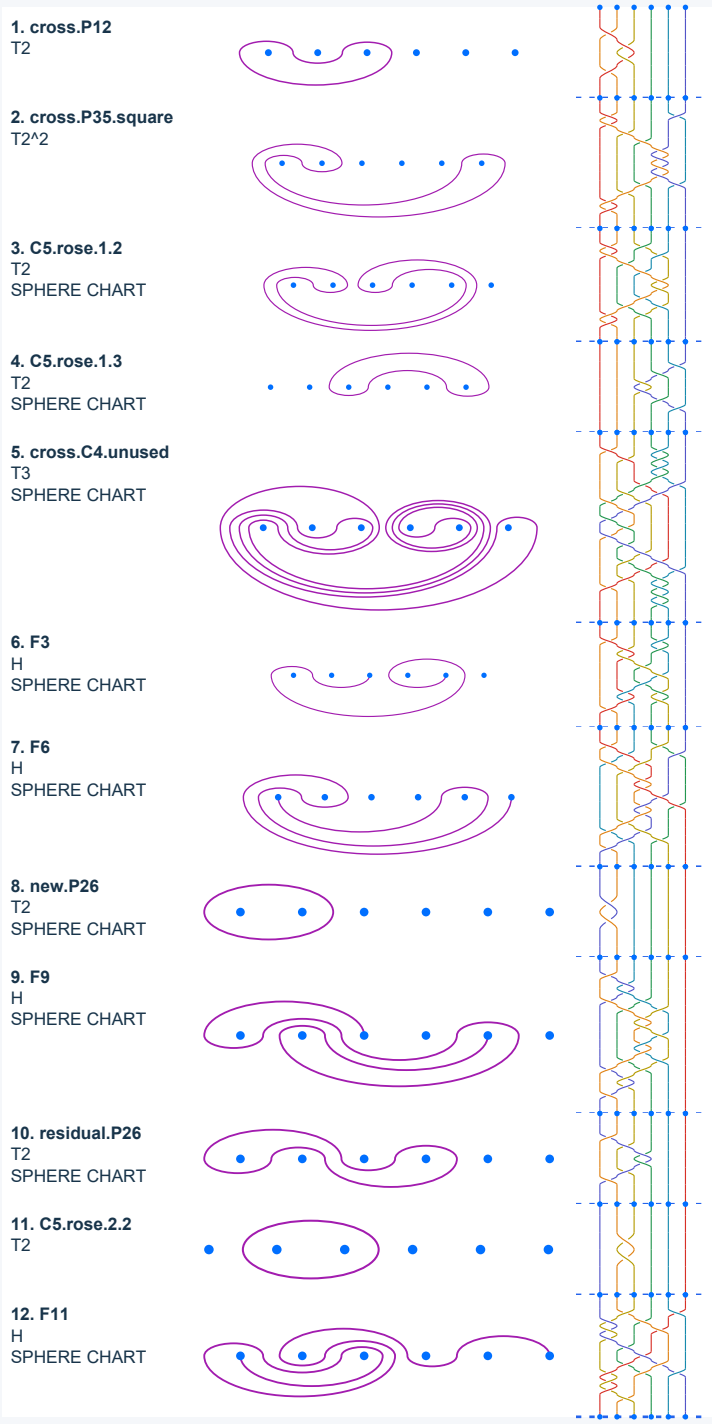
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -2 -2 1 2 -4 -2 5; after g=-2 1 2 -4 -2 5.

Before



After



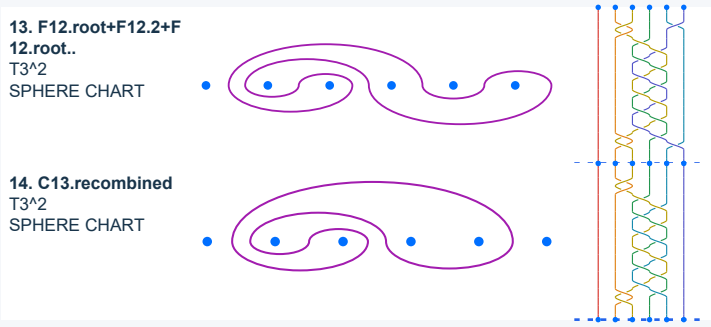
II - step 126 - continued

C5.rose.2.2 up past F11

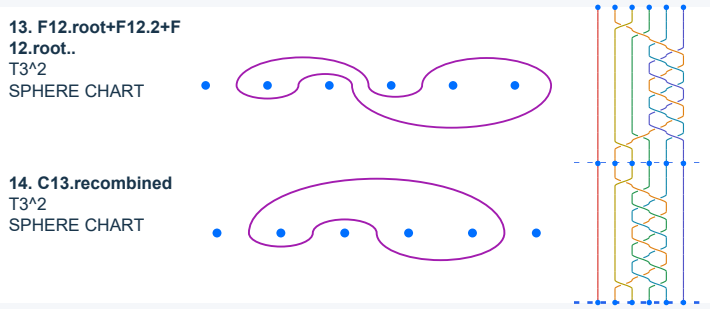
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=4 -2 -2 1 2 -4 -2 5; after g=-2 1 2 -4 -2 5.

Before



After



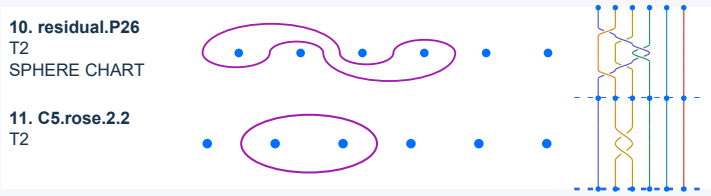
II - step 127

C5.rose.2.2 up past residual.P26

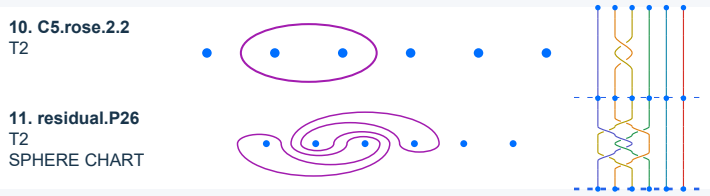
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 3 below. Only the changed interval is shown. Display frames: before g=-2 1 2 -4 -2 5; after g=-2 1 2 -4 -2 5.

Before



After



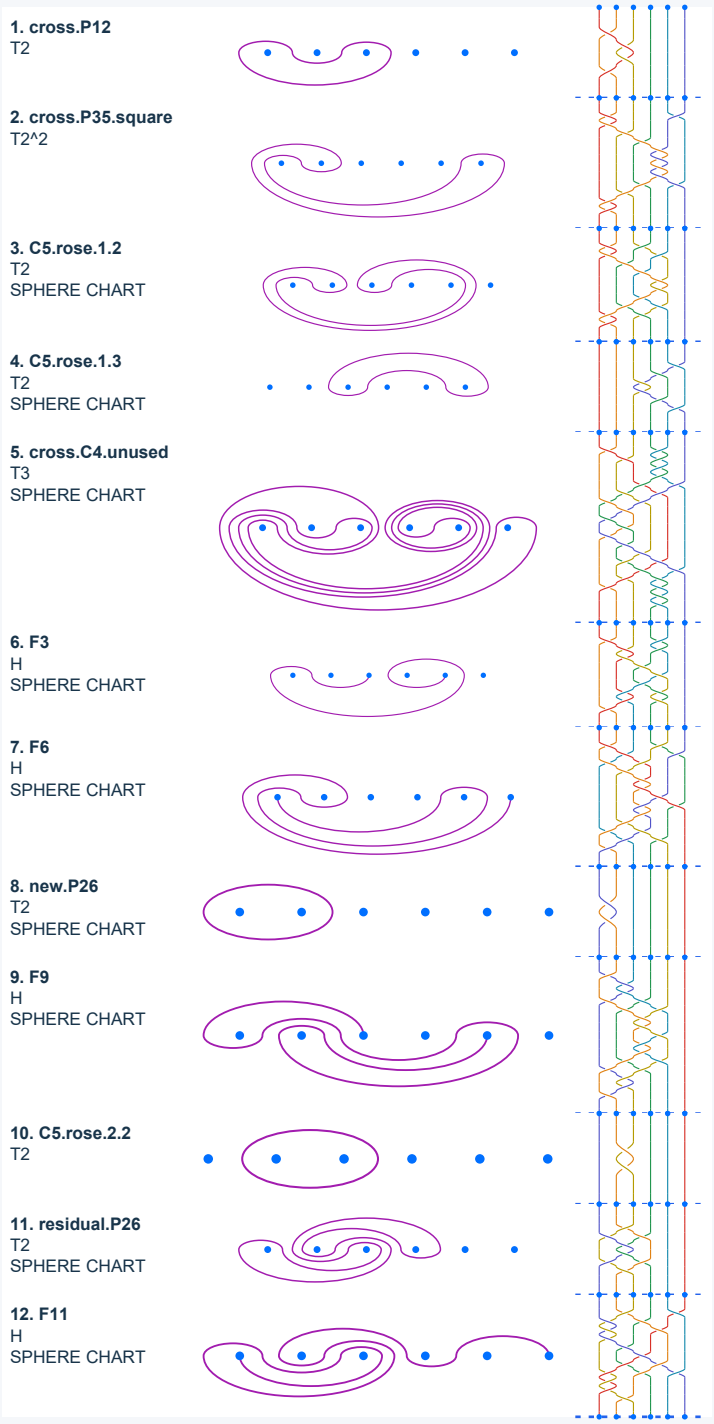
II - step 128

C5.rose.2.2 up past F9

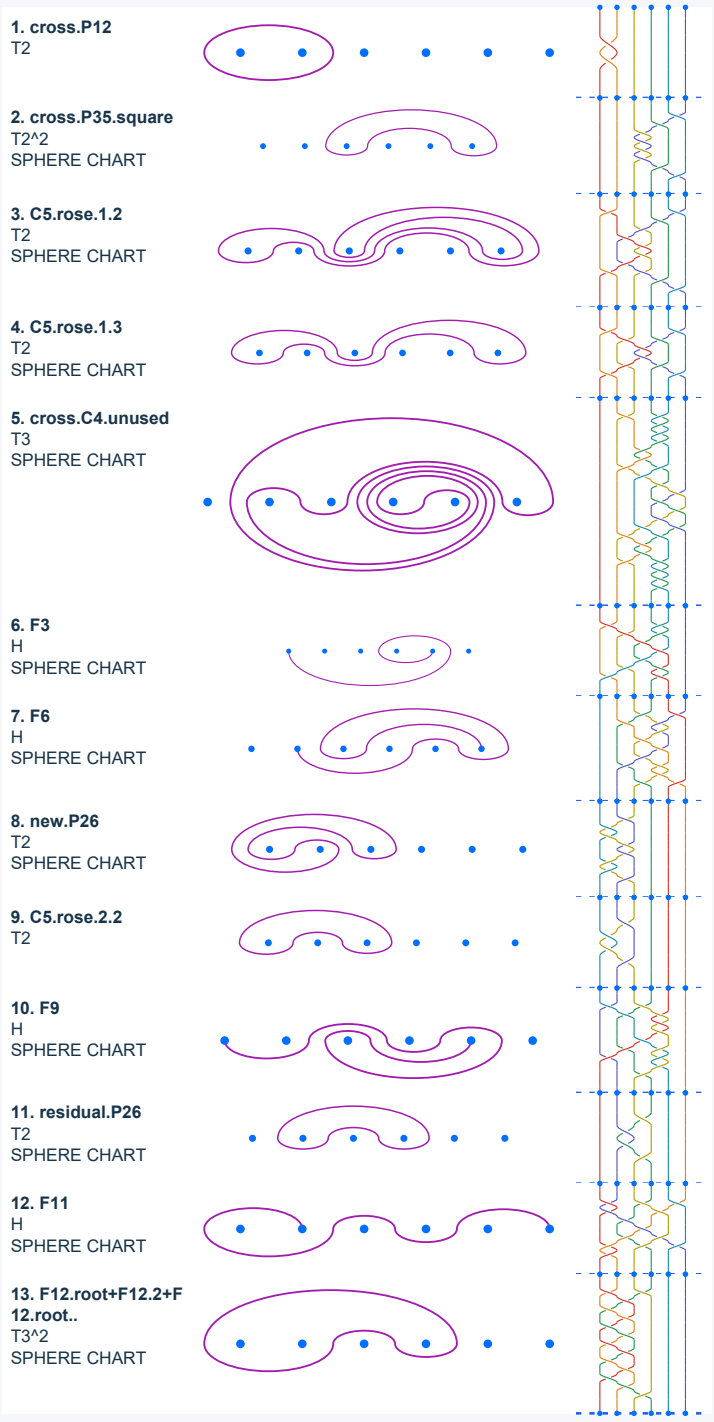
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 1 2 -4 -2 5; after g=2 -4 -2 5.

Before



After



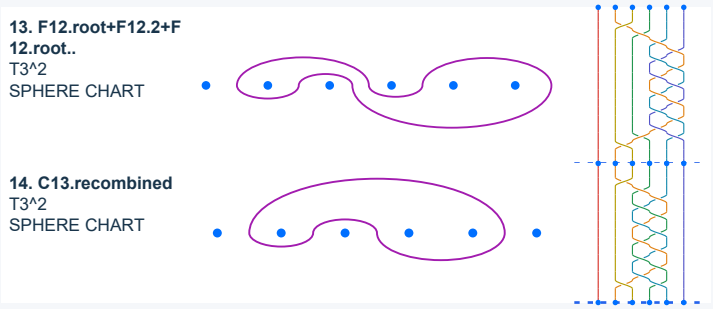
II - step 128 - continued

C5.rose.2.2 up past F9

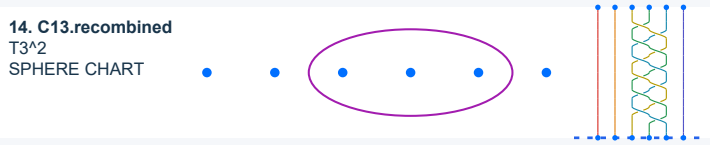
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 1 2 -4 -2 5; after g=2 -4 -2 5.

Before



After



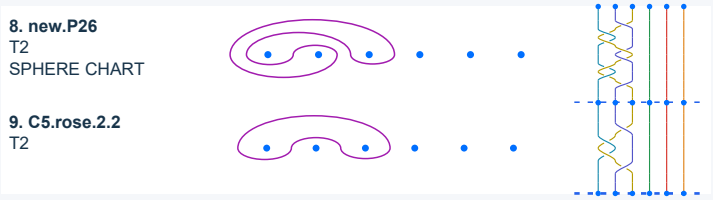
II - step 129

C5.rose.2.2 up past new.P26

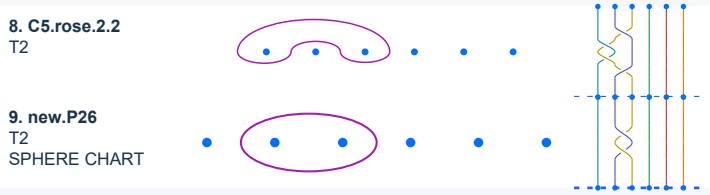
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 5 below. Only the changed interval is shown. Display frames: before g=2 -4 -2 5; after g=2 -4 -2 5.

Before



After



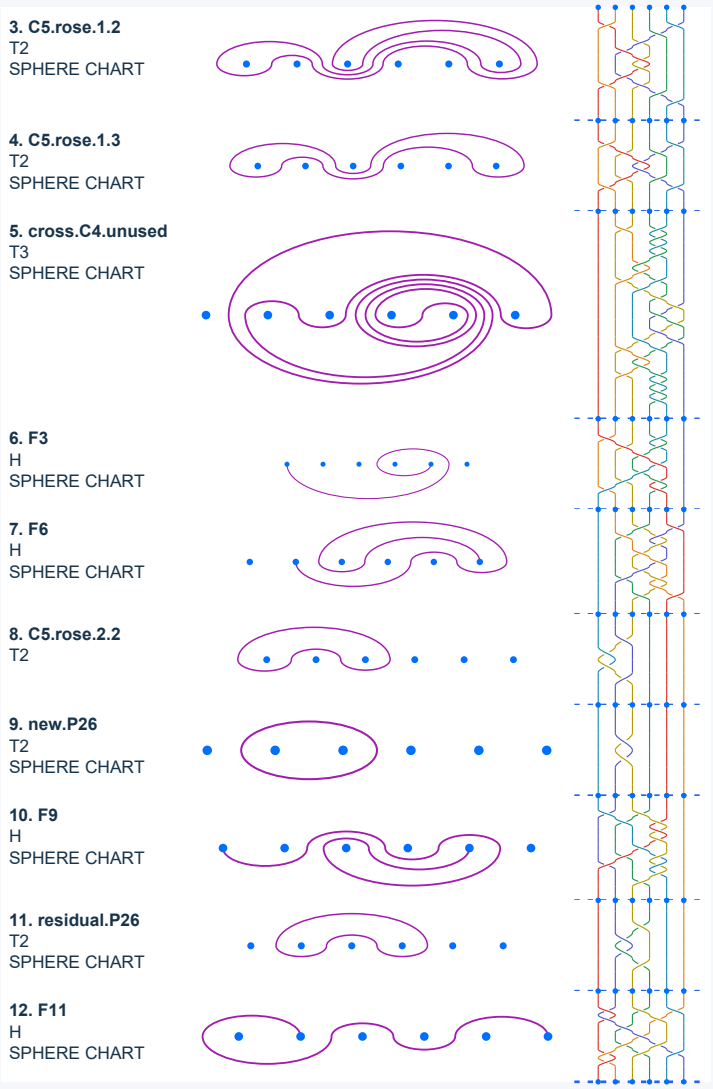
II - step 130

C5.rose.2.2 up past F6

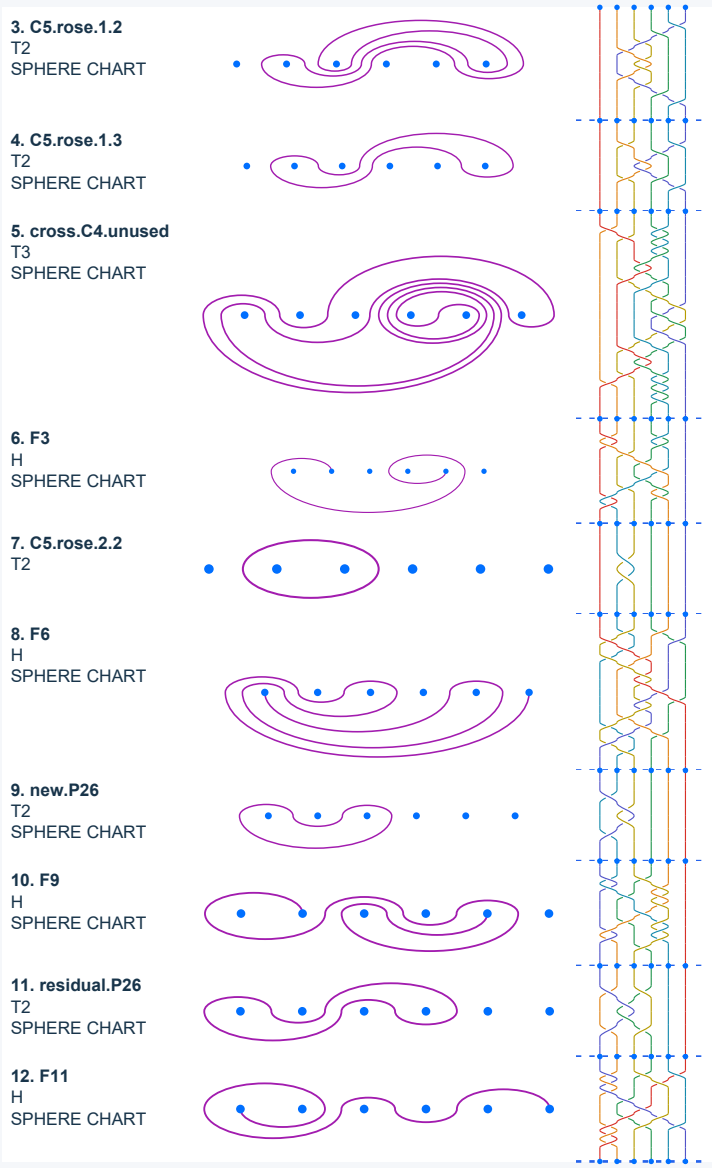
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=2 -4 -2 5; after g=1 2 -4 -2 5.

Before



After



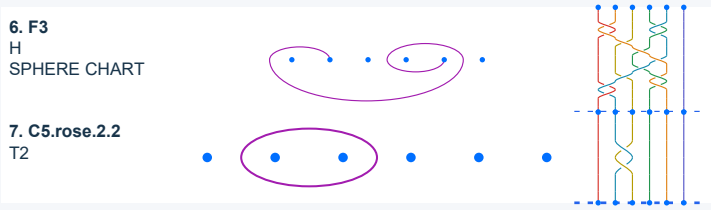
II - step 131

C5.rose.2.2 up past F3

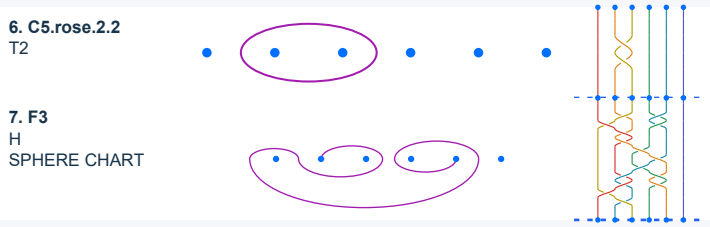
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 7 below. Only the changed interval is shown. Display frames: before g=1 2 -4 -2 5; after g=1 2 -4 -2 5.

Before



After



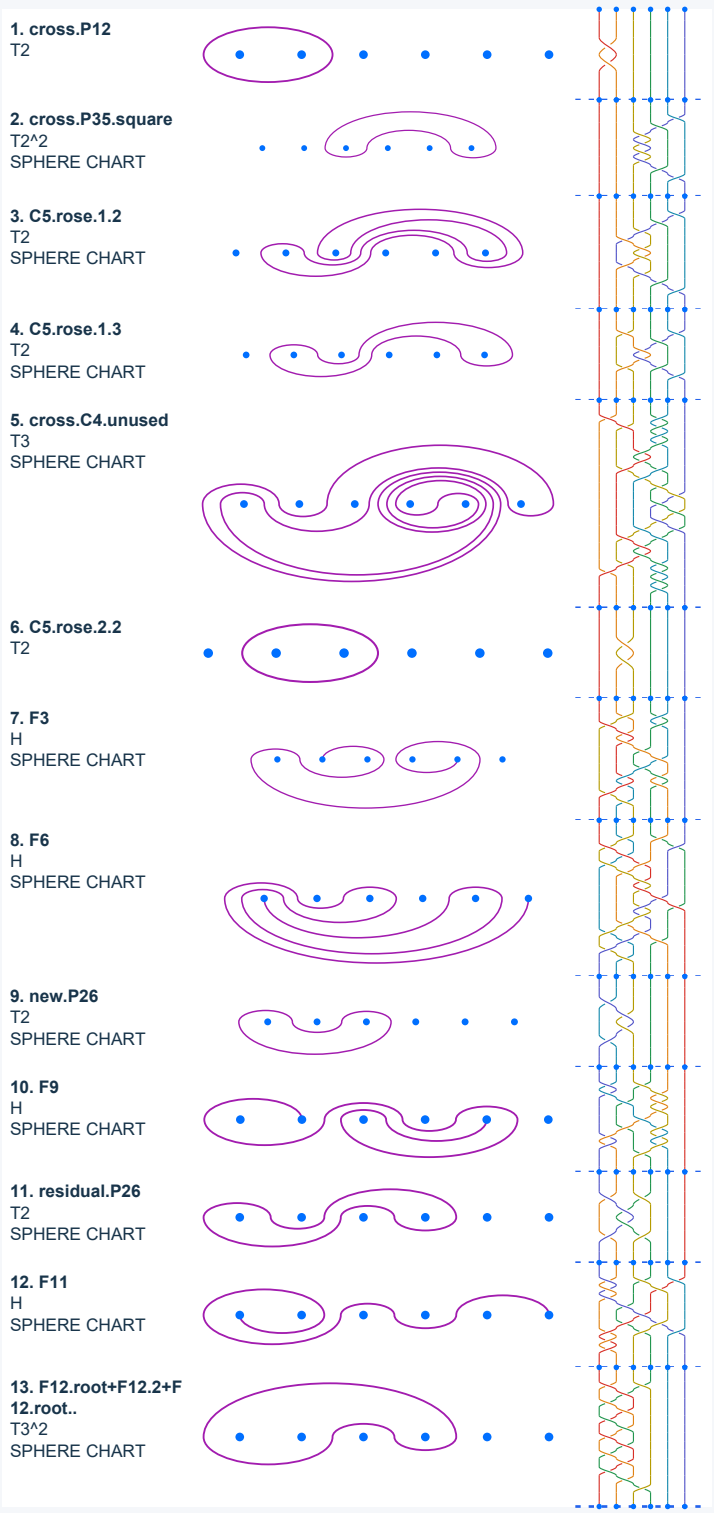
II - step 132

C5.rose.2.2 up past cross.C4.unused

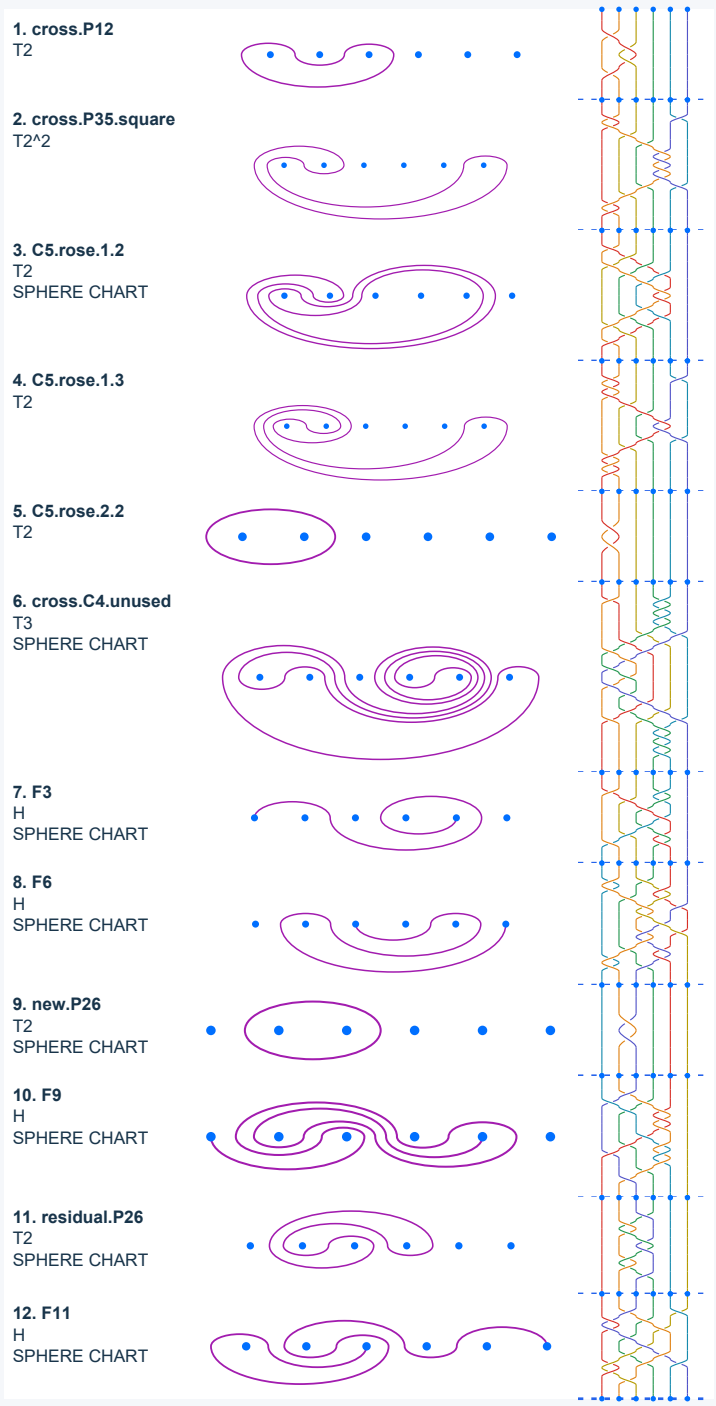
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 -2 5; after g=-4 -2 5.

Before



After



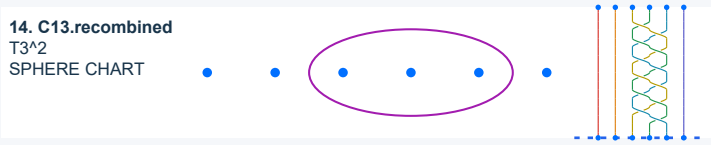
II - step 132 - continued

C5.rose.2.2 up past cross.C4.unused

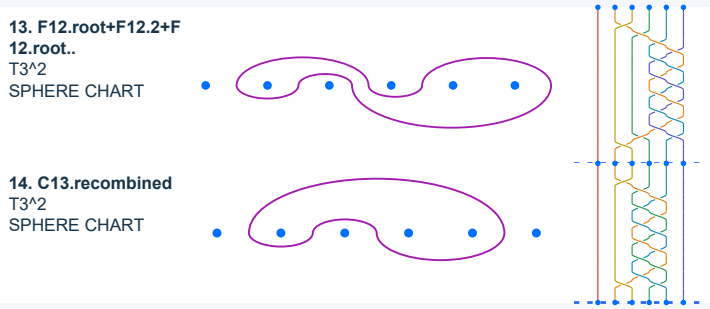
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 2 -4 -2 5; after g=-4 -2 5.

Before



After



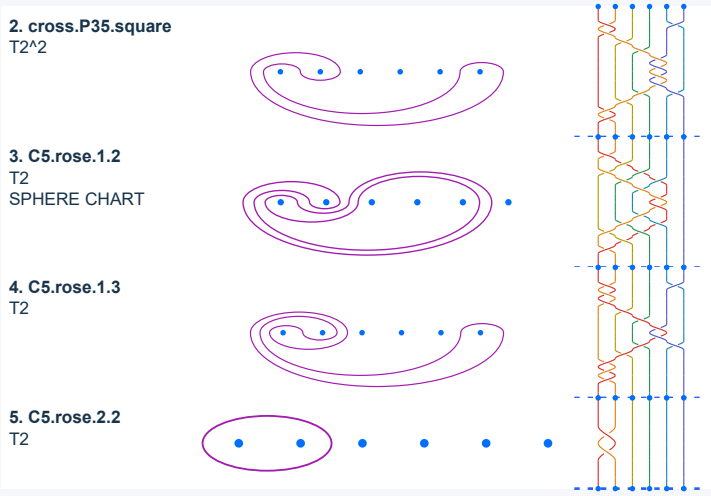
II - step 133

Compress the five 135 petals and retain inverse P15

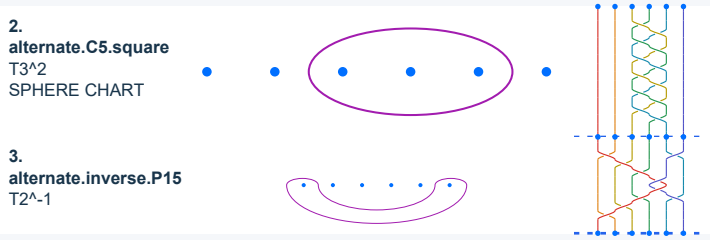
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=-4 -2 5.

Before



After



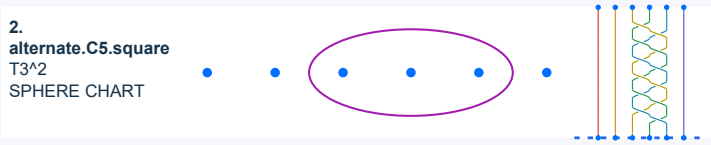
II - step 134

Use the half-rotated PC half lantern twice

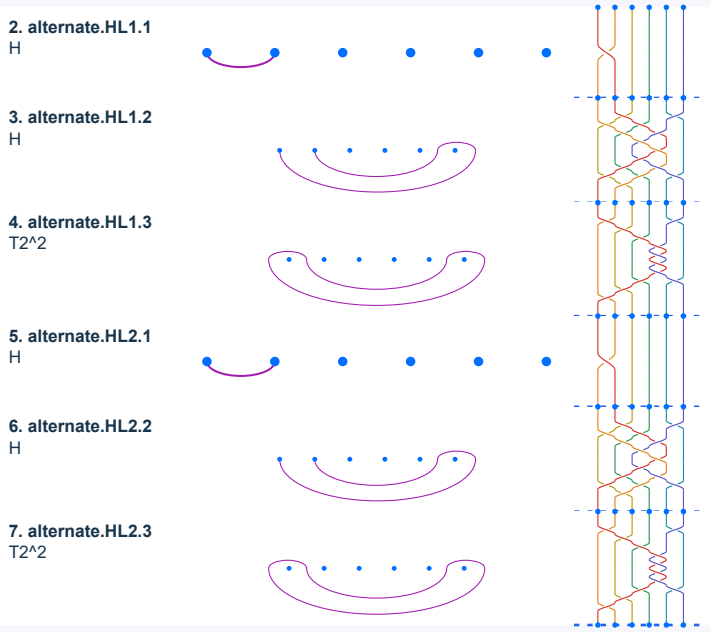
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 10 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=-4 -2 5.

Before



After



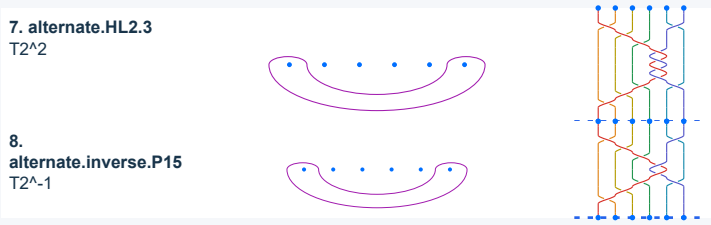
II - step 135

Cancel inverse P15 against one unit of the last squared pair

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 9 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=-4 -2 5.

Before



After



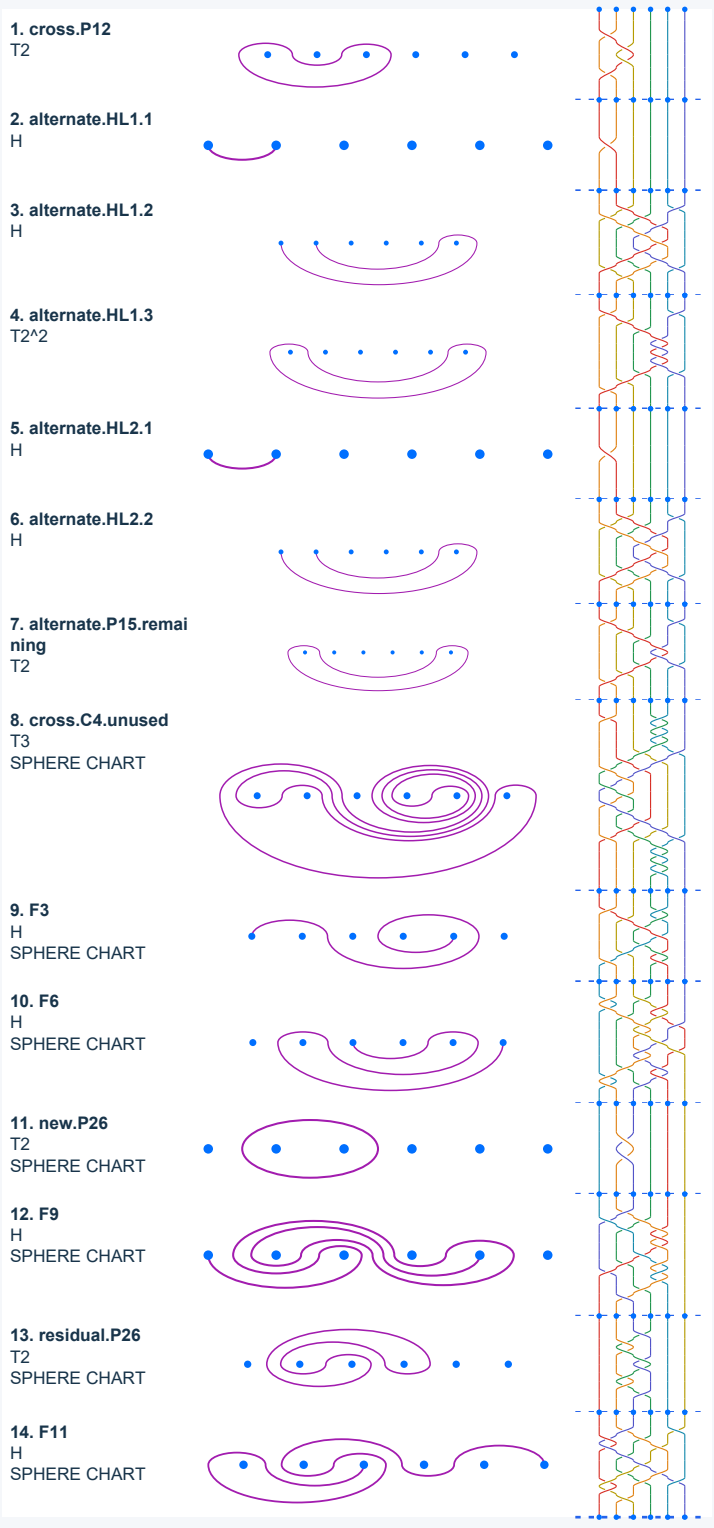
II - step 136

Expand the last ordinary C4 root by a PC rose

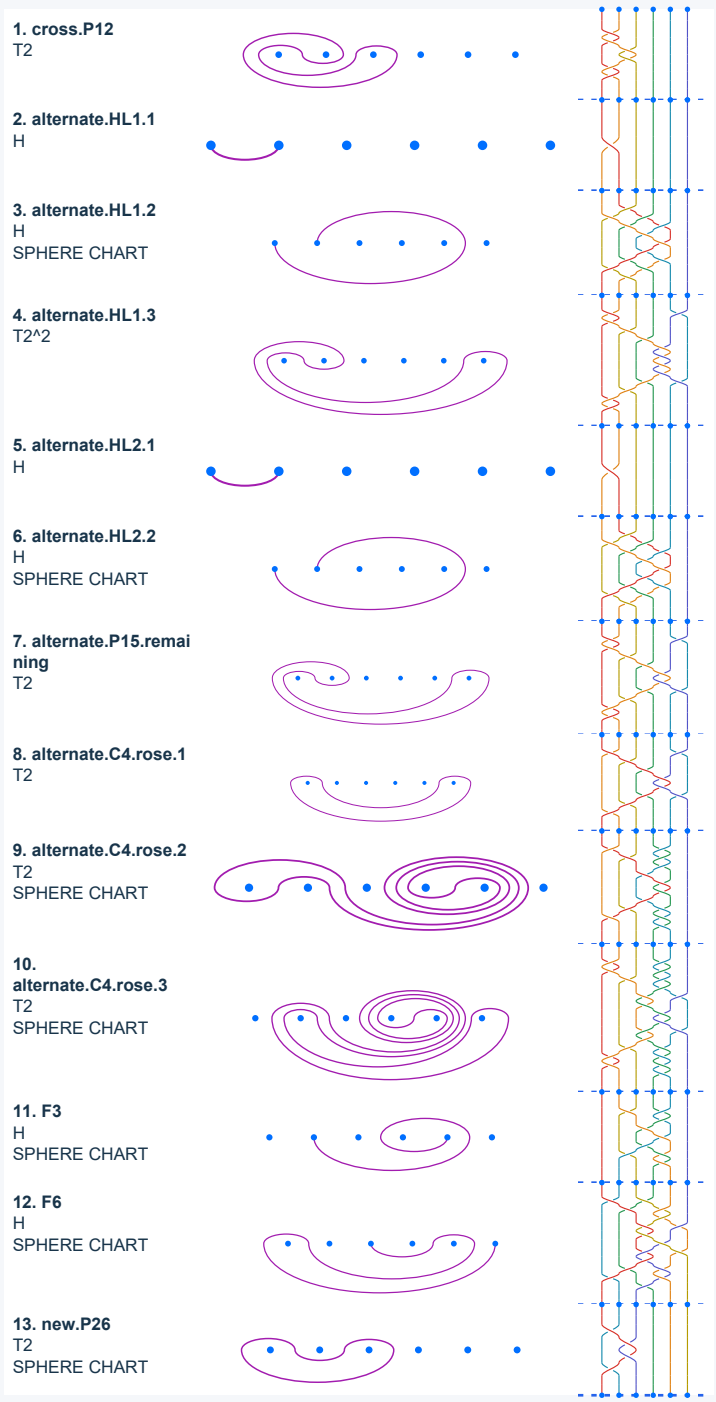
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=1 -4 -2 5.

Before



After



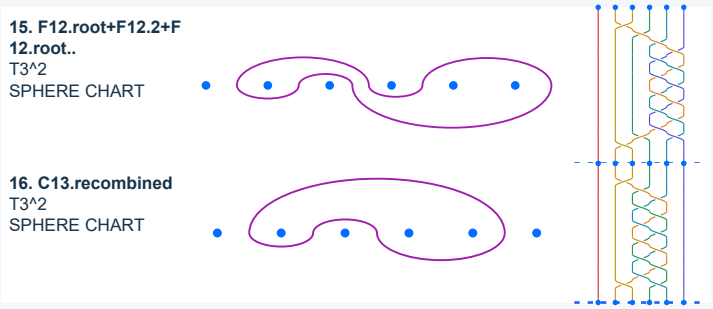
II - step 136 - continued

Expand the last ordinary C4 root by a PC rose

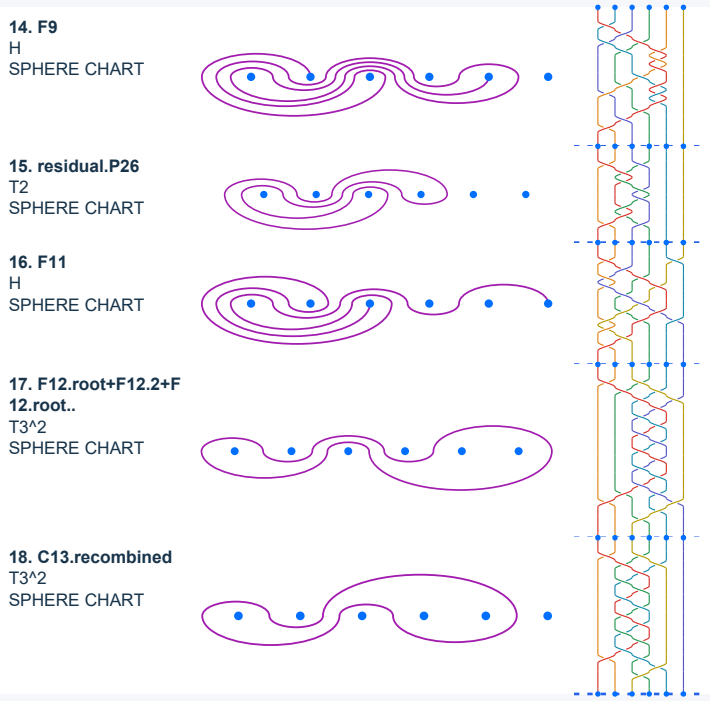
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-4 -2 5; after g=1 -4 -2 5.

Before



After



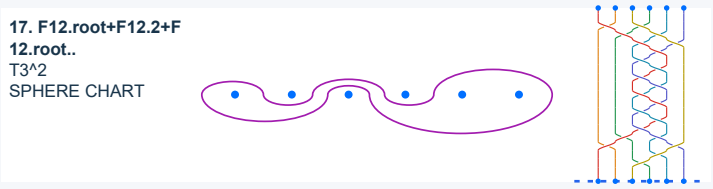
II - step 137

Draw the whole C12 square as H56 inverse applied to C13

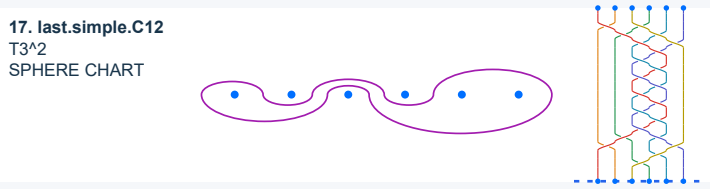
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

Before



After



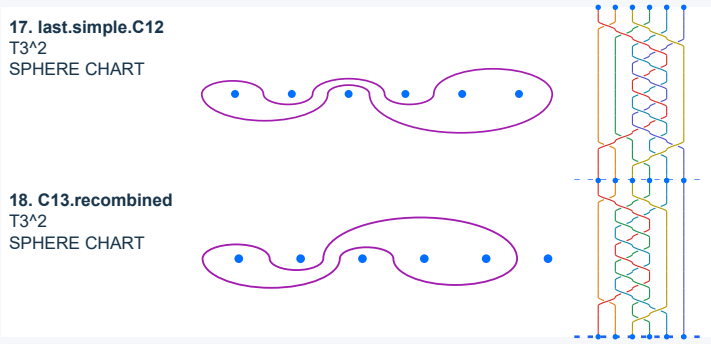
II - step 138

Apply the ordered two-lantern consequence; retain both inverses

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

Before



After



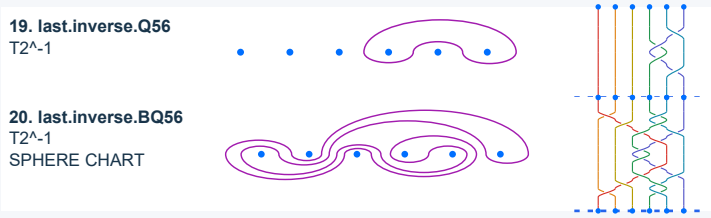
II - step 139

Transport the second inverse up across last.inverse.Q56

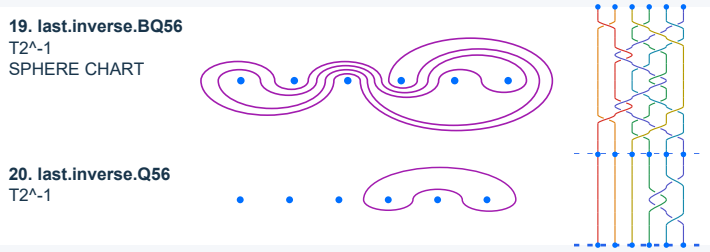
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

18 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

Before



After



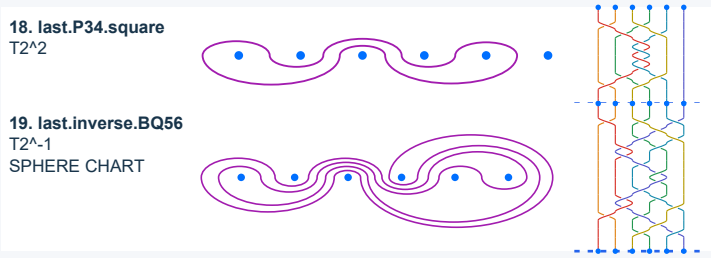
II - step 140

Transport the second inverse up across last.P34.square

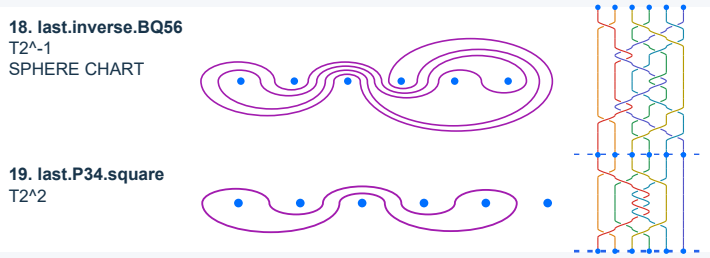
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

17 unchanged displayed factors above; 1 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

Before



After



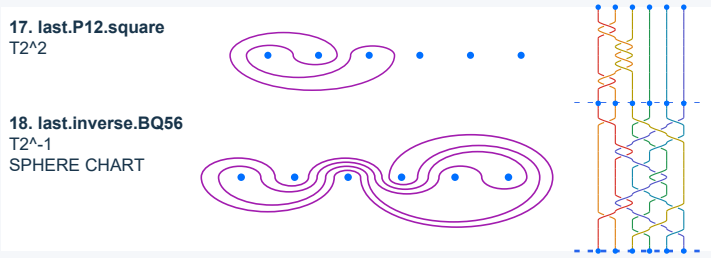
II - step 141

Transport the second inverse up across last.P12.square

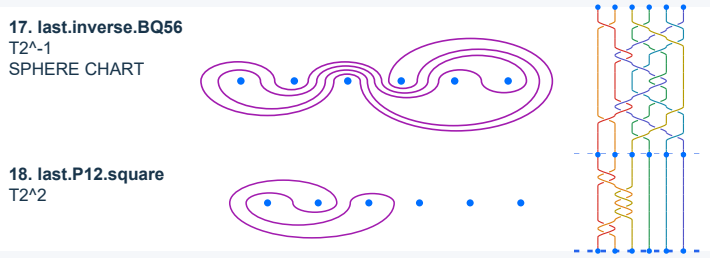
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 2 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=1 -4 -2 5.

Before



After



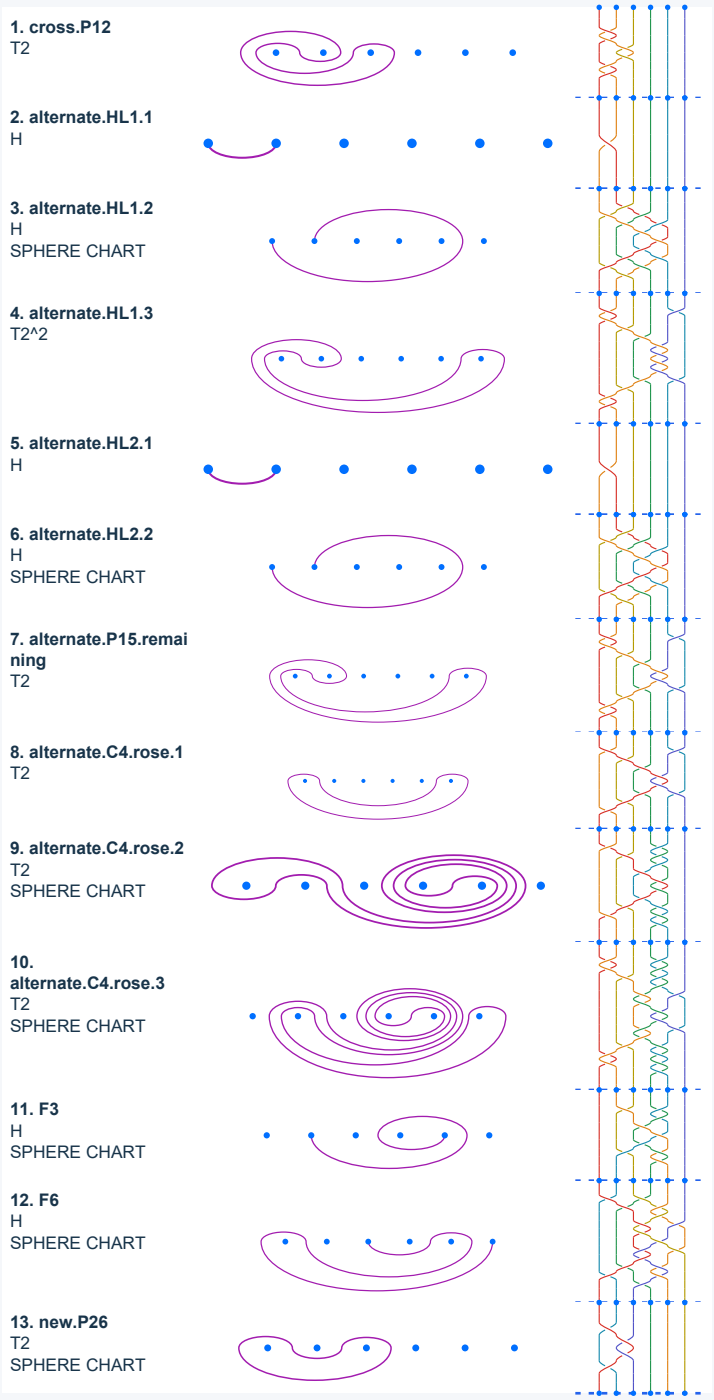
II - step 142

Transport the second inverse up across F11

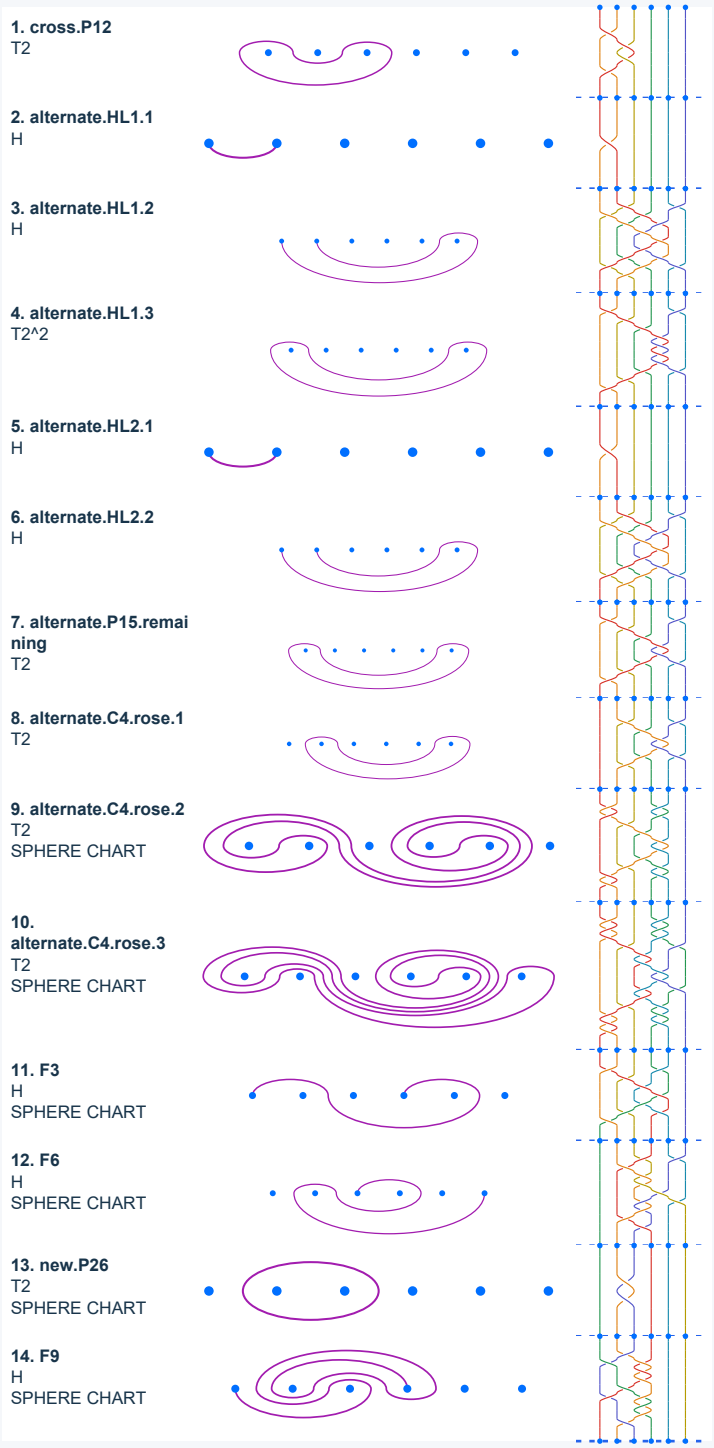
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=-2 5.

Before



After



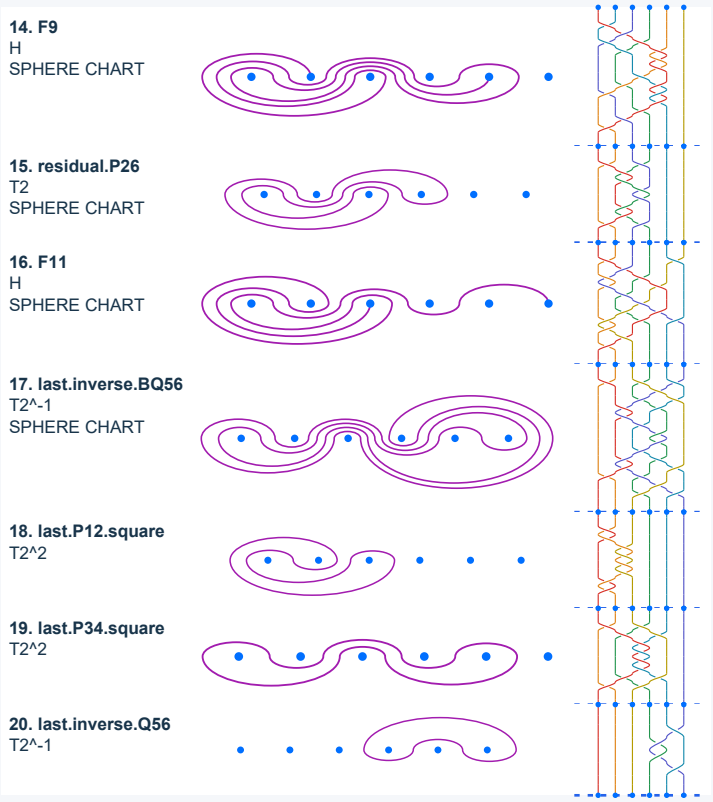
II - step 142 - continued

Transport the second inverse up across F11

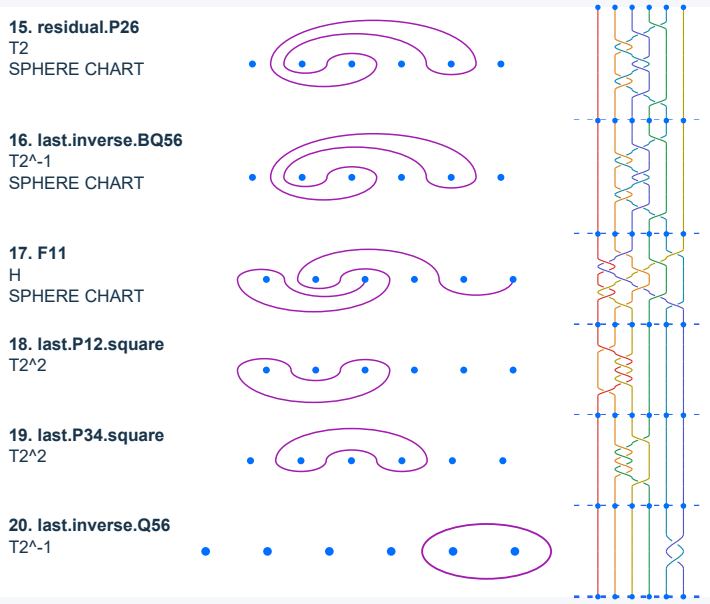
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=1 -4 -2 5; after g=-2 5.

Before



After



II - step 143

Use the exact sphere match with residual.P26

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

20 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 5; after g=-2 5.

No displayed row changes. The exact certificate records the change of source presentations.

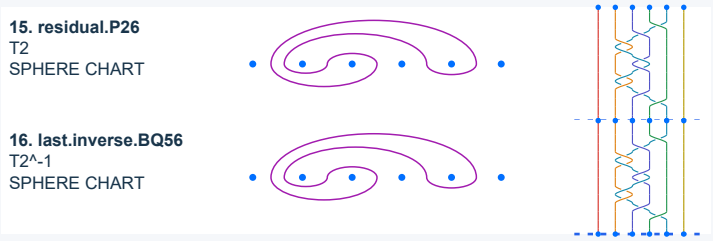
II - step 144

Cancel the matched positive and inverse pair exactly in the disk

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 4 below. Only the changed interval is shown. Display frames: before g=-2 5; after g=-2 5.

Before



After

Identity / no factors in this interval.

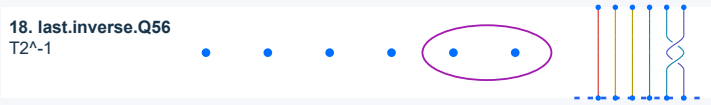
II - step 145

Replace inverse P56 using the squared sphere relation

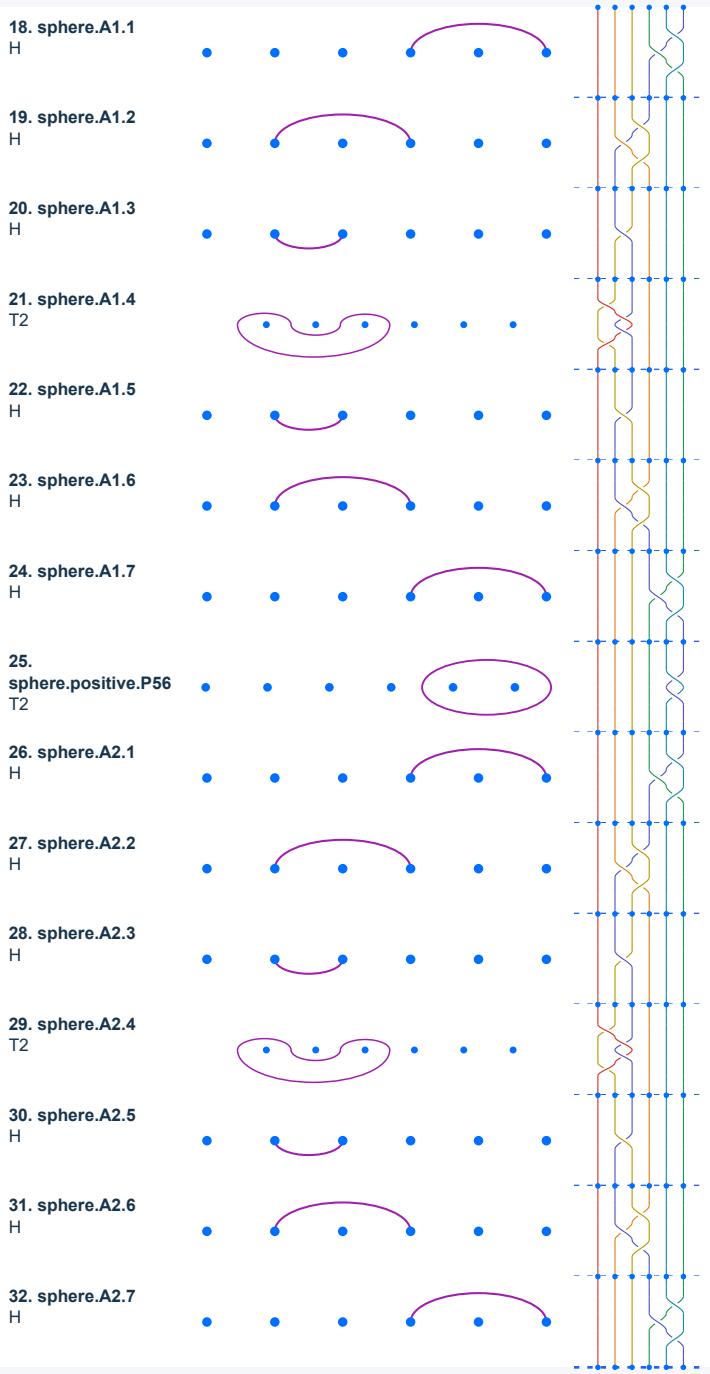
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

17 unchanged displayed factors above; 0 below. Only the changed interval is shown. Display frames: before g=-2 5; after g=-2 5.

Before



After



The actual positive fifty: complete factorization

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

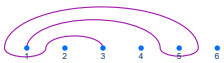
1. cross.P12
T2



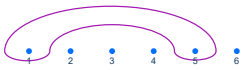
2. alternate.HL1.1
H



3. alternate.HL1.2
H
SPHERE CHART



4. alternate.HL1.3
T2^2
SPHERE CHART



5. alternate.HL2.1
H



6. alternate.HL2.2
H
SPHERE CHART



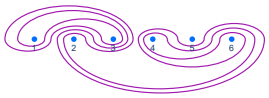
7. alternate.P15.remaining
T2
SPHERE CHART



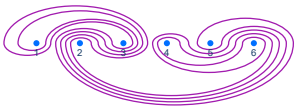
8. alternate.C4.rose.1
T2
SPHERE CHART



9. alternate.C4.rose.2
T2
SPHERE CHART



10. alternate.C4.rose.3
T2
SPHERE CHART



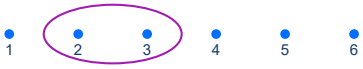
11. F3
H
SPHERE CHART



12. F6
H
SPHERE CHART



13. new.P26
T2
SPHERE CHART



14. F9
H
SPHERE CHART

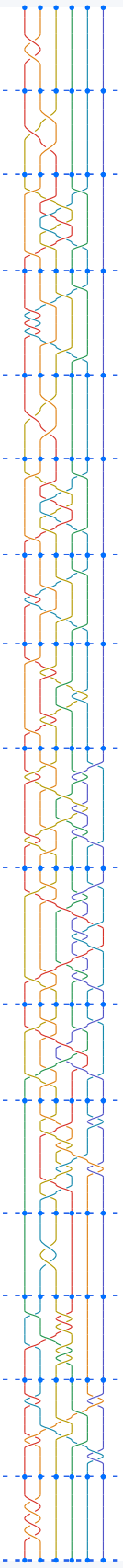


15. F11
H
SPHERE CHART



16. last.P12.square
T2^2





The actual positive fifty: complete factorization - continued

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

17. last.P34.square
T2^2



18. sphere.A1.1
H



19. sphere.A1.2
H



20. sphere.A1.3
H



21. sphere.A1.4
T2



22. sphere.A1.5
H



23. sphere.A1.6
H



24. sphere.A1.7
H



25. sphere.positive.P56
T2



26. sphere.A2.1
H



27. sphere.A2.2
H



28. sphere.A2.3
H



29. sphere.A2.4
T2



30. sphere.A2.5
H

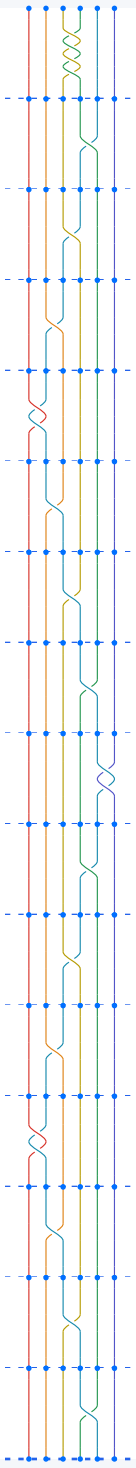


31. sphere.A2.6
H



32. sphere.A2.7
H





III. Isolate an actual standard twenty

Seven disk Hurwitz moves and one explicit sphere rechart isolate the twenty at the bottom. It is a disk conjugate of W0. The remaining thirty is not assumed to be W1.

III - step 001

Move F3 up: transport the adjacent {5,6} pair

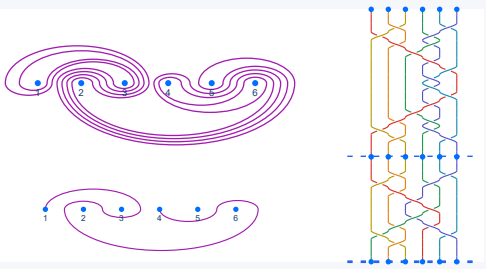
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 21 below. Only the changed interval is shown.

Before

10.
alternate.C4.rose.3
T2
SPHERE CHART

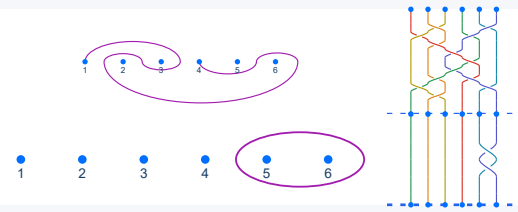
11. F3
H
SPHERE CHART



After

10. F3
H
SPHERE CHART

11.
alternate.C4.rose.3
T2
SPHERE CHART



III - step 002

SPHERE MOVE: rechart the transported pair as the round T56

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

10 unchanged displayed factors above; 21 below. Only the changed interval is shown.

Before

11.
alternate.C4.rose.3
T2
SPHERE CHART



After

11.
alternate.C4.rose.3
T2



III - step 003

Move the round T56 down, conjugating the crossed factor

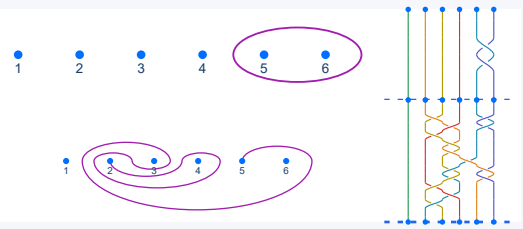
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 20 below. Only the changed interval is shown.

Before

11.
alternate.C4.rose.3
T2

12. F6
H
SPHERE CHART



After

11. F6
H
SPHERE CHART

12.
alternate.C4.rose.3
T2



III - step 004

Move the round T56 down, conjugating the crossed factor

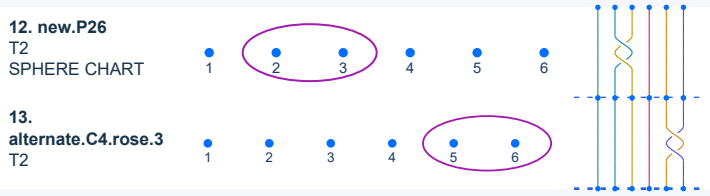
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 19 below. Only the changed interval is shown.

Before



After



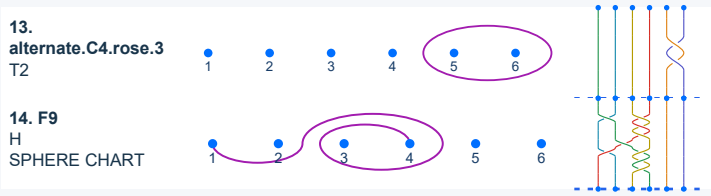
III - step 005

Move the round T56 down, conjugating the crossed factor

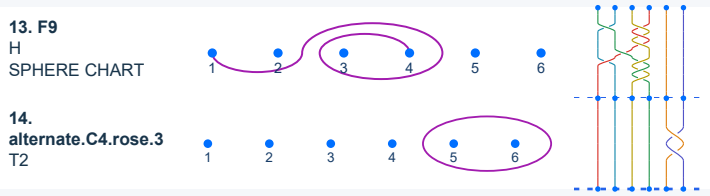
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 18 below. Only the changed interval is shown.

Before



After



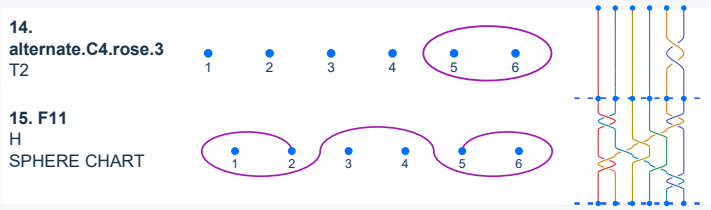
III - step 006

Move the round T56 down, conjugating the crossed factor

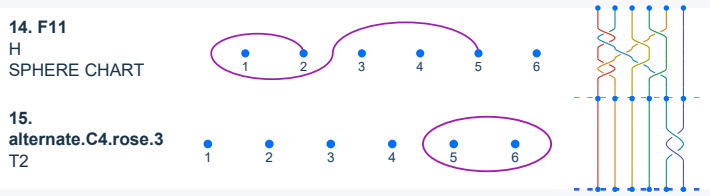
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 17 below. Only the changed interval is shown.

Before



After



III - step 007

Move the round T56 down, conjugating the crossed factor

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 16 below. Only the changed interval is shown.

Before



After



III - step 008

Move the round T56 down, conjugating the crossed factor

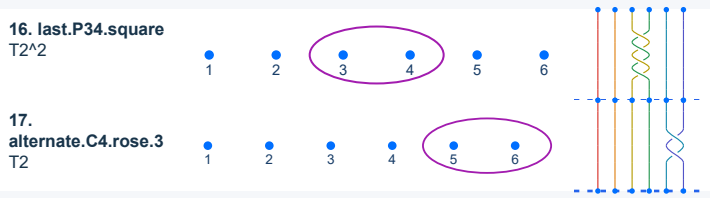
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 15 below. Only the changed interval is shown.

Before



After



The isolated standard twenty

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. alternate.C4.rose.3
T2



2. sphere.A1.1
H



3. sphere.A1.2
H



4. sphere.A1.3
H



5. sphere.A1.4
T2



6. sphere.A1.5
H



7. sphere.A1.6
H



8. sphere.A1.7
H



9. sphere.positive.P56
T2



10. sphere.A2.1
H



11. sphere.A2.2
H



12. sphere.A2.3
H



13. sphere.A2.4
T2



14. sphere.A2.5
H

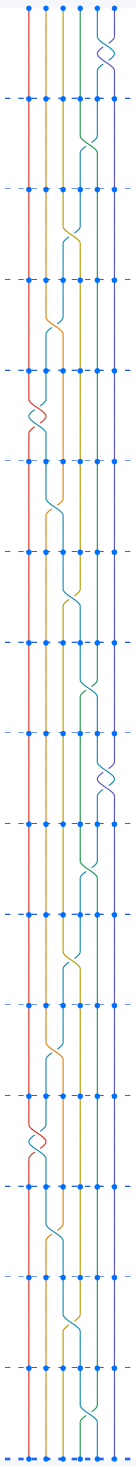


15. sphere.A2.6
H



16. sphere.A2.7
H





IV. The thirty and its exact disk-boundary ending

First remove the independently verified standard twenty in an explicit sphere move. The remaining proof does not assume the thirty product is identity. Every chart change is checked locally, the ordered disk product is checked on all five based meridians, and the final double cap is explicit.

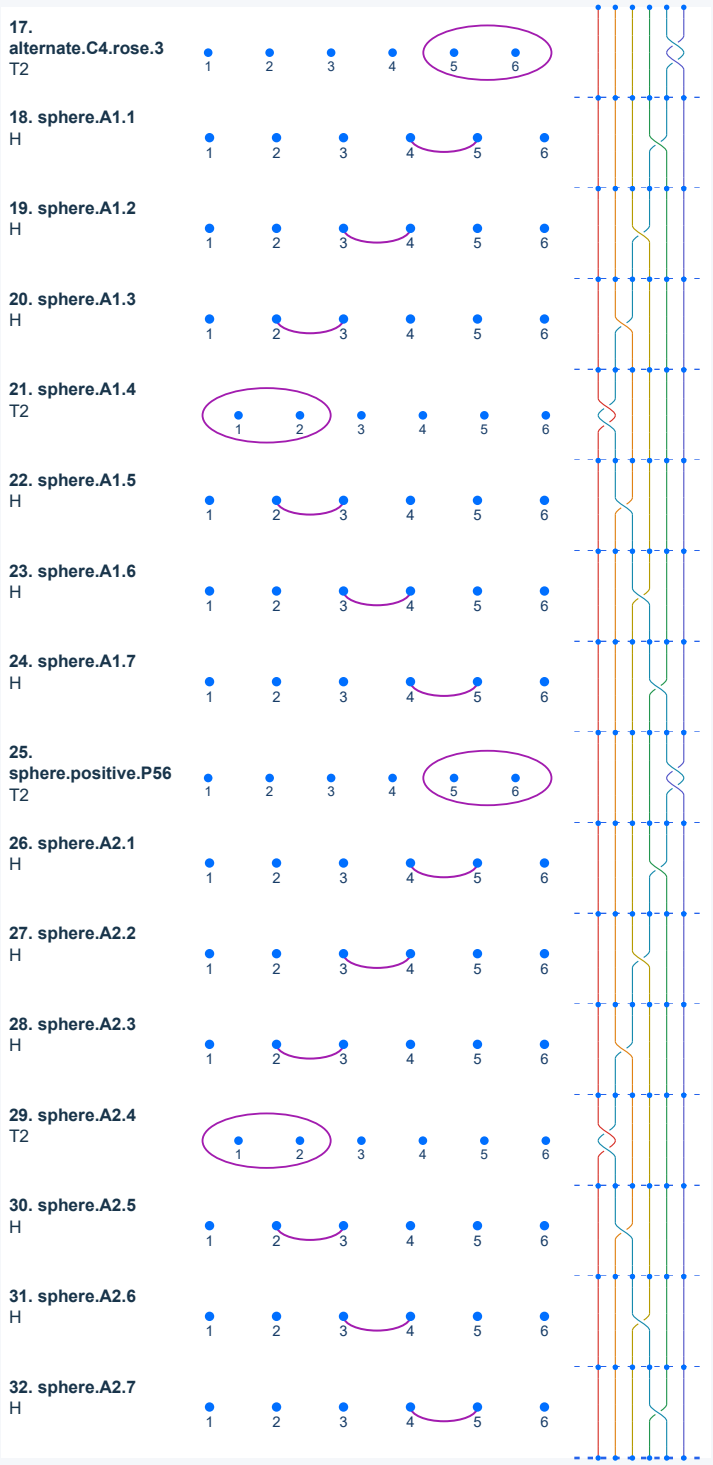
IV - step 001

SPHERE MOVE: close the isolated standard twenty (W0)

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After

Identity / no factors in this interval.

IV - step 002

Use a disk-equal five-point representative of cross.P12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 003

Use a disk-equal five-point representative of alternate.HL1.1

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 004

SPHERE MOVE: put point 6 at infinity for alternate.HL1.2

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

2 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before

3. alternate.HL1.2
H
SPHERE CHART



After

3. alternate.HL1.2
H



IV - step 005

SPHERE MOVE: put point 6 at infinity for alternate.HL1.3

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

3 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before

4. alternate.HL1.3
T2^2
SPHERE CHART



After

4. alternate.HL1.3
T2^2



IV - step 006

Use a disk-equal five-point representative of alternate.HL2.1

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 007

SPHERE MOVE: put point 6 at infinity for alternate.HL2.2

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

5 unchanged displayed factors above; 10 below. Only the changed interval is shown.

Before

6. alternate.HL2.2
H
SPHERE CHART



After

6. alternate.HL2.2
H



IV - step 008

SPHERE MOVE: put point 6 at infinity for alternate.P15.remaining

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

6 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before

7. alternate.P15.remaining
T2
SPHERE CHART



After

7. alternate.P15.remaining
T2



IV - step 009

SPHERE MOVE: put point 6 at infinity for alternate.C4.rose.1

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

7 unchanged displayed factors above; 8 below. Only the changed interval is shown.

Before

8. alternate.C4.rose.1
T2
SPHERE CHART



After

8. alternate.C4.rose.1
T2



IV - step 010

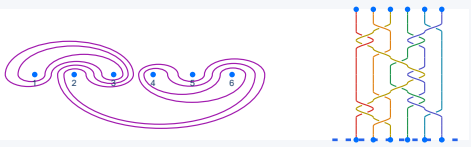
SPHERE MOVE: put point 6 at infinity for alternate.C4.rose.2

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown.

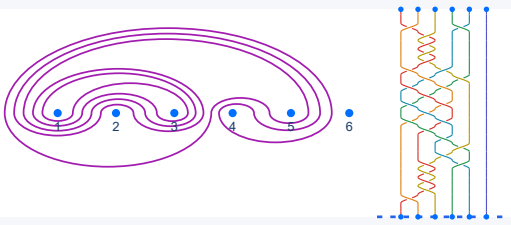
Before

9. alternate.C4.rose.2
T2
SPHERE CHART



After

9. alternate.C4.rose.2
T4



IV - step 011

SPHERE MOVE: put point 6 at infinity for F3

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 012

SPHERE MOVE: put point 6 at infinity for F6

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

10 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before

11. F6
H
SPHERE CHART



After

11. F6
H



IV - step 013

SPHERE MOVE: put point 6 at infinity for new.P26

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

11 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before

12. new.P26
T2
SPHERE CHART



After

12. new.P26
T2



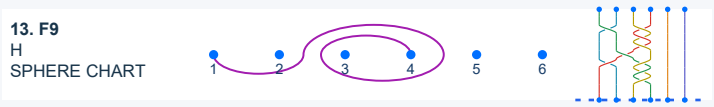
IV - step 014

SPHERE MOVE: put point 6 at infinity for F9

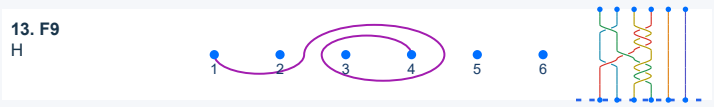
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



IV - step 015

SPHERE MOVE: put point 6 at infinity for F11

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

13 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



IV - step 016

Use a disk-equal five-point representative of last.P12.square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

IV - step 017

Use a disk-equal five-point representative of last.P34.square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown.

No displayed row changes. The exact certificate records the change of source presentations.

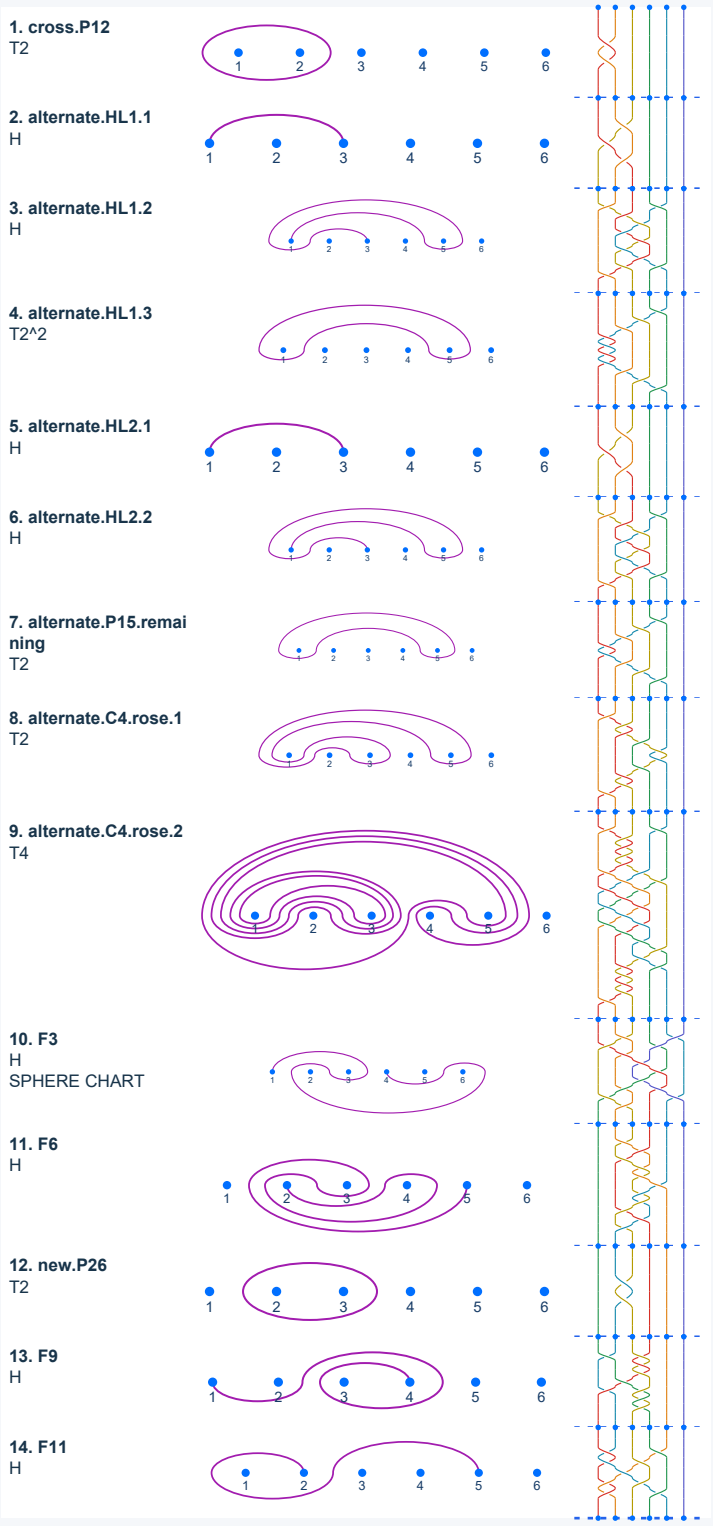
IV - step 018

Exact five-point disk relation: double outer-boundary twist

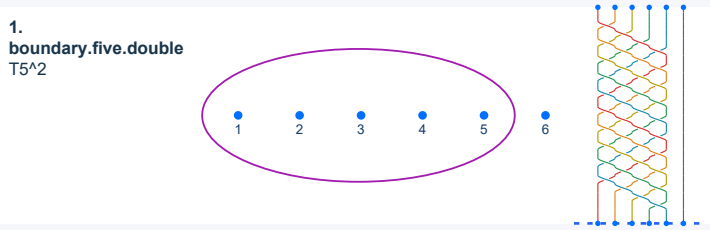
EXACT DISK BOUNDARY RELATION: independently calculated based Artin actions agree.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



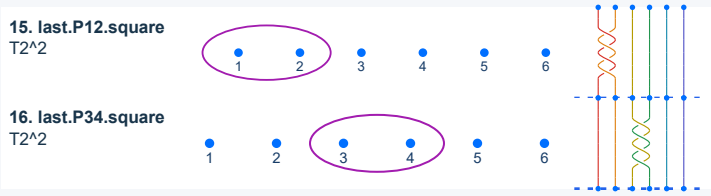
IV - step 018 - continued

Exact five-point disk relation: double outer-boundary twist

EXACT DISK BOUNDARY RELATION: independently calculated based Artin actions agree.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After

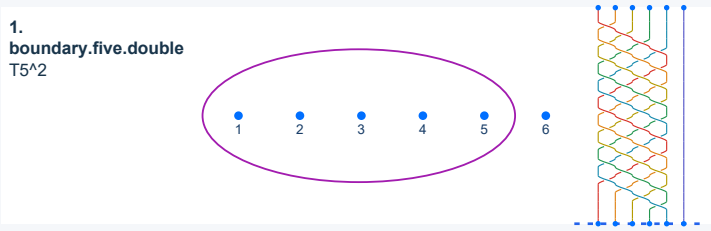
IV - step 019

SPHERE MOVE: cap the double boundary around point 6

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After

Identity / no factors in this interval.

V. The six-point sphere cut-system certificate

The visible comparison uses all six marked points and five consecutive-point arcs. There is no boundary-to-first-point arc. The independent faithful five-point disk calculation is retained as algebraic backup, with point 6 at infinity.

Calculated product, on the sphere



Reference six-point sphere chain



All five adjacent-point arcs are recovered from the action and jointly certified. The separate disk calculation has the following equal actual/reference meridian images:

x1 -> 1 2 3 4 5 1 2 3 4 5 1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1

x2 -> 1 2 3 4 5 1 2 3 4 5 2 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1

x3 -> 1 2 3 4 5 1 2 3 4 5 3 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1

x4 -> 1 2 3 4 5 1 2 3 4 5 4 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1

x5 -> 1 2 3 4 5 1 2 3 4 5 -4 -3 -2 -1 -5 -4 -3 -2 -1

SPHERE MOVE: a five-point curve is complementary to a singleton curve.

The singleton twist is trivial in the marked-sphere mapping class group, so (sigma1 sigma2 sigma3 sigma4)^5 caps to that identity. Spherical braids retain a central order-two rotation, which is quotiented out when passing to sphere mapping classes. The chosen genus-two lift of this single twist is -I. Its square caps with +I, preserving the lifted identity in this proof. A framed inner boundary cannot be discarded this way. Both disk discrepancies and lift matrices are saved in sphere-cut-certificate.json.

VI. Sources, limitations and downloadable certificates

The new ending is an independent exact boundary-action proof. The classification theorem is not used to infer identity. A shorter geometric derivation and an elementary full W0 W1 Hurwitz route remain separate improvements.

Graphical working seed and saved substitution route:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-compact/index.html

New twenty extraction, thirty boundary proof and certificates:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/index.html

Auroux, Fiber sums of genus 2 Lefschetz fibrations, https://arxiv.org/pdf/math/0204285. Lemmas 6-7 concern standard-block invariance; Theorem 9 recognizes transitive positive identity factorizations. Its identity hypothesis must not be reversed.

Scope: sphere mapping-class action and the separately checked closed genus-two lift. No hidden disk equality, cumulative joint support drawing, or unperformed elementary Hurwitz normal form is claimed.

Source SHA256: 05dbc99bc8137ce2a716bef51080ef25abfd13fa29f08c95f9bde68ffbe8c417

All 177 atomic transitions in the linked chain are retained: 5 working-chart joins, 145 saved substitutions, 8 twenty-extraction moves, 1 explicit twenty closure, and 18 thirty-closure steps.